Main points of applied avian histology

1st part

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Digital Veterinary Histology

LEARNING OBJECTIVES (aims)

The information in this chapter should enable you to do the following:

1. Identify &define key structural and histological features of avian tissues &organs .

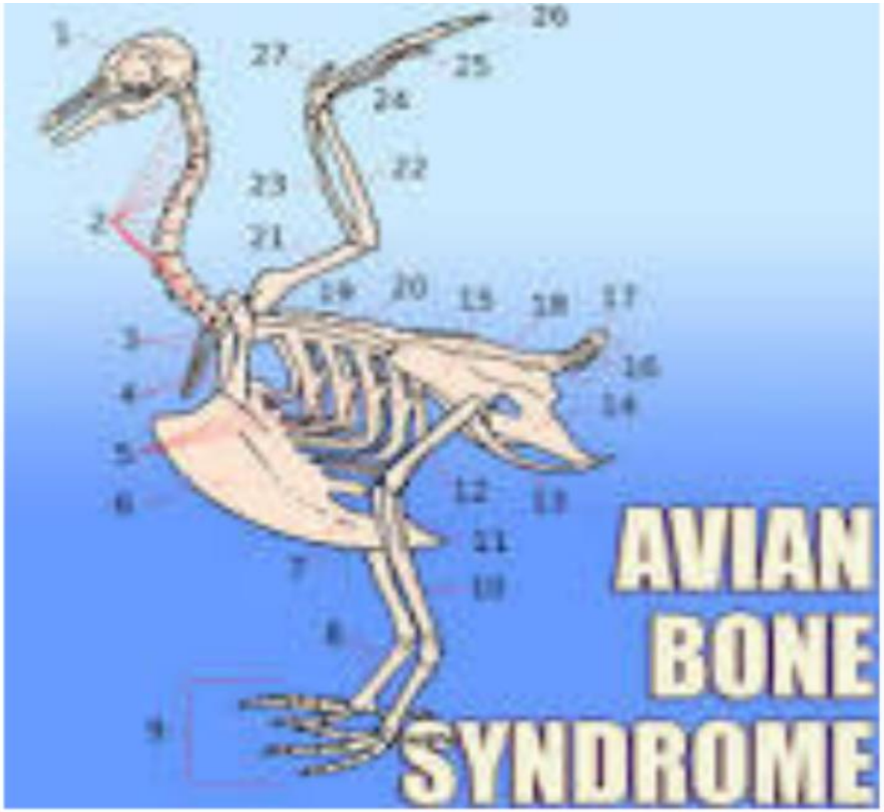
2. Explain &identify normal avian histology tissues in comparison with mammalian tissues.

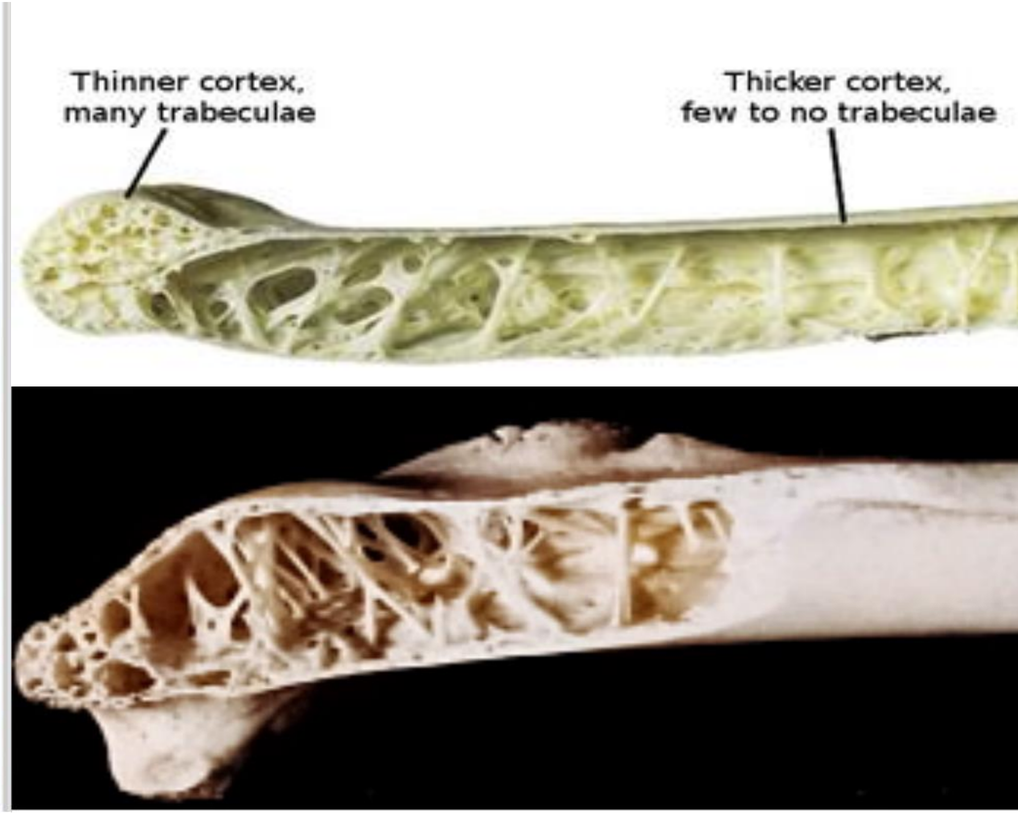
3. Understand using a comparative approach, key species differences in avian histology.

4. Describe & explain mechanisms of tissue/organ repair, by which cells, organs, tissues, &avian species in attempt to maintain homeostasis.

Introduction: This chapter describes only major differences in histology from that of the mammals. As, avian histology is consider a scientific base for avian histopathology &patholophysiology.

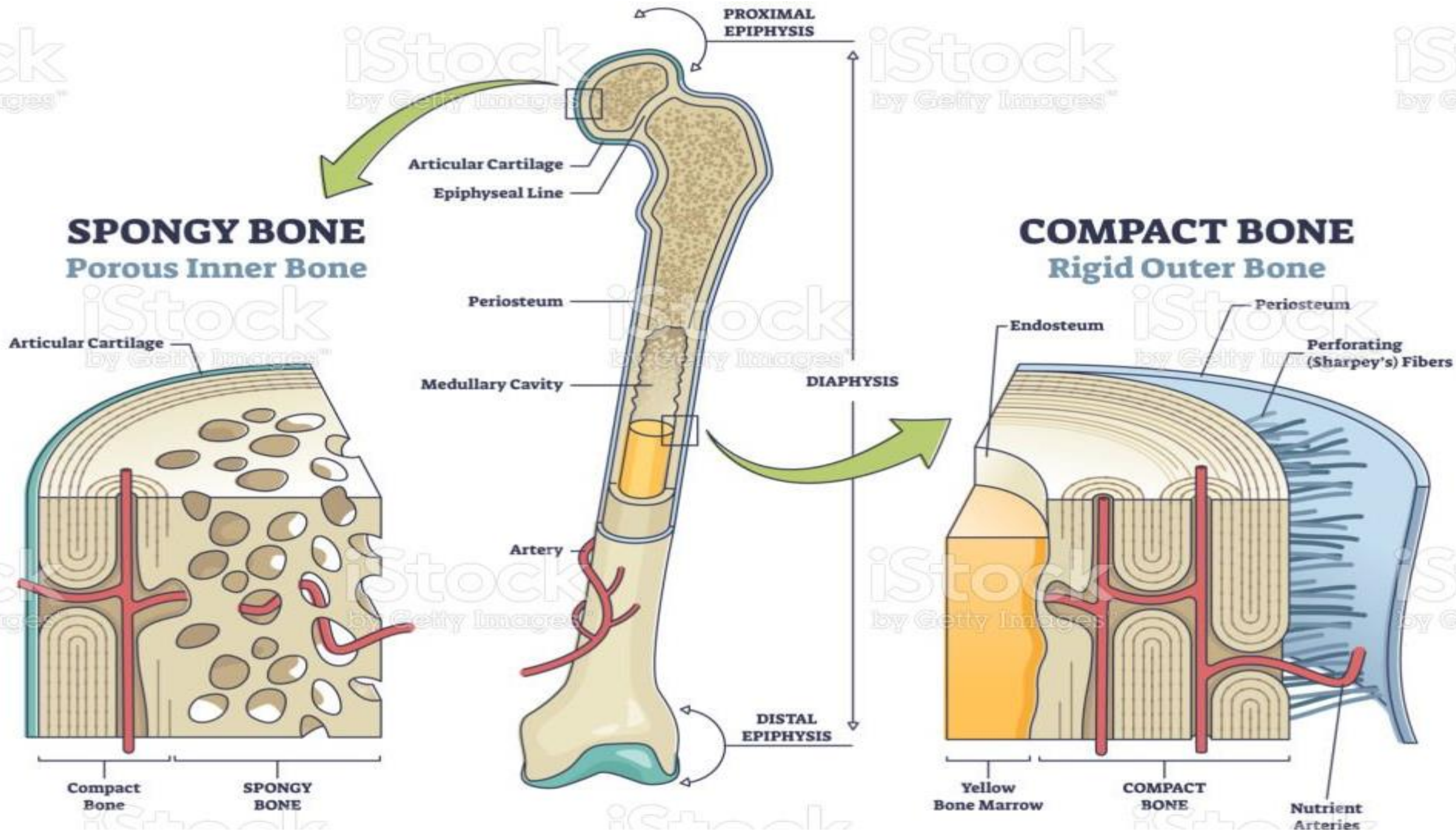
Avian bone tissue contains more inorganic compounds (84%)





Avian bone tissue contains more inorganic compounds (84% minerals)than mammalian bone tissue (70% minerals). Avian bone therefore is hard &poorly elastic &splinters easily.

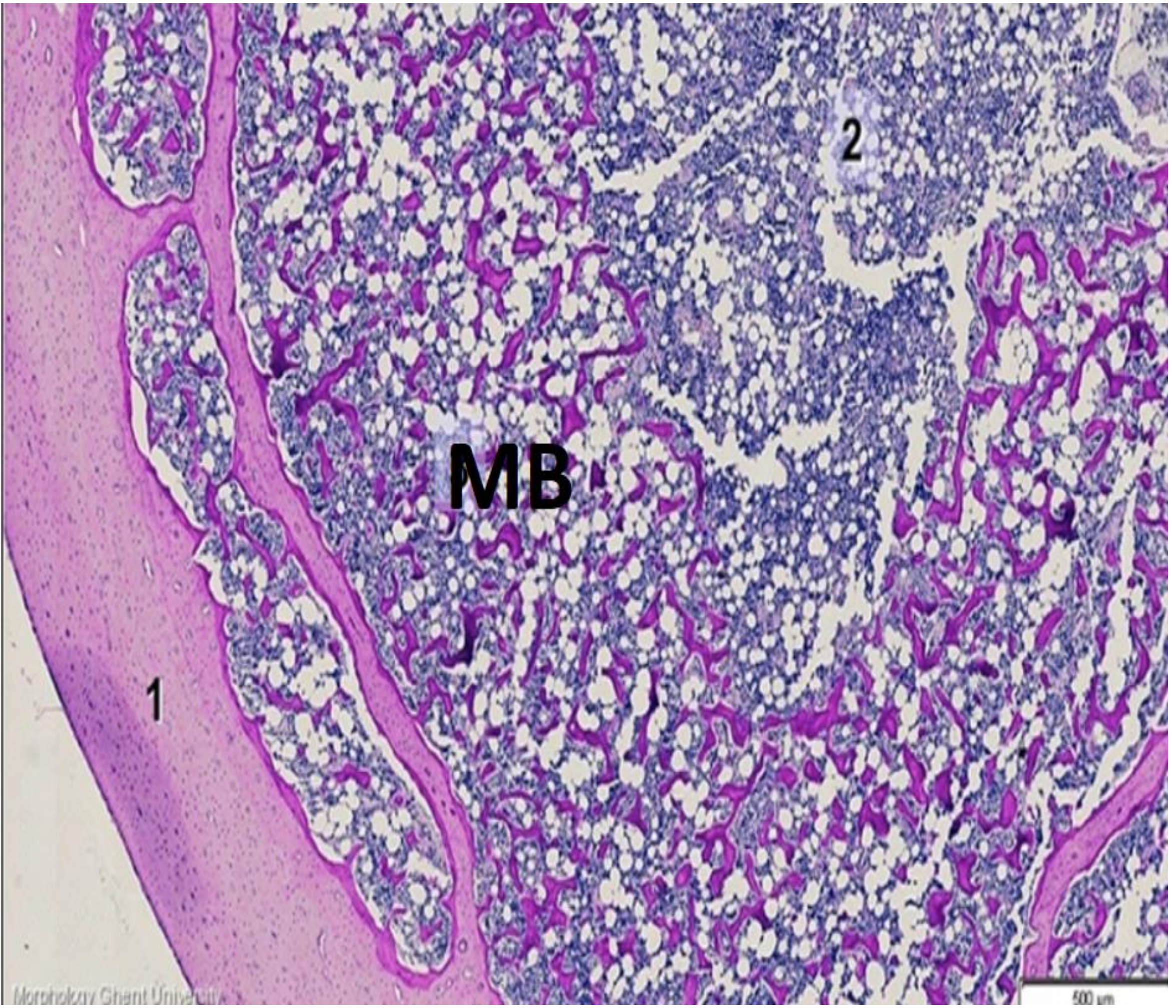
Types of avian bone tissue

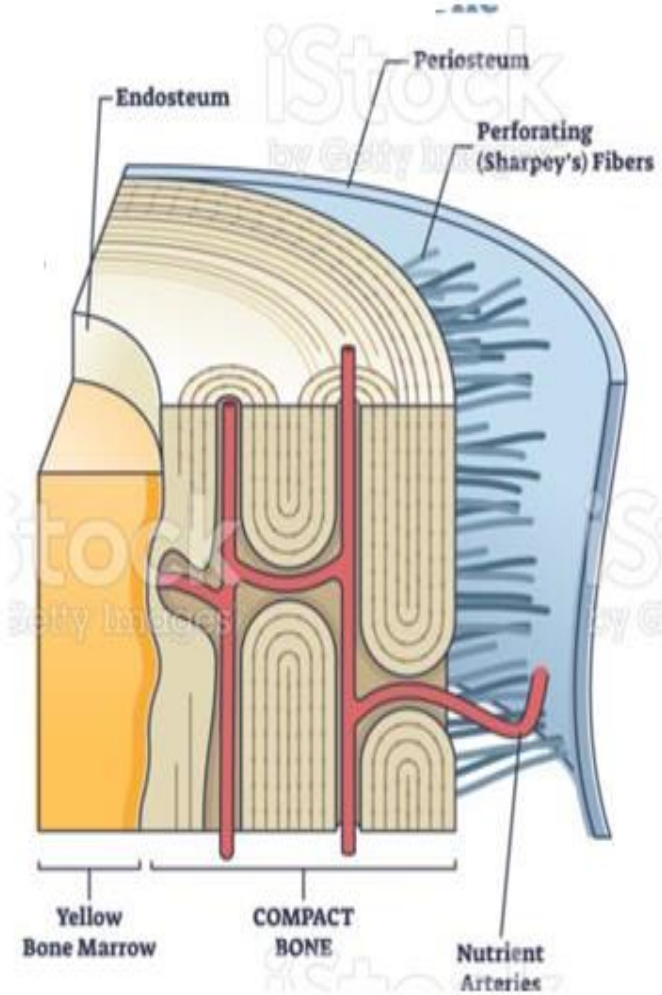


1-Spongy bone

2-Compact bone

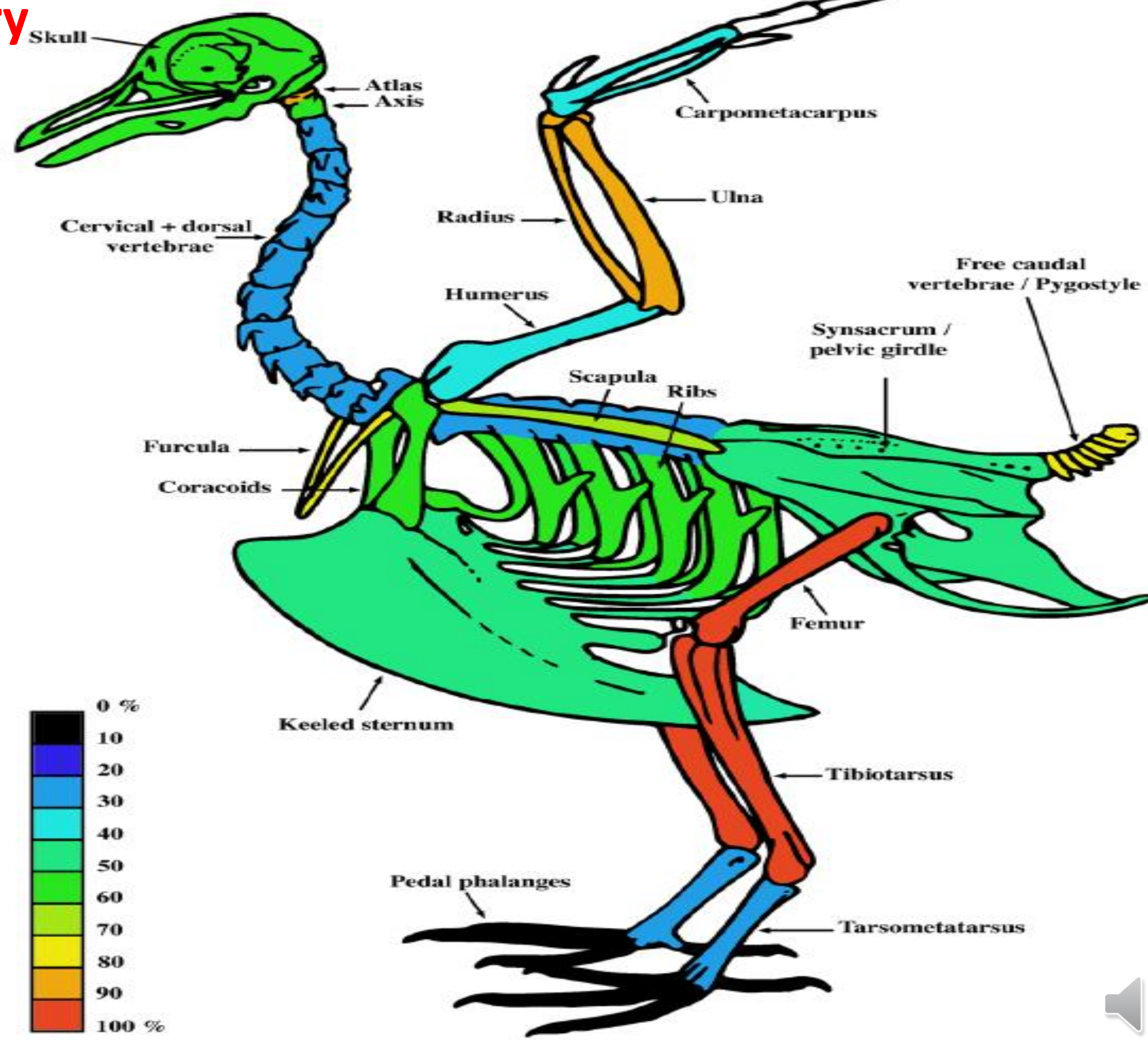
Medullary bone(MB)





A special endosteal tissue forming in the bones of female birds during egg laying to serve as a labile calcium reservoir for building the hard eggshell.

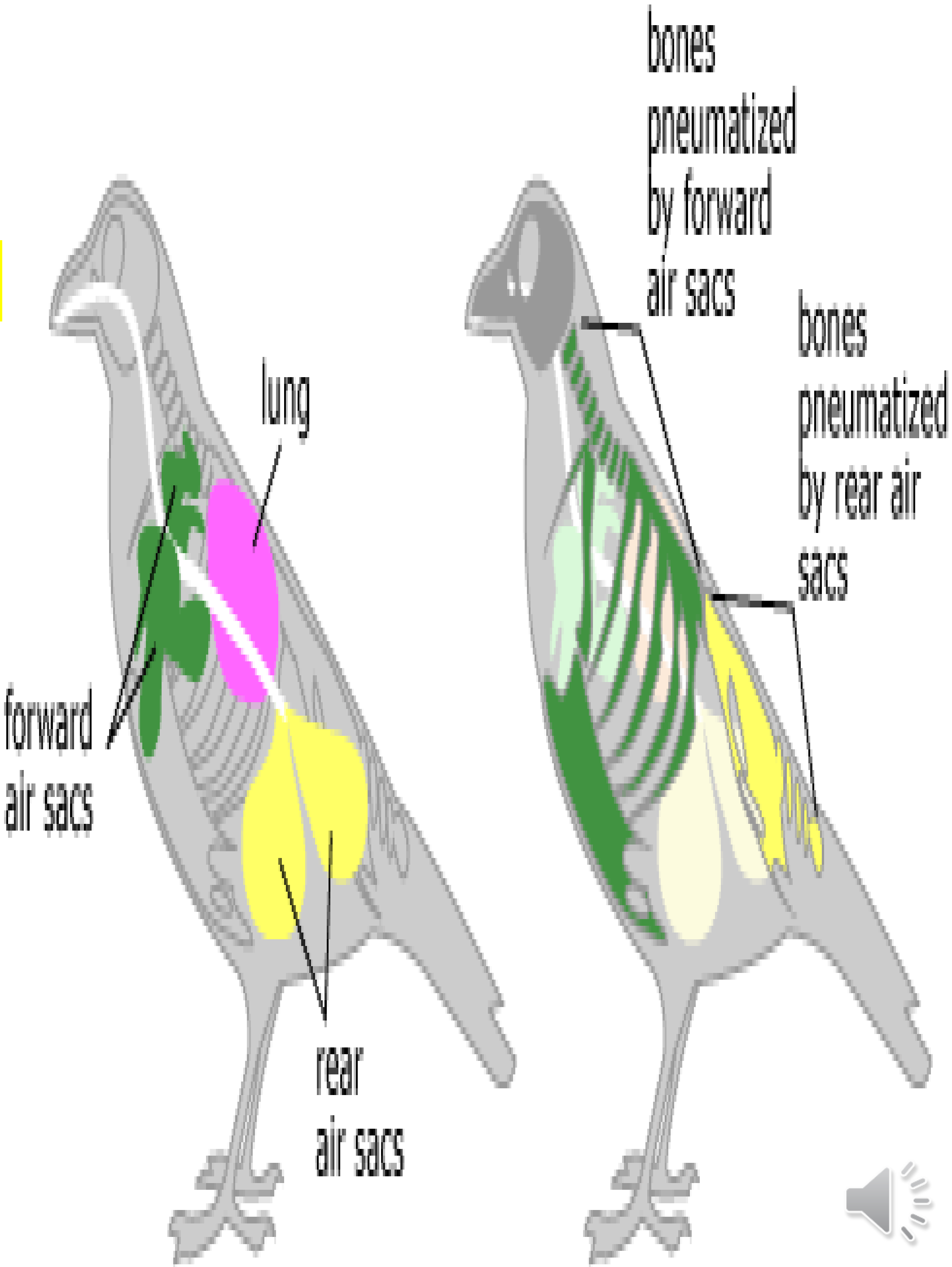
Medullary Bone (MB)



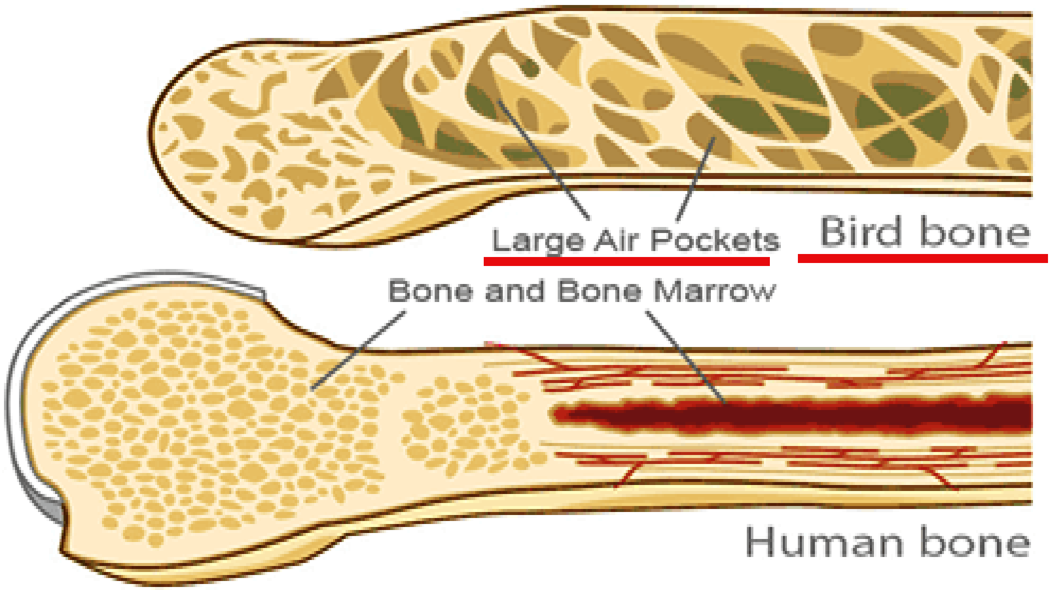
Pneumatic Bones

In adult birds, especially good flying birds, some bones are hollow&connected to the respiratory system through the air sacs. eg

1- Skull
2- Humerus
3- Clavicle
4- Keel
5- Lumbar vertebra
6- Sacral vertebra



Pneumatic bone



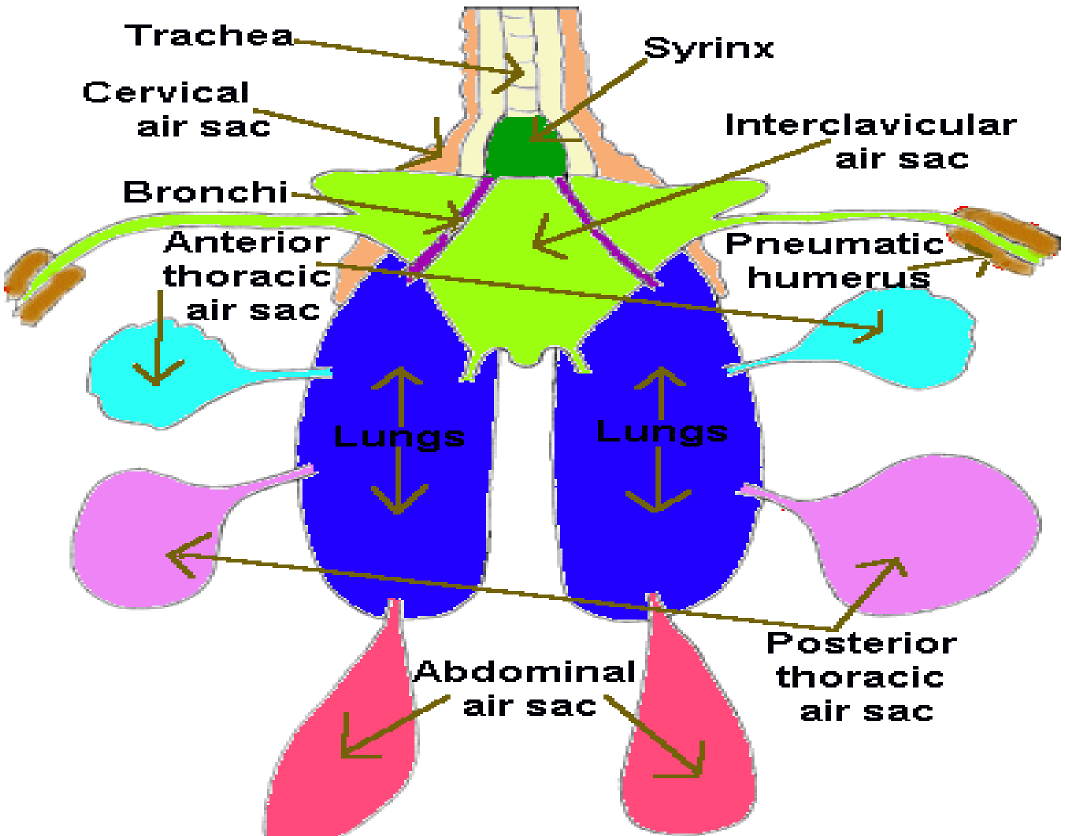
Pneumatic Avian Bone

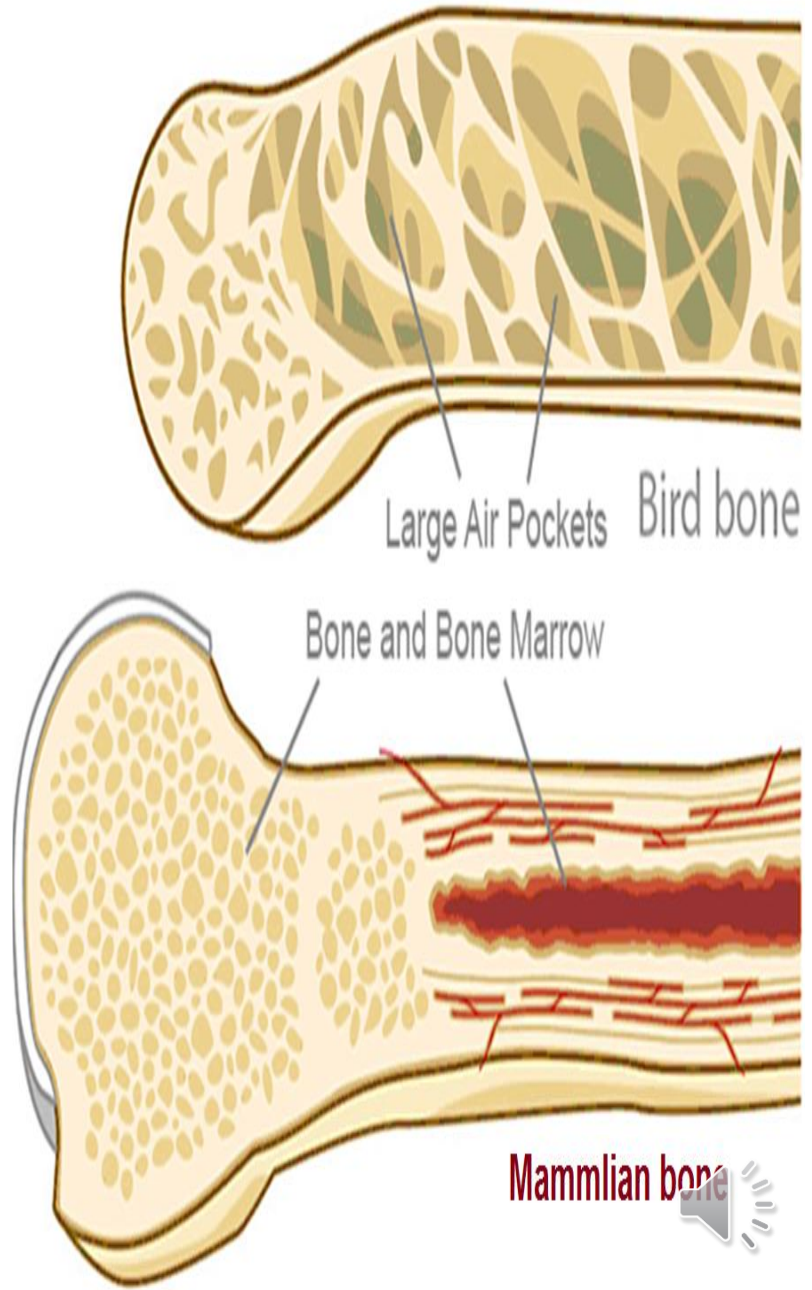
Pneumatic bone has a **honeycomb structure** No. **4** (air pockets)on cross-section.

Pneumatic bone is much more extensive in birds that fly for long distances, e.g. songbirds.

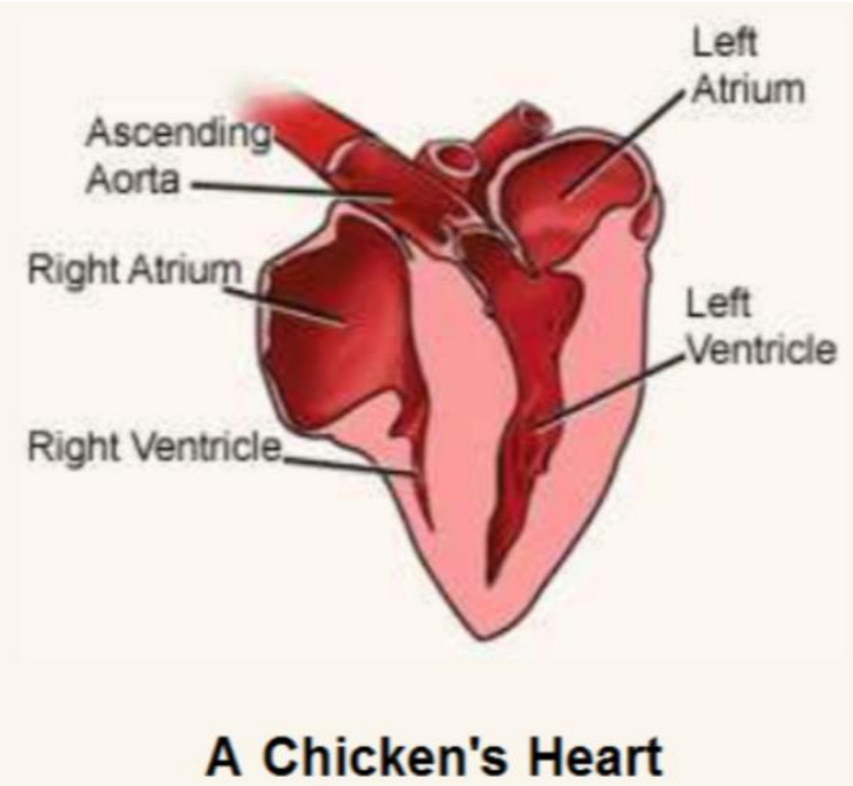
Pneumatic Bones

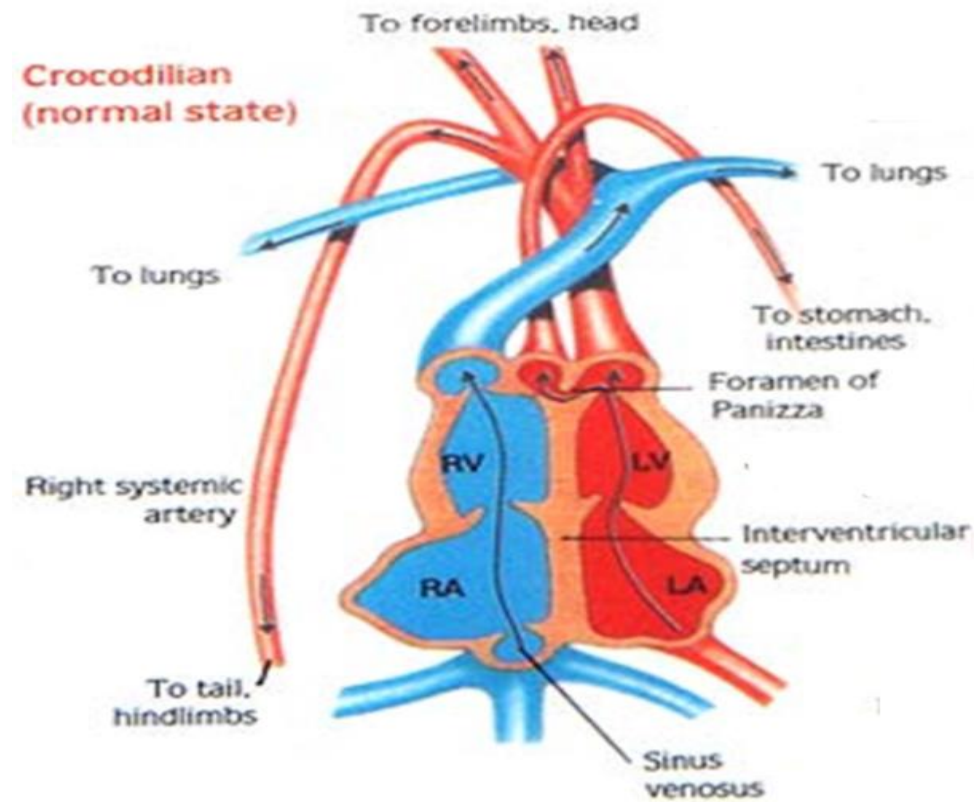
In adult birds, especially good **flying birds**, some **bones** are **hollow&**connected to **respiratory system** through **air sacs.**





Cardiovascular system

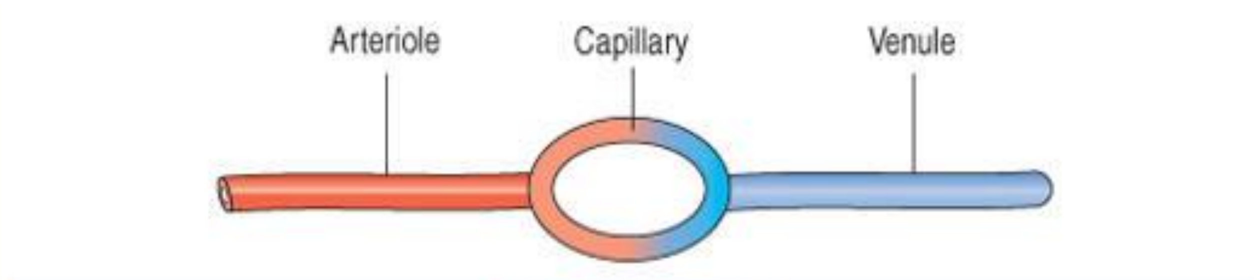




In the birds, the **right &left sides** of the heart are completely **separated** as in mammalian model, but the separation develops differently from that of the mammals &***resemble*** more the development in the ***crocodile.***

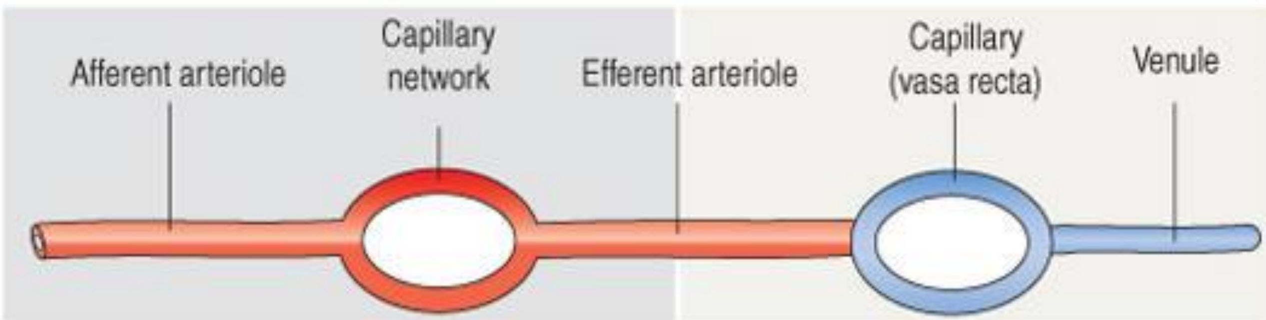
Types of portal circulation

In general, a capillary network is interposed between an arteriole and a venule.



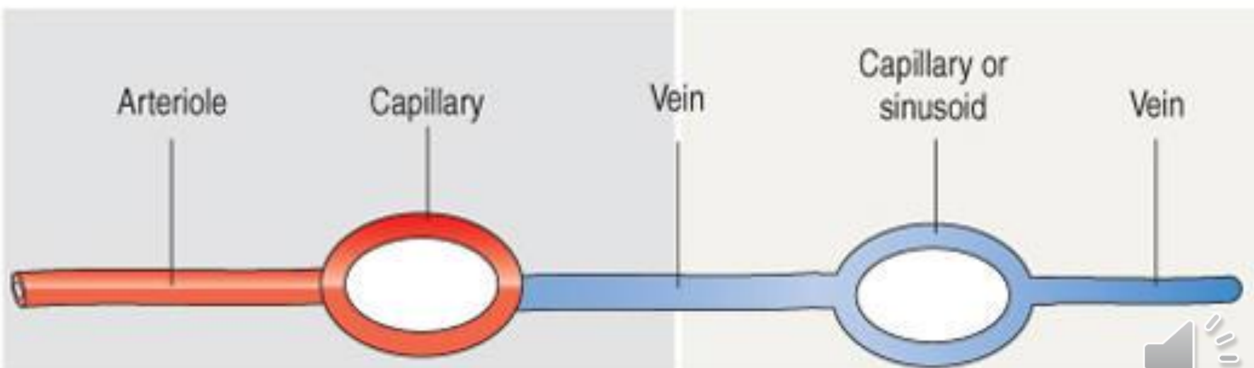
Typical arrangement

In the kidney, an arteriole is interposed between two capillary networks. An afferent arteriole gives rise to a mass of capillaries, the **glomerulus**. These capillaries coalesce to form an efferent arteriole, which gives rise to capillary networks (peritubular capillary network and the vasa recta) surrounding the nephrons.



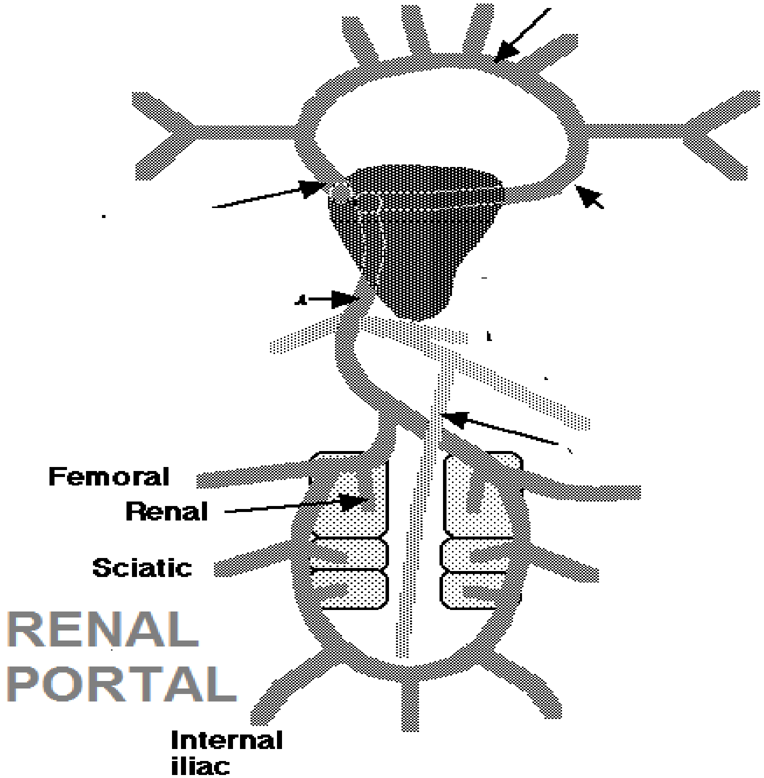
Arterial portal system

In the liver and hypophysis, veins feed into an extensive capillary or sinusoid network draining into a vein. This distribution is called the **venous portal system**.

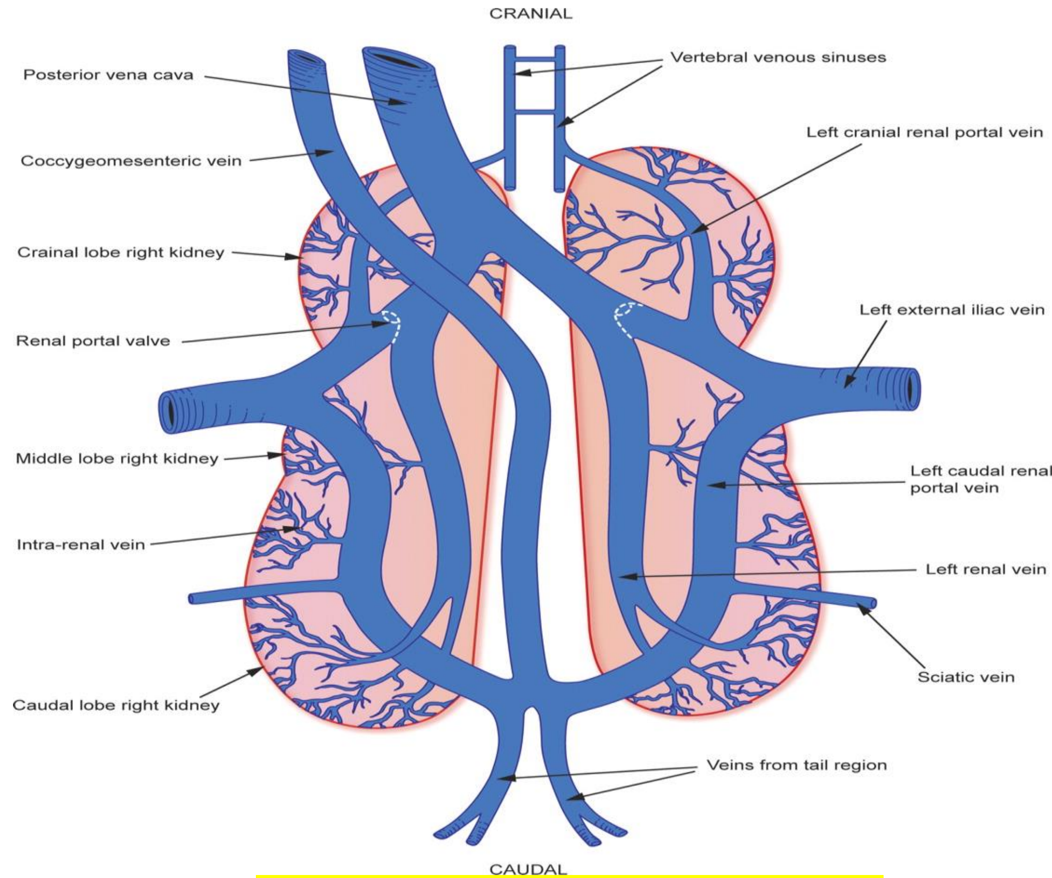


Venous portal system

RENAL PORTAL



Renal Portal Circulation



In the venous side, there are *renal portal veins* but also a *shunt*, guraded by a *muscular valve*. When this valve is open, the resistance of the kidneys &liver sends *blood directly* back to the *heart* allowing rapid circulation during *exercise*.

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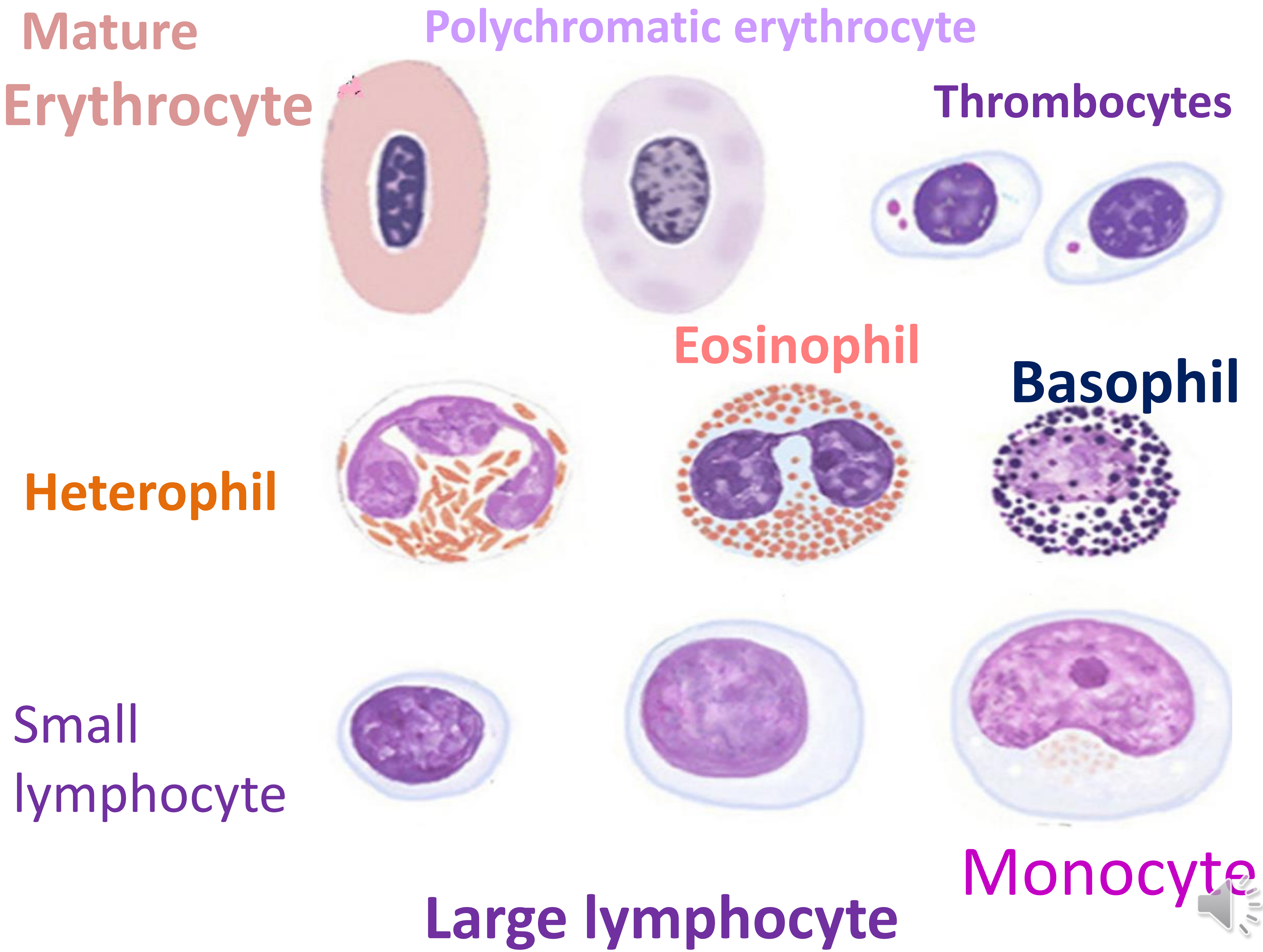
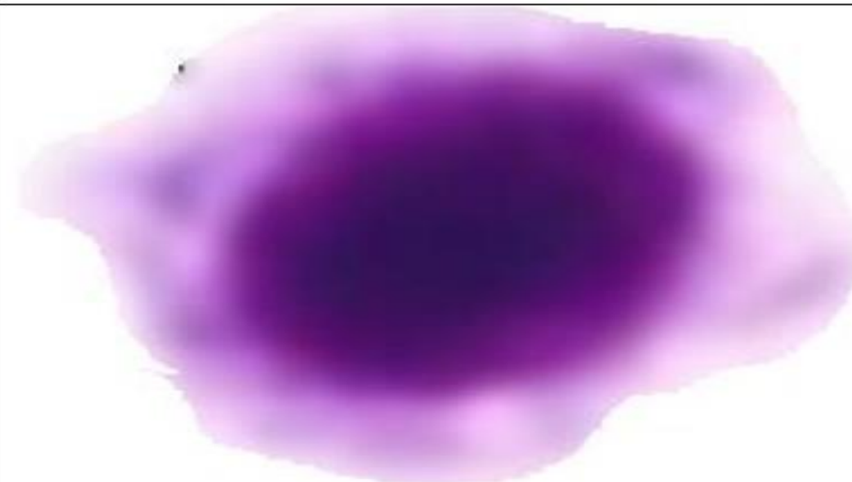

Mature Erythrocyte
Polychromatic erythrocyte
Thrombocytes
Eosinophil
Basophil
Heterophil
Small lymphocyte
Large lymphocyte
Monocyte

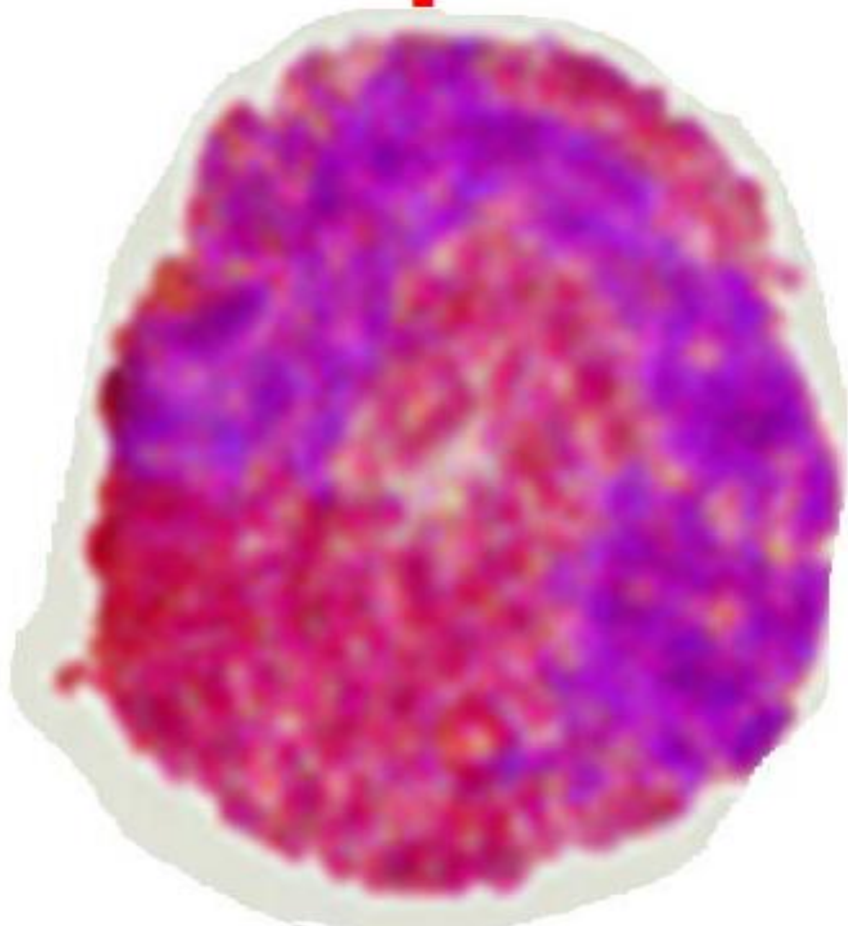
Avian blood cells

Cell	**No. in Thousands per cubic mm**	**Histological features**
Erythrocyte	3,000	Elongate & nucleated
Thrombocytes	30	Nucleated - no multinucleated megakaryocytes
Total Leukocytes	20-30	-------

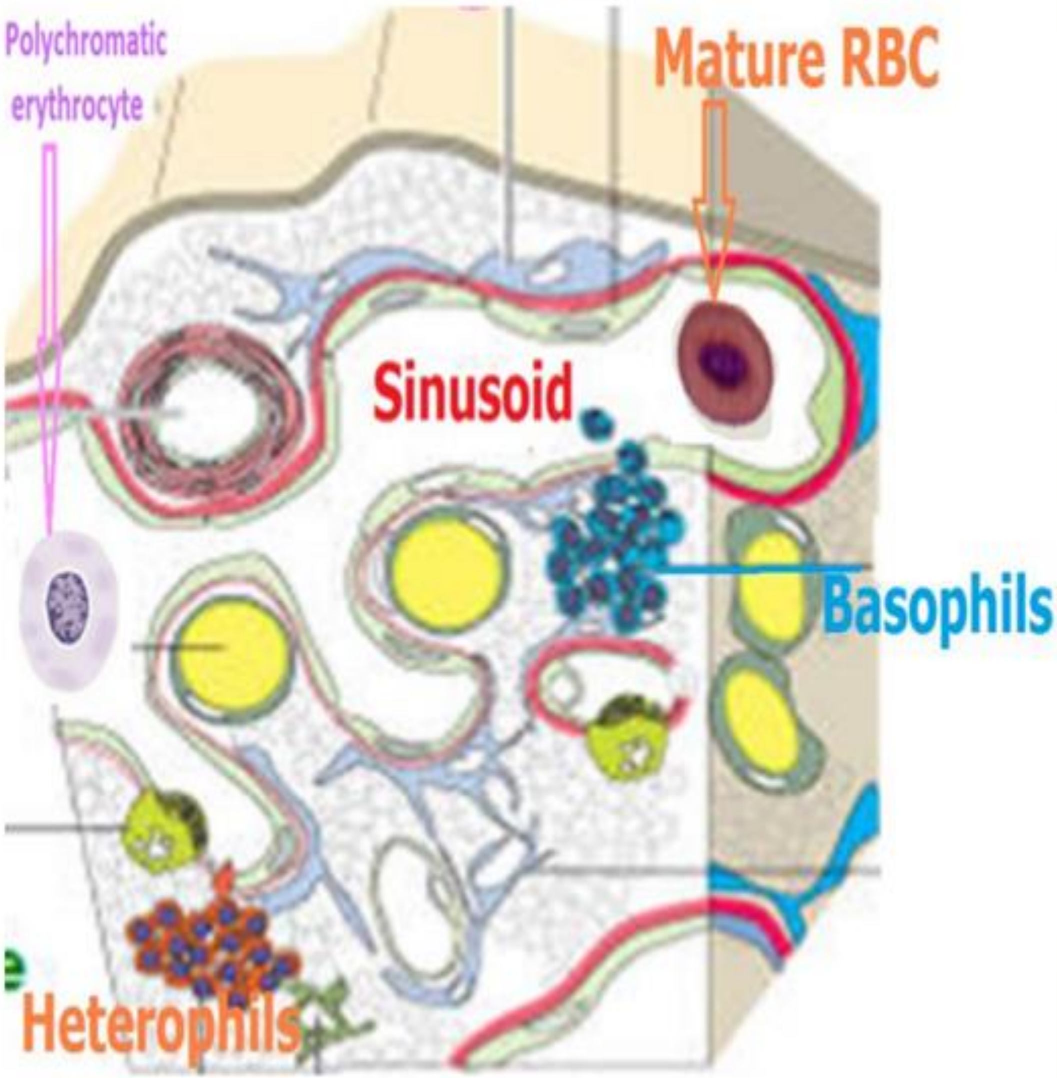
Avian blood

Erythrocytes	Thrombocytes
	
Large, elongated, *flat cells* with oval *nucleus*	***Smaller &less* elongated more spherical nucleus.**

Pseudo-eosinophil=heterophil

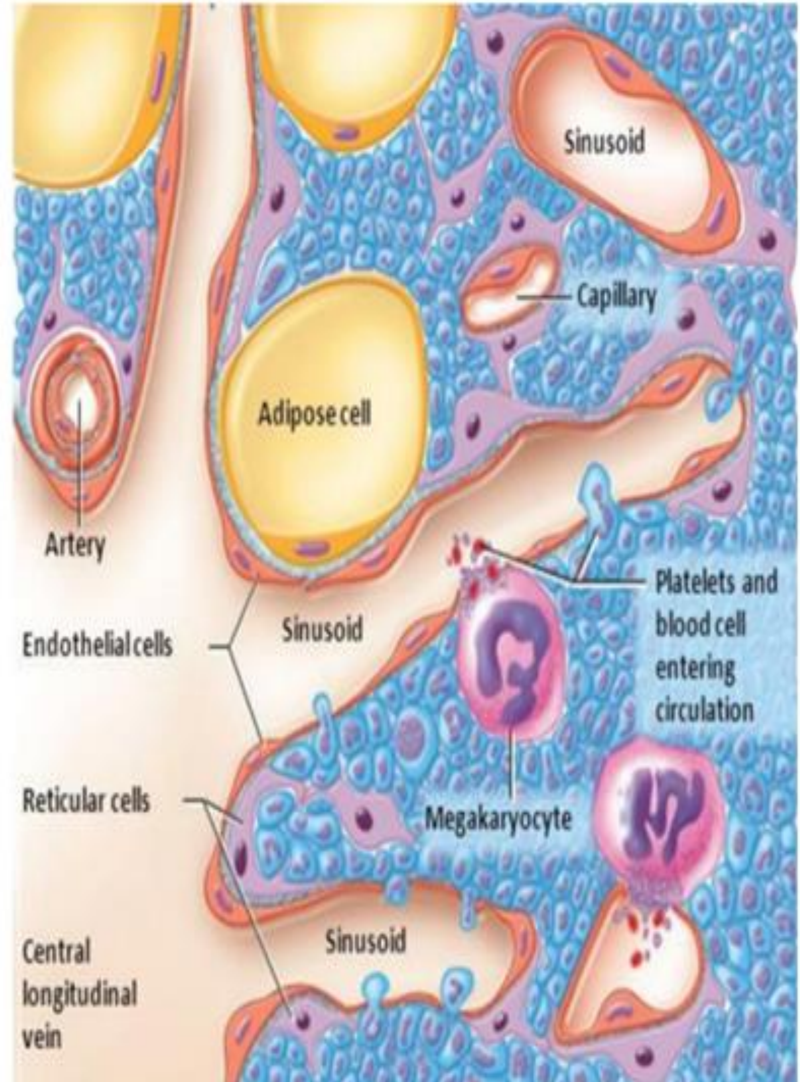
Heterophil	Eosinophil
	
Acidophilic, specific granules.	*Acidophilic, specific granules.*
Rod-shaped granules	Spheric granules

Avian bone marrow



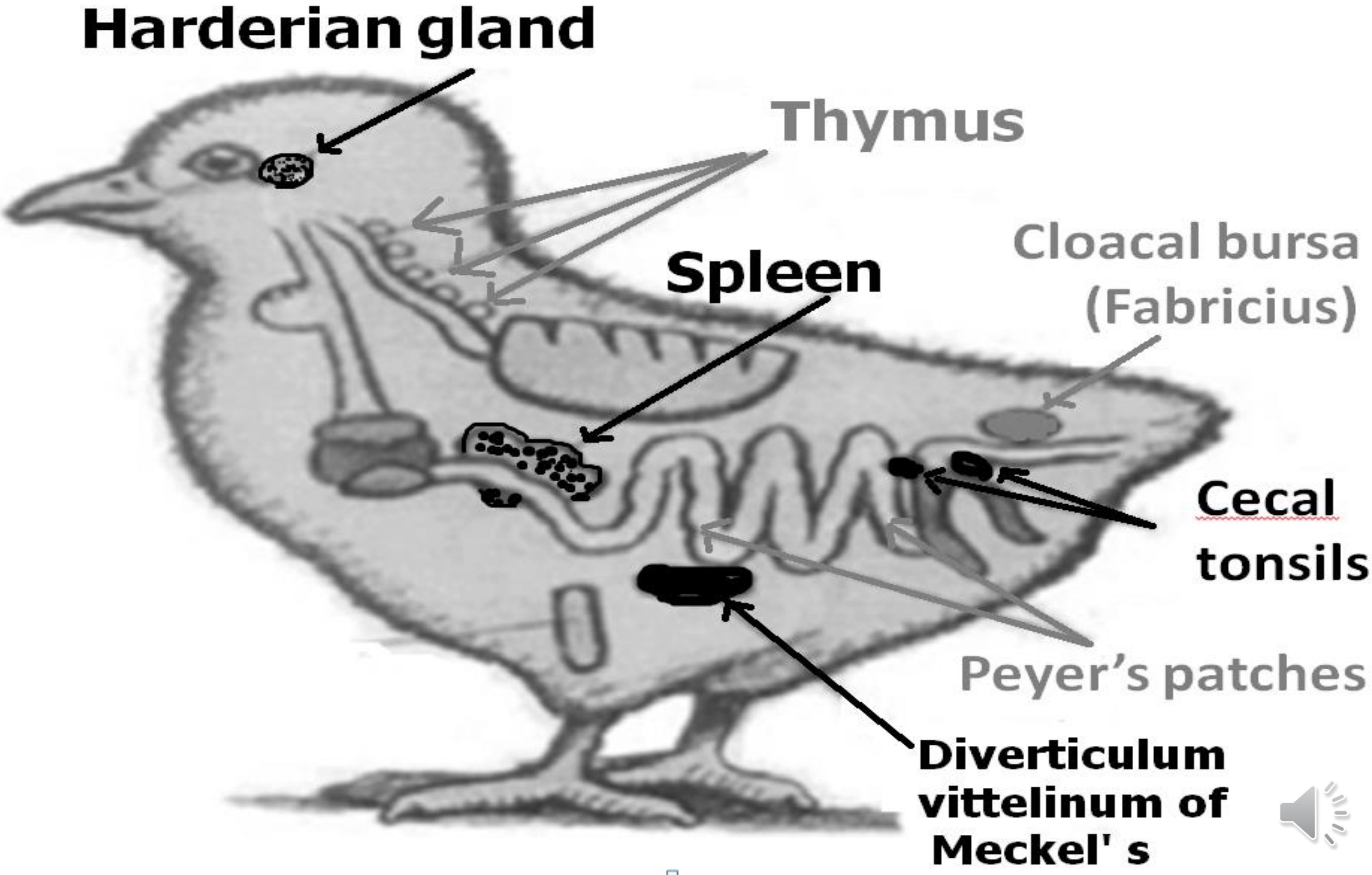
Erythropoiesis takes place ***within*** the vascular **sinusoids**

Mammalian bone marrow



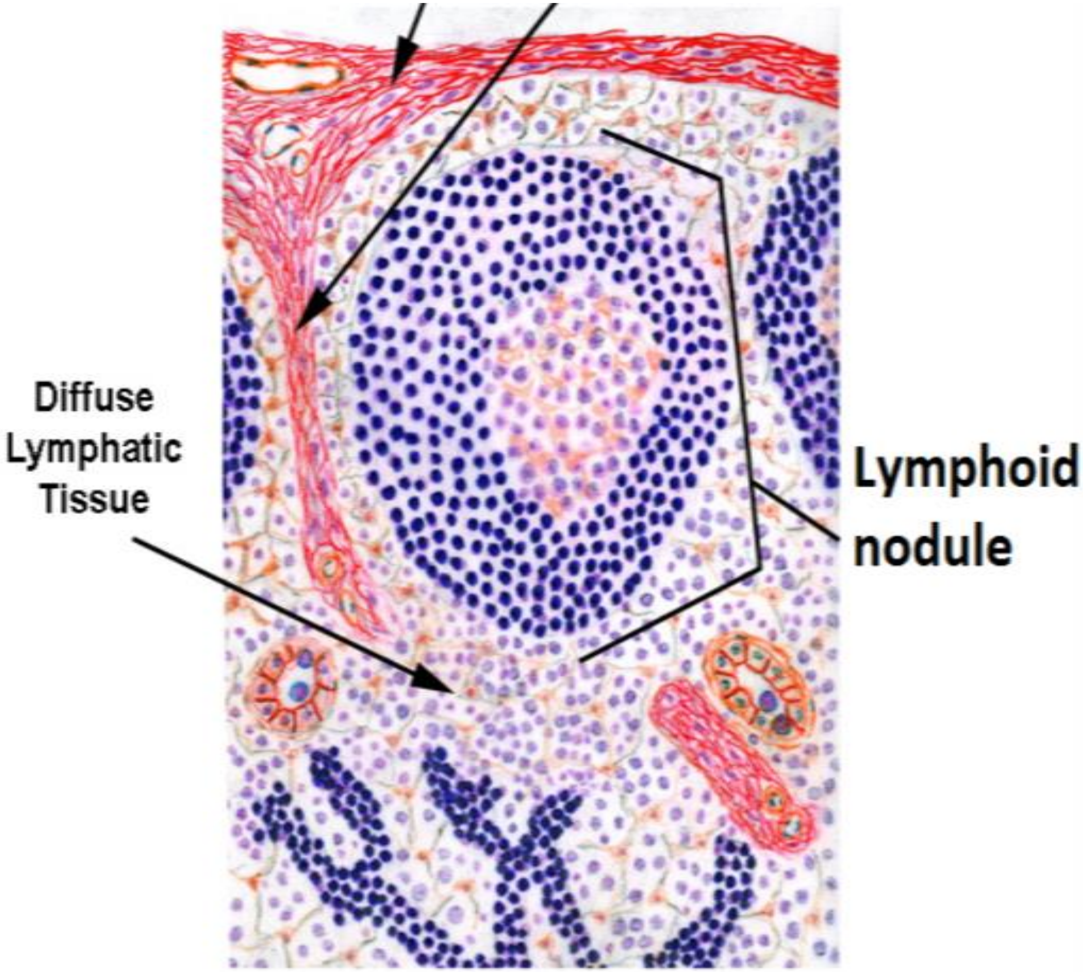
Erythropoiesis takes place ***extravascular*** **spaces of marrow.**

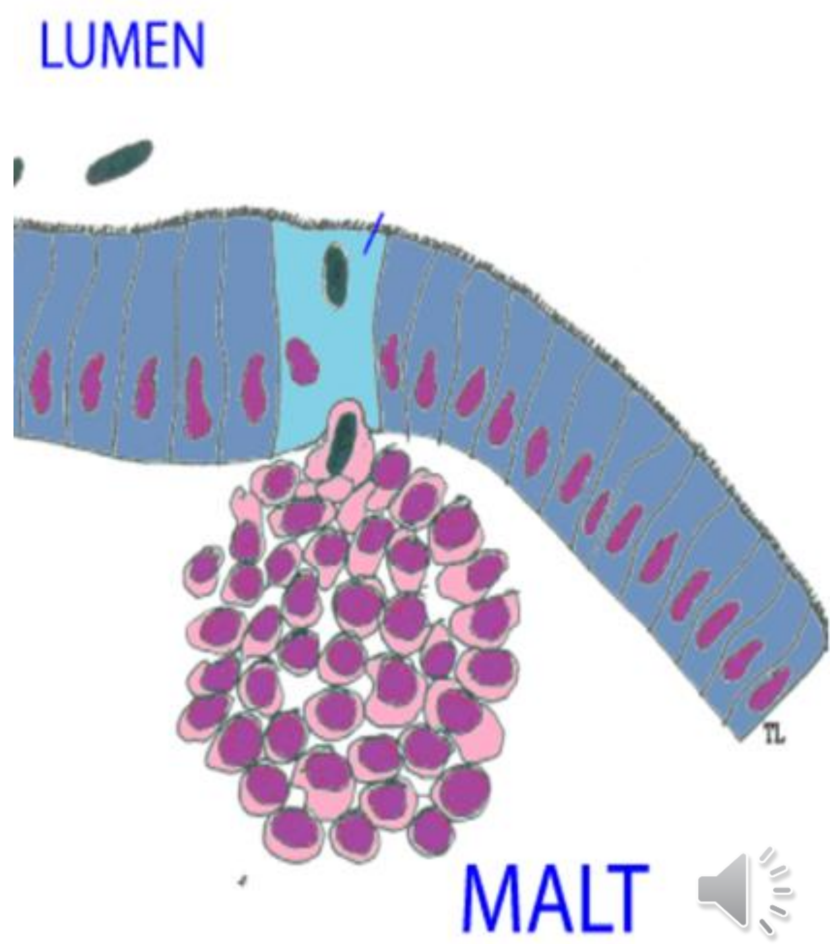
Immune system



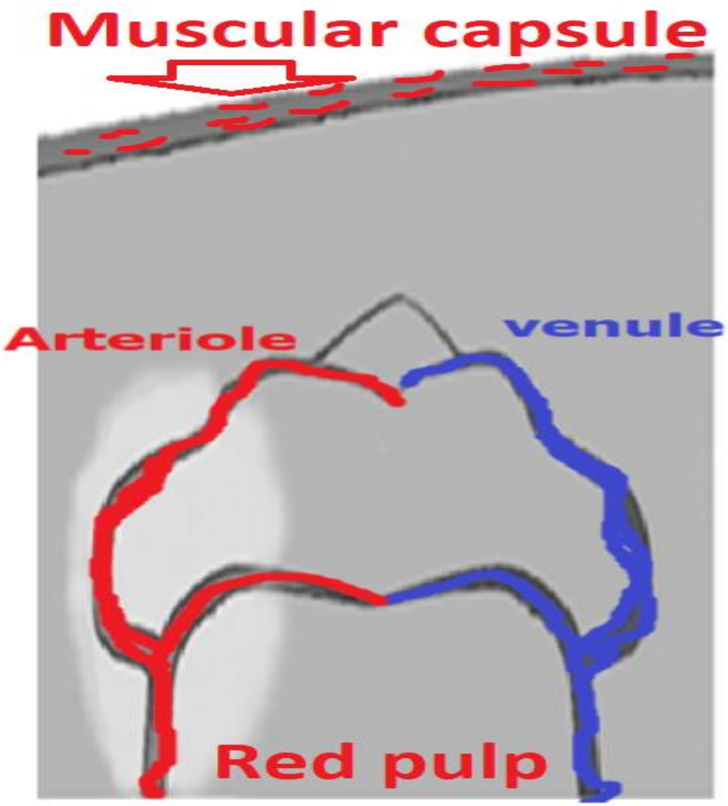
Immune system

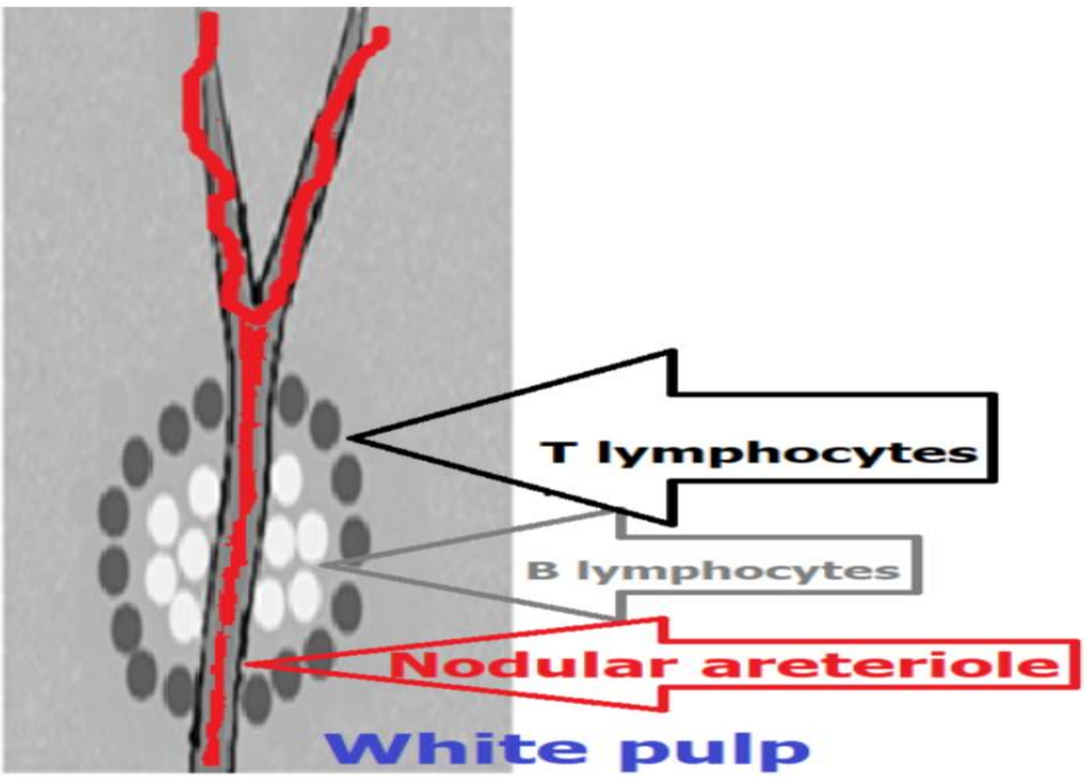
1. **Lymph nodes** are *absent* in the chicken, meanwhile, **diffuse lymphatic tissue** &**lymphoid nodules** are present.





2-Avian spleen is characterized by:





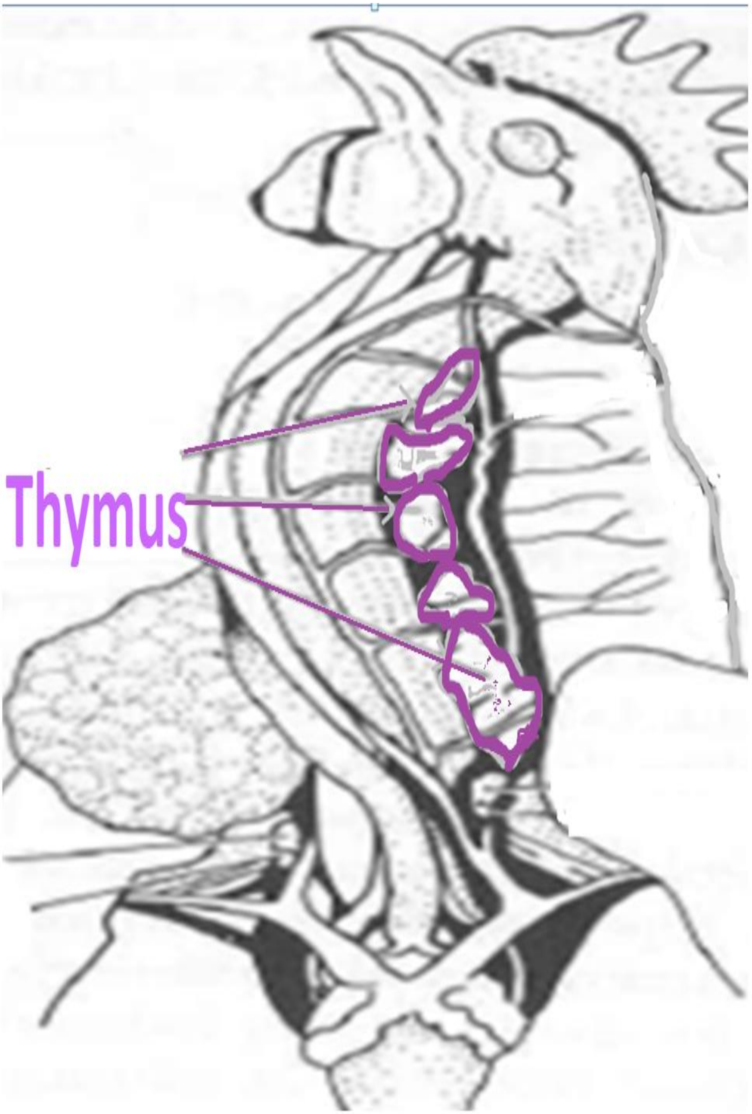
1-Muscular capsule

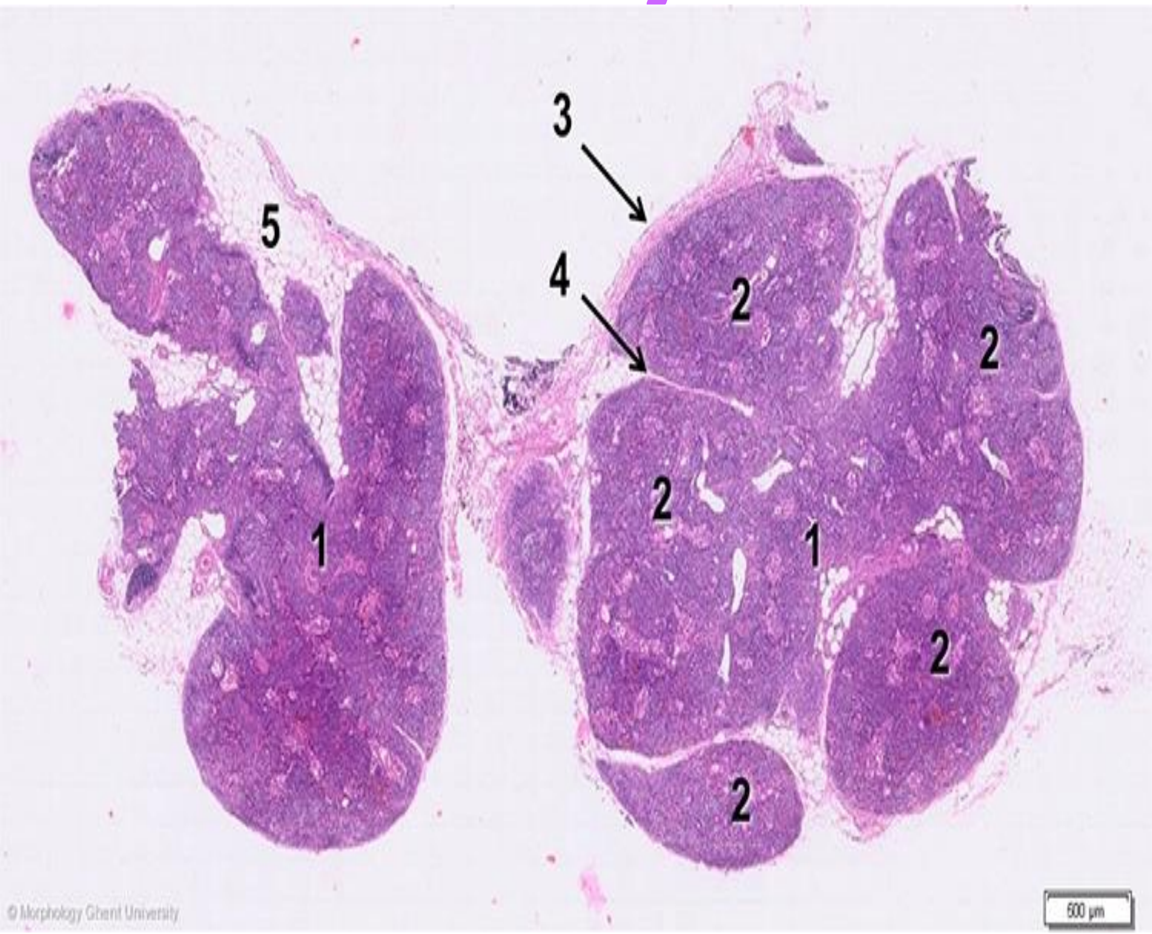
2-Absence of trabeculae.

3- Poor demarcation between red &white pulps.

4-Direct connections between arteriole &venule.

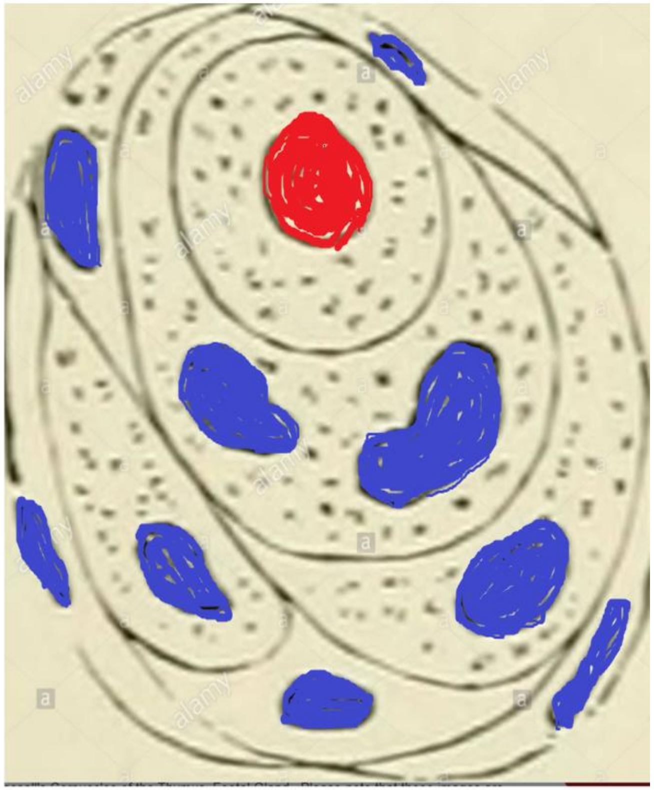
3-Avian thymus

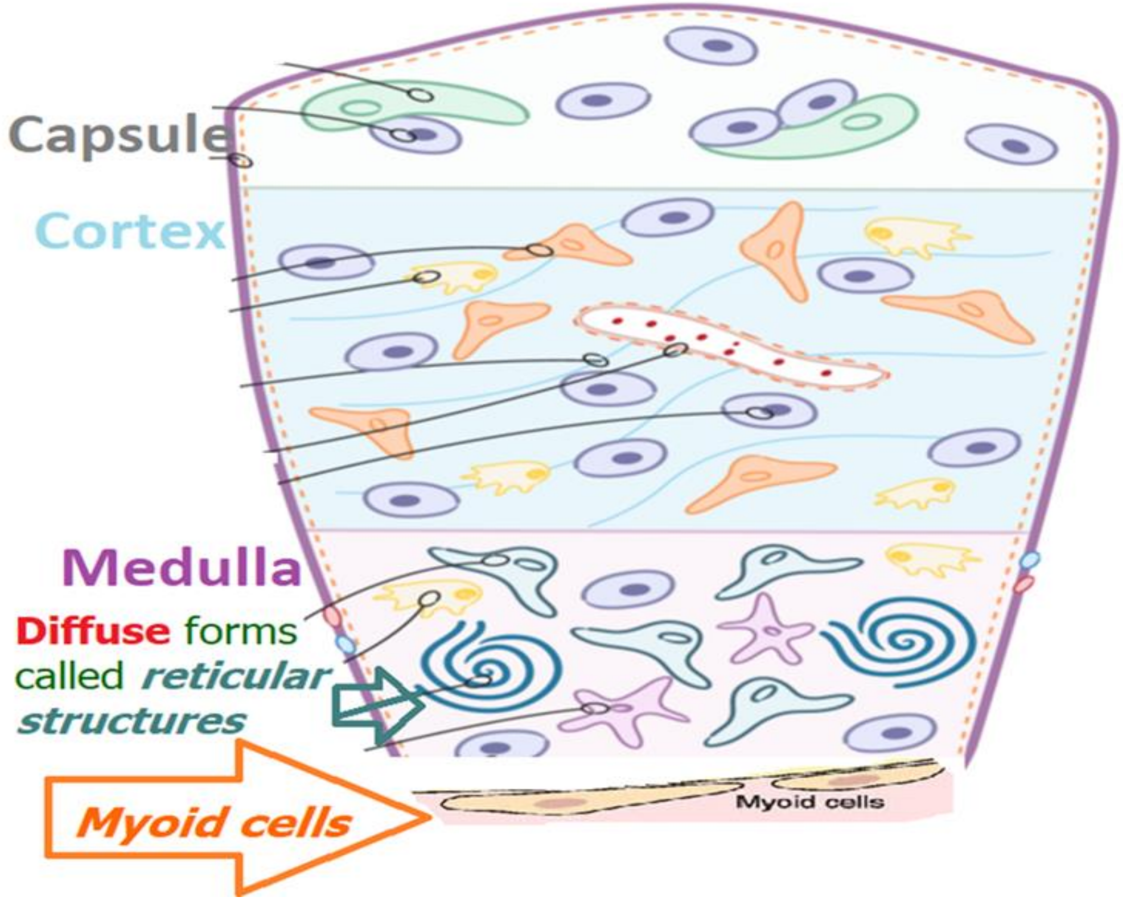




The thymus is arranged into **incompletely separated lobules** (1)of cortical &medullary tissue.

Avian thymic medulla





Thymic(Hassall's)corpuscles

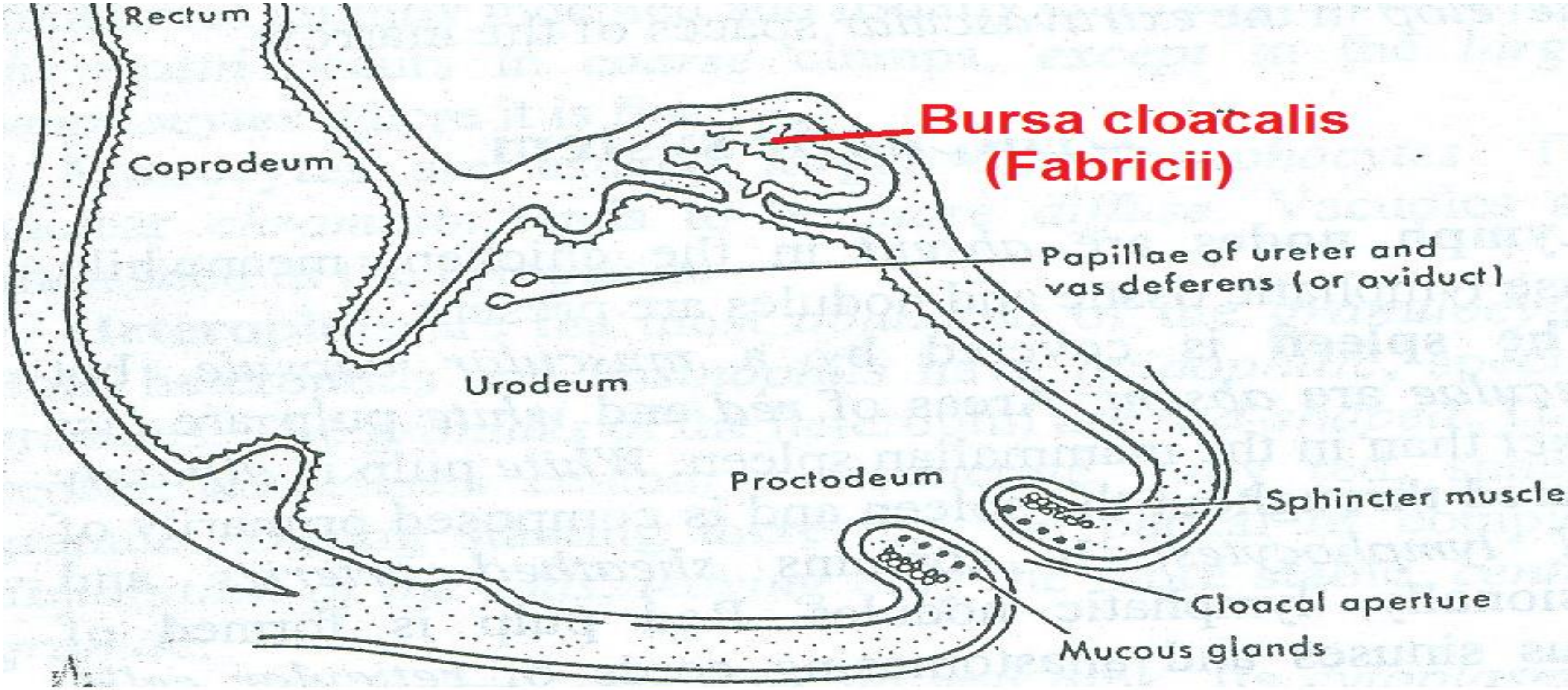
Reticular corpuscles & myoid cells

1-Typical thymic (Hassall's)corpuscles

2- **Diffuse** forms called ***reticular structures*** **(right).**

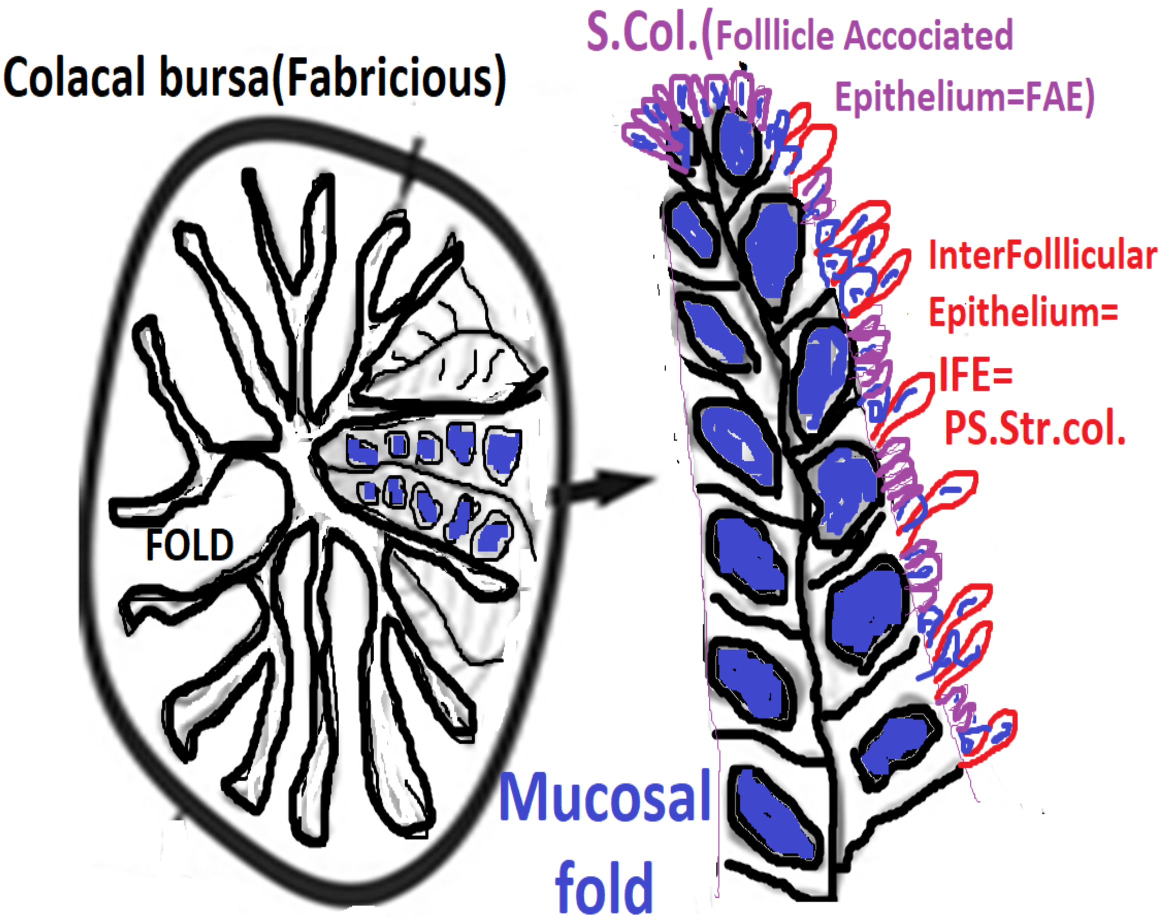
3- ***Myoid cells*** have fibrous cytoplasm

4- Cloacal bursa (of Fabricii)

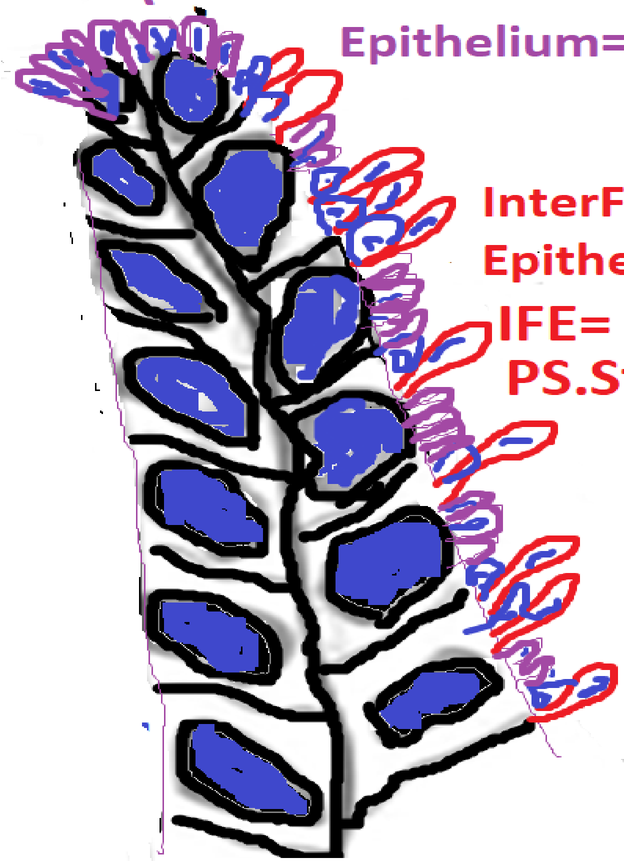


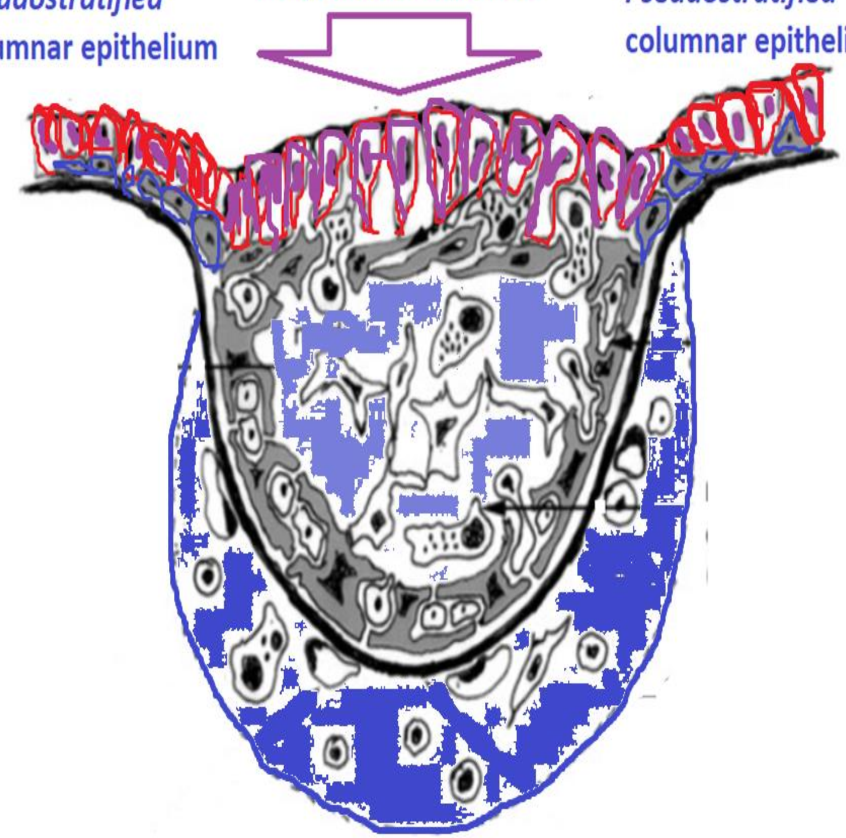
The bursa of Fabricius is a saclike **dorsal diverticulum of the proctodeum** that is ***unique to birds*** .

Cloacal bursa (Fabricius)



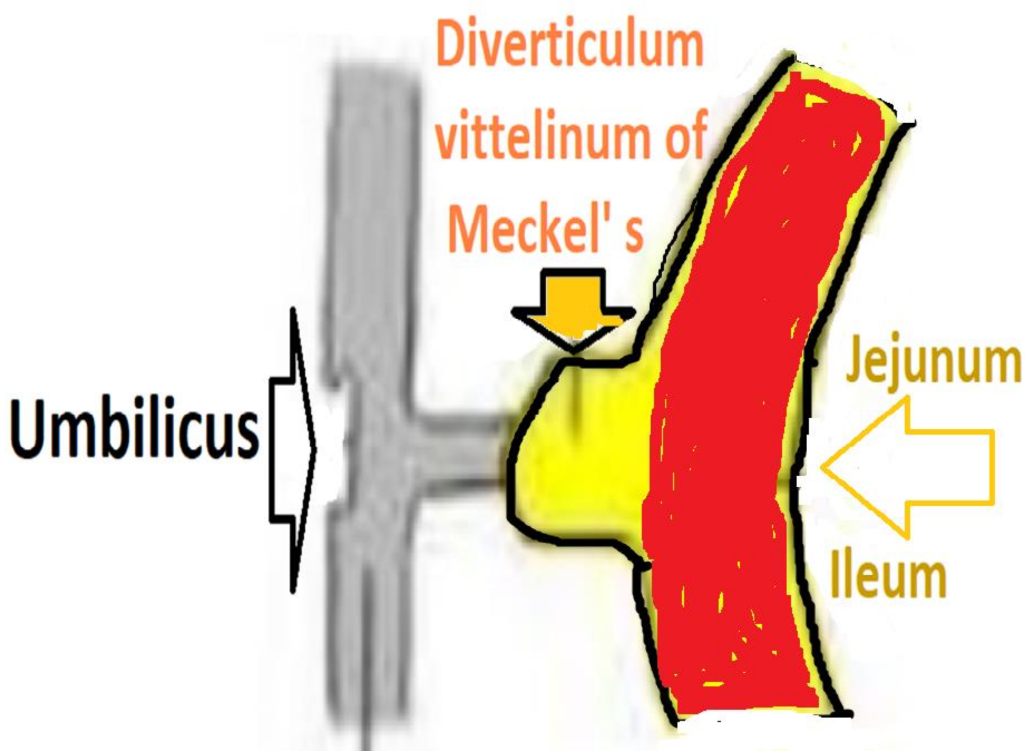
Bursa has tall, thick mucosal *folds* (plicae) filled with **numerous polyhedral lymphoid nodules(3)**.





The *bursa* is lined by a ***pseud.str.* col. Epith.**, except at the *apex* of each *follicle*, which is covered by an *epithelial tuft* of **simple *columnar* cells.**

5-Diverticulum vittelinum of Meckel' s



Diverticulum **vittelinum** is **remnant** of **yolk sac** that gets **infiltrated with lymphoid tissue.**

Acting as a secondary lymphoid organ

Functions of diverticulum vittelinum of Meckel' s :

1-lumen of jejunum

2-wall of the jejunum

3- diverticulum vittelinum

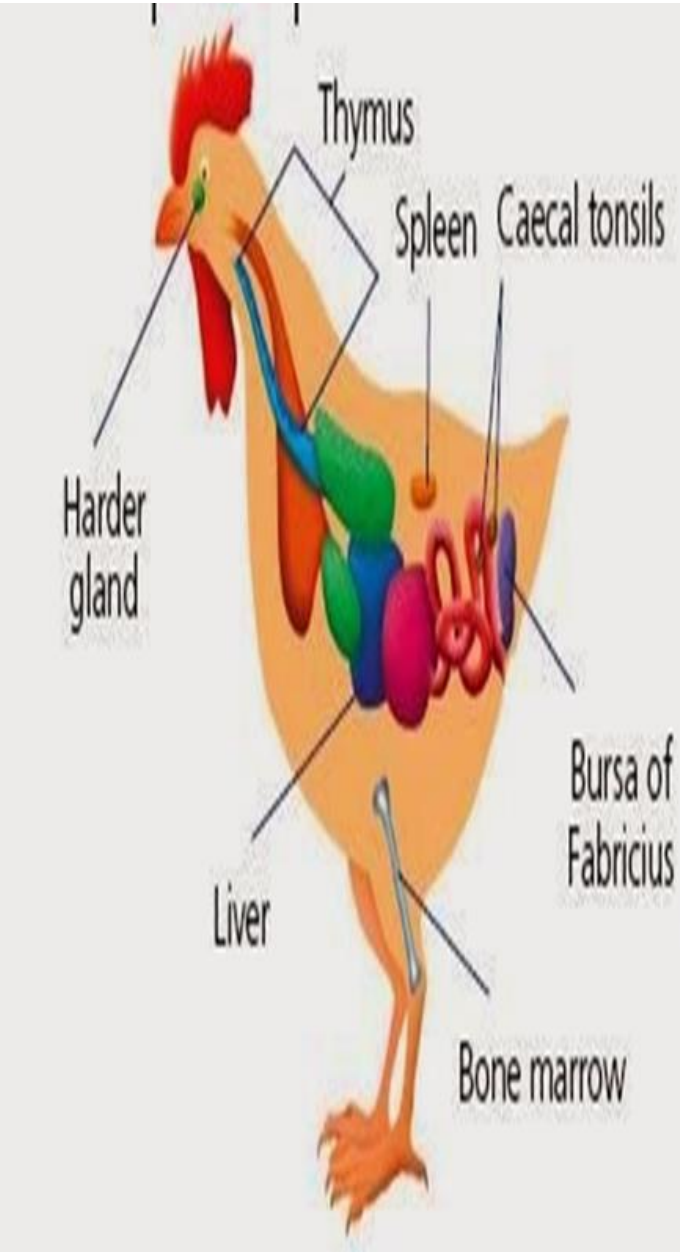
4-adipose tissue

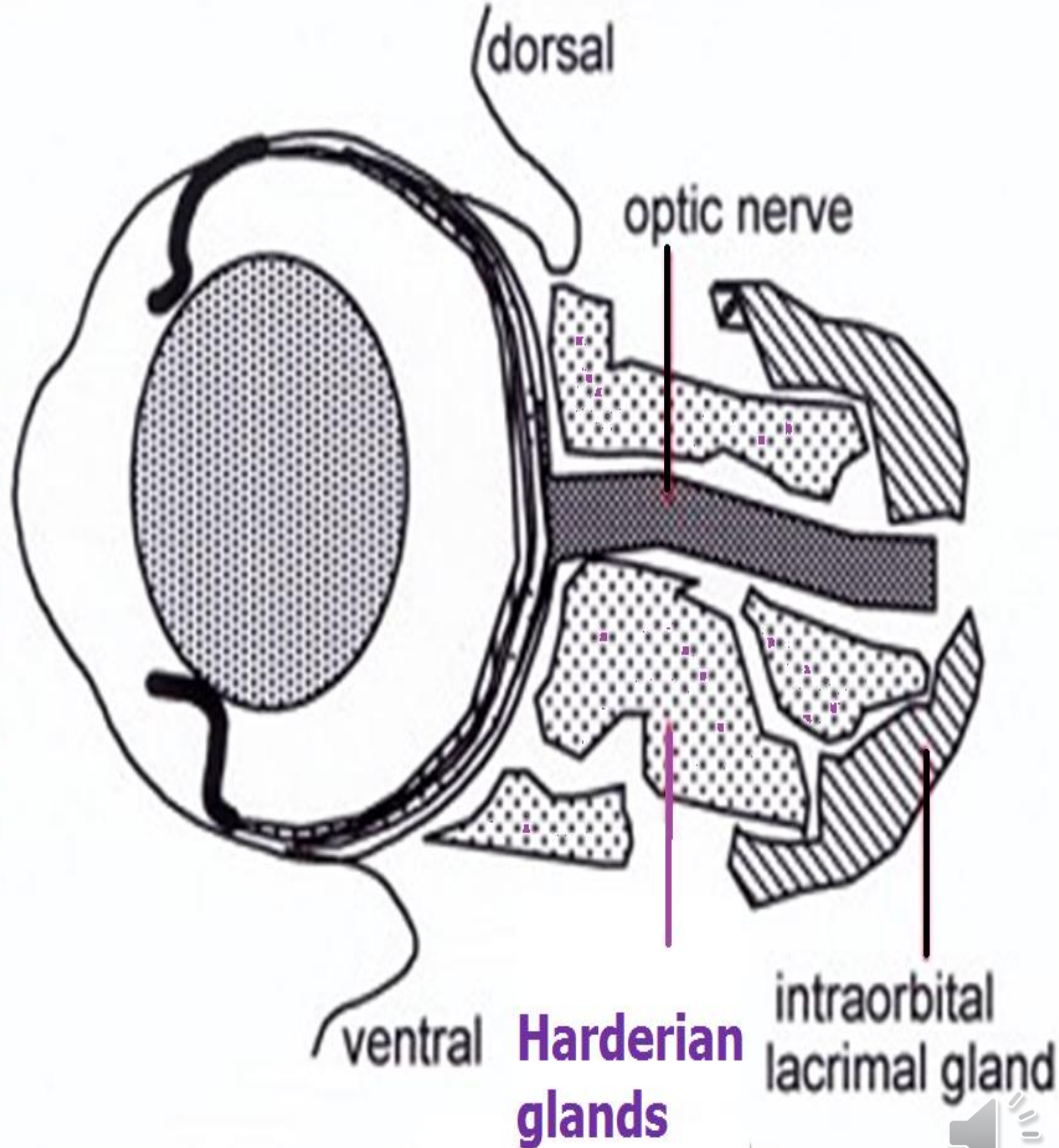
1-supplying nutrients first days after hatching

2-acting as a secondary lymphoid organ.

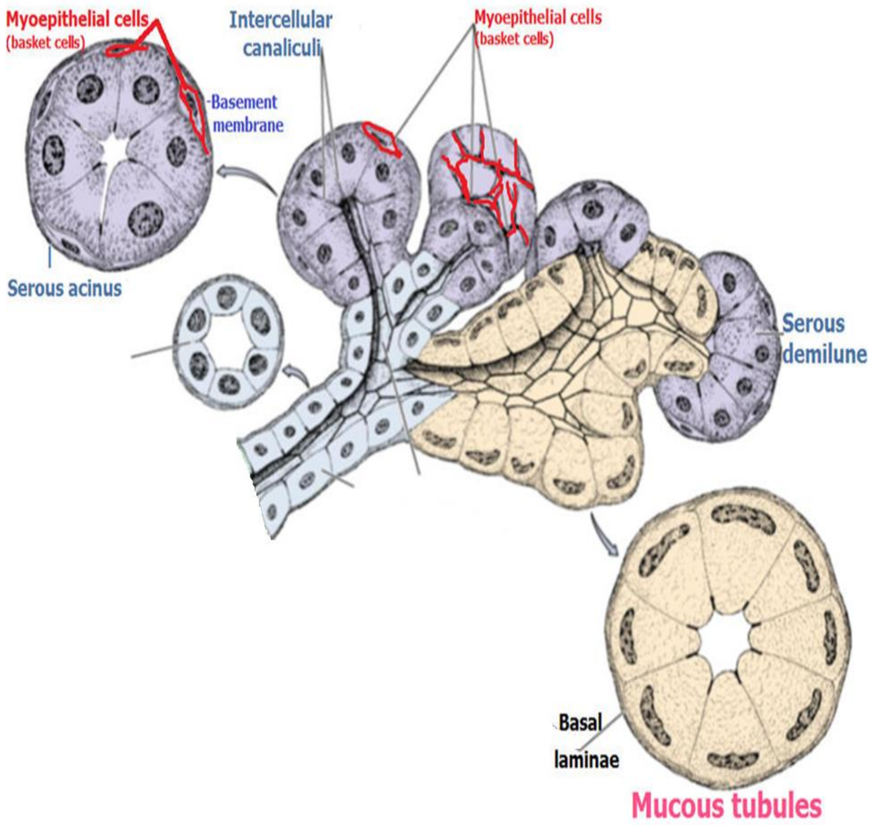
In chickens, pigeons, ducks and geese, Meckel’s diverticulum persists through life but in most birds it disappears early.

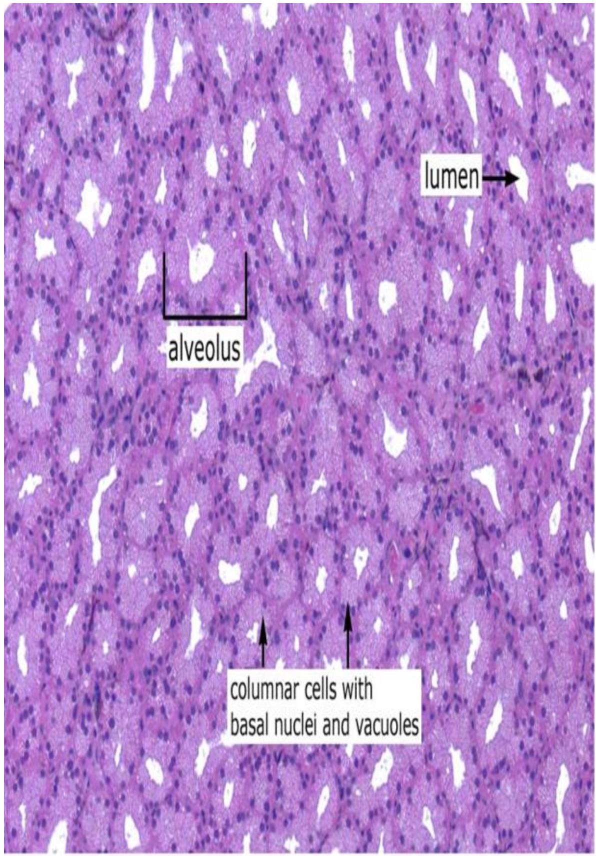
Orbital Harderian glands





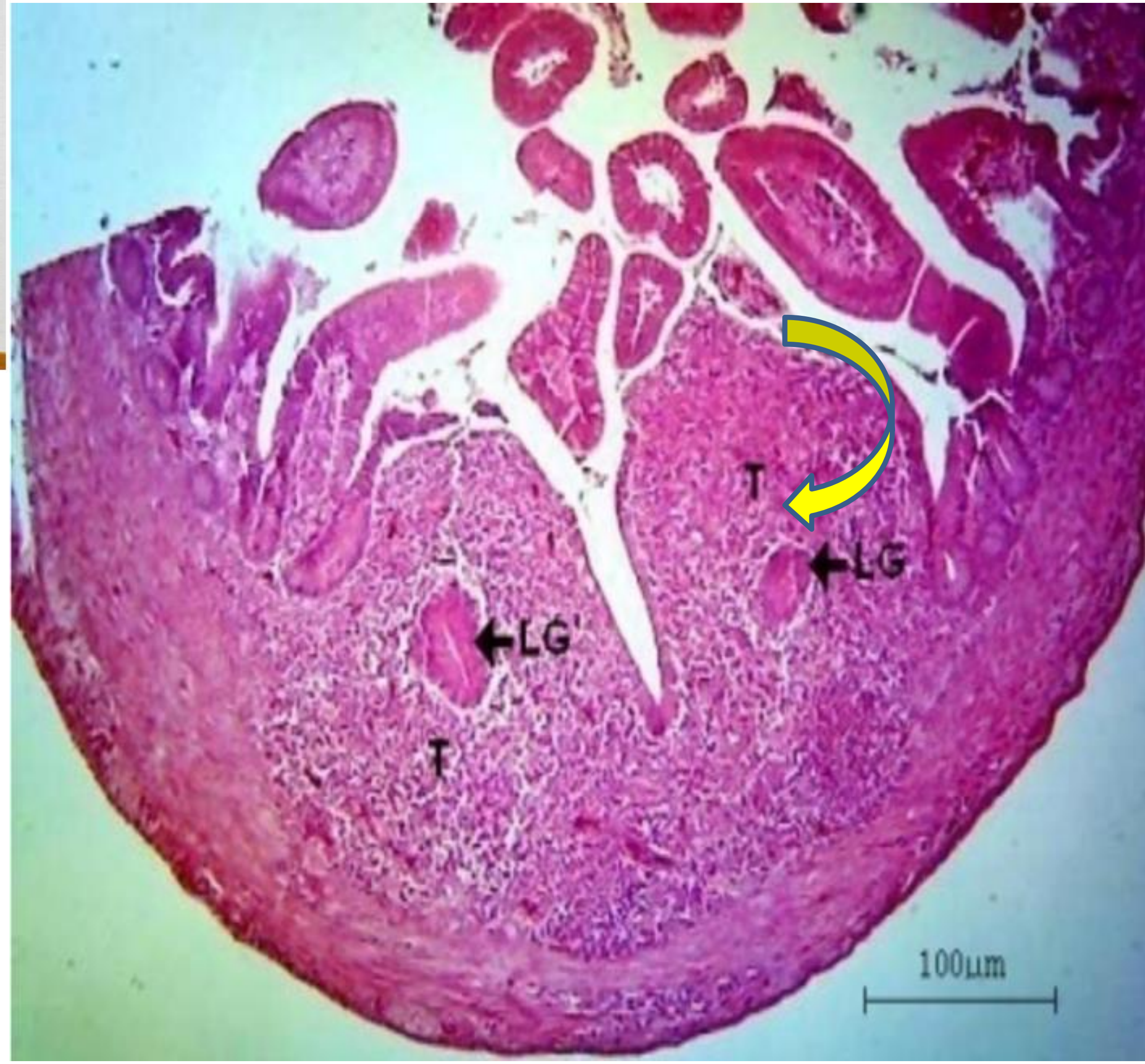
6-Lacrimal gland of Harder (Harderian gland)





Harderian glands are tubuloalveolar,secrete seromucous secretion. **Their acini** were filled with many lymphocytes &plasma cells.

Caecal tonsil (T)



LG=Intestinal gland (cross sectioned)

2-Main points of applied avian histology
2nd part

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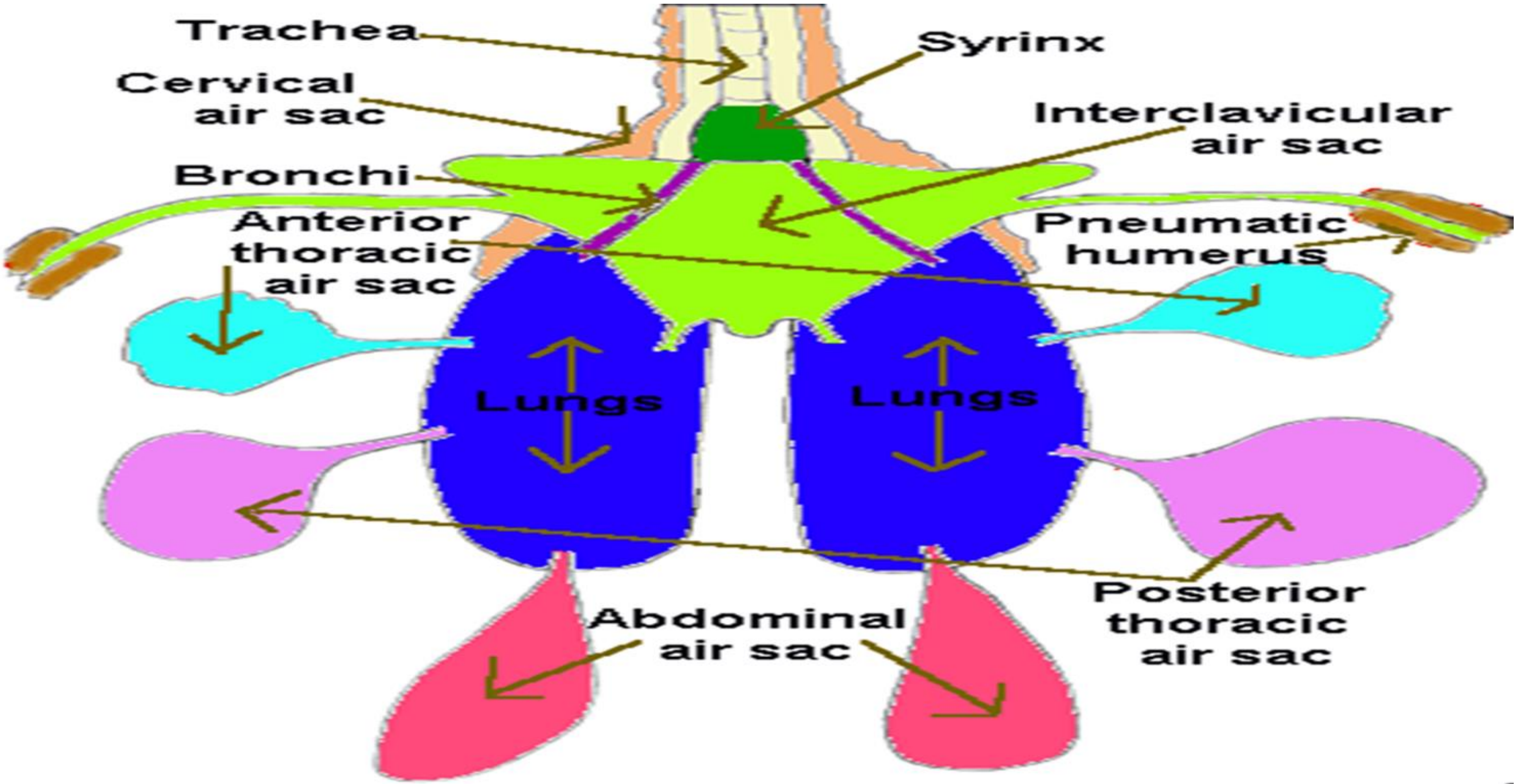
Digital Veterinary Histology

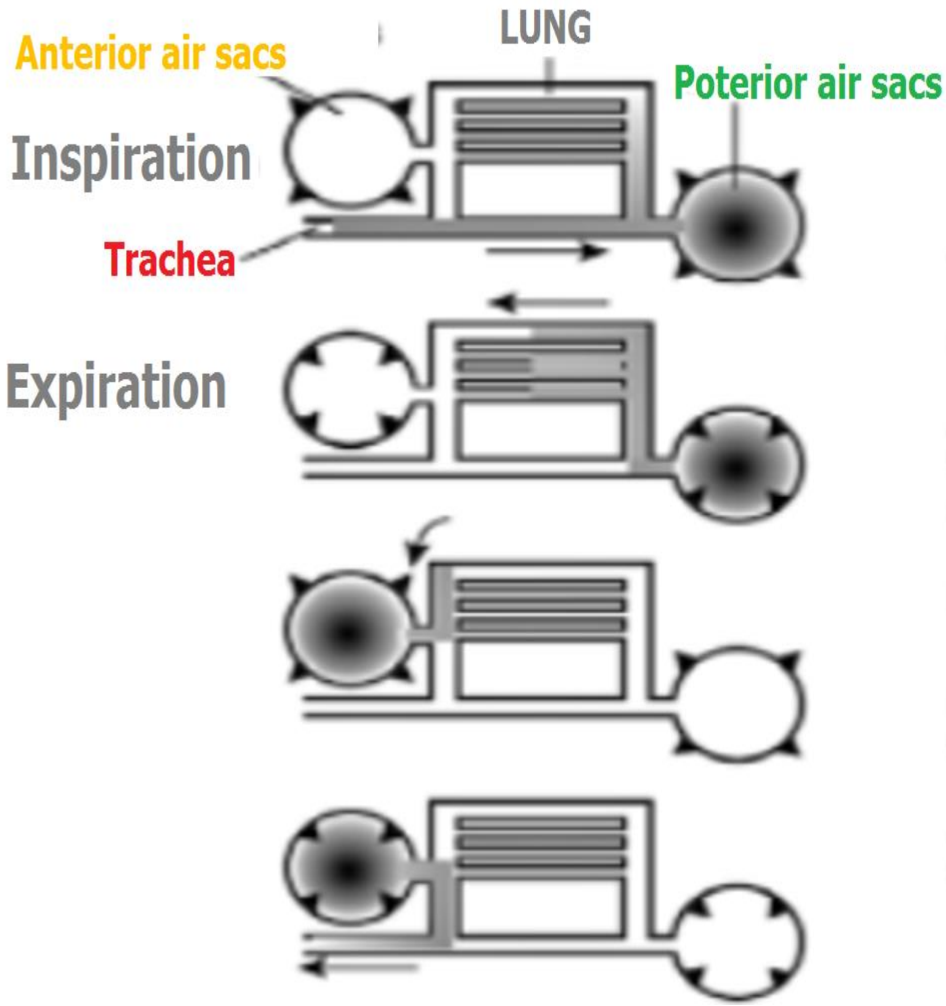
Avian lung &air sacs

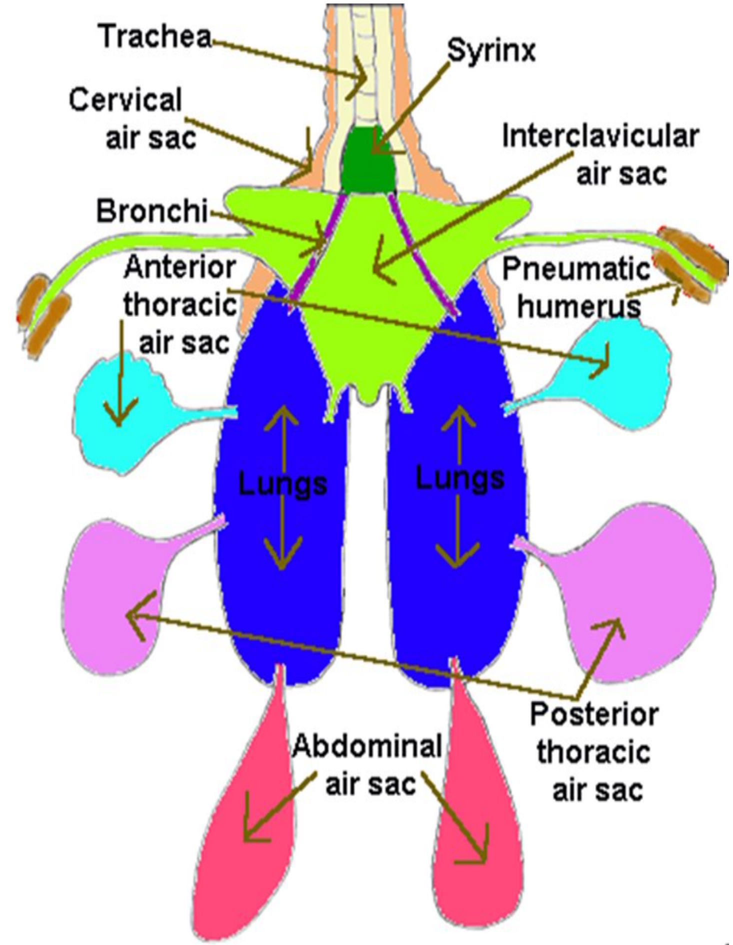


-Non-expandable avian *lungs* do *not* have *alveoli*

-*Expandable air sacs*

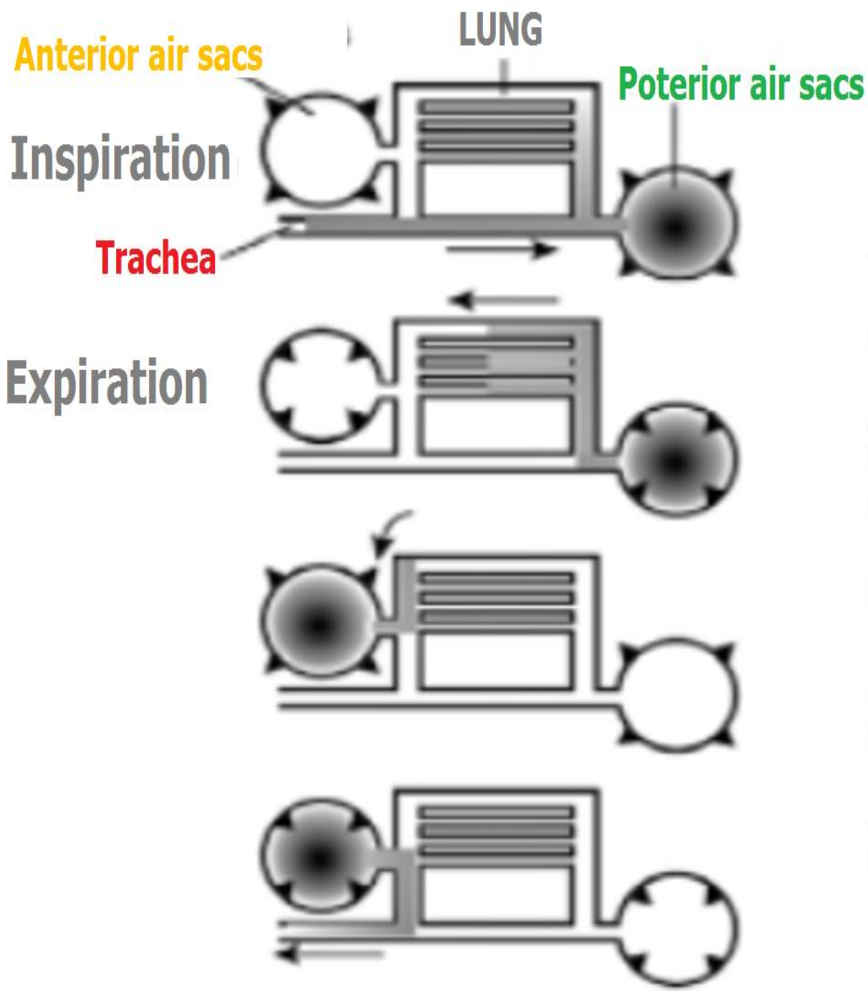
Air sacs

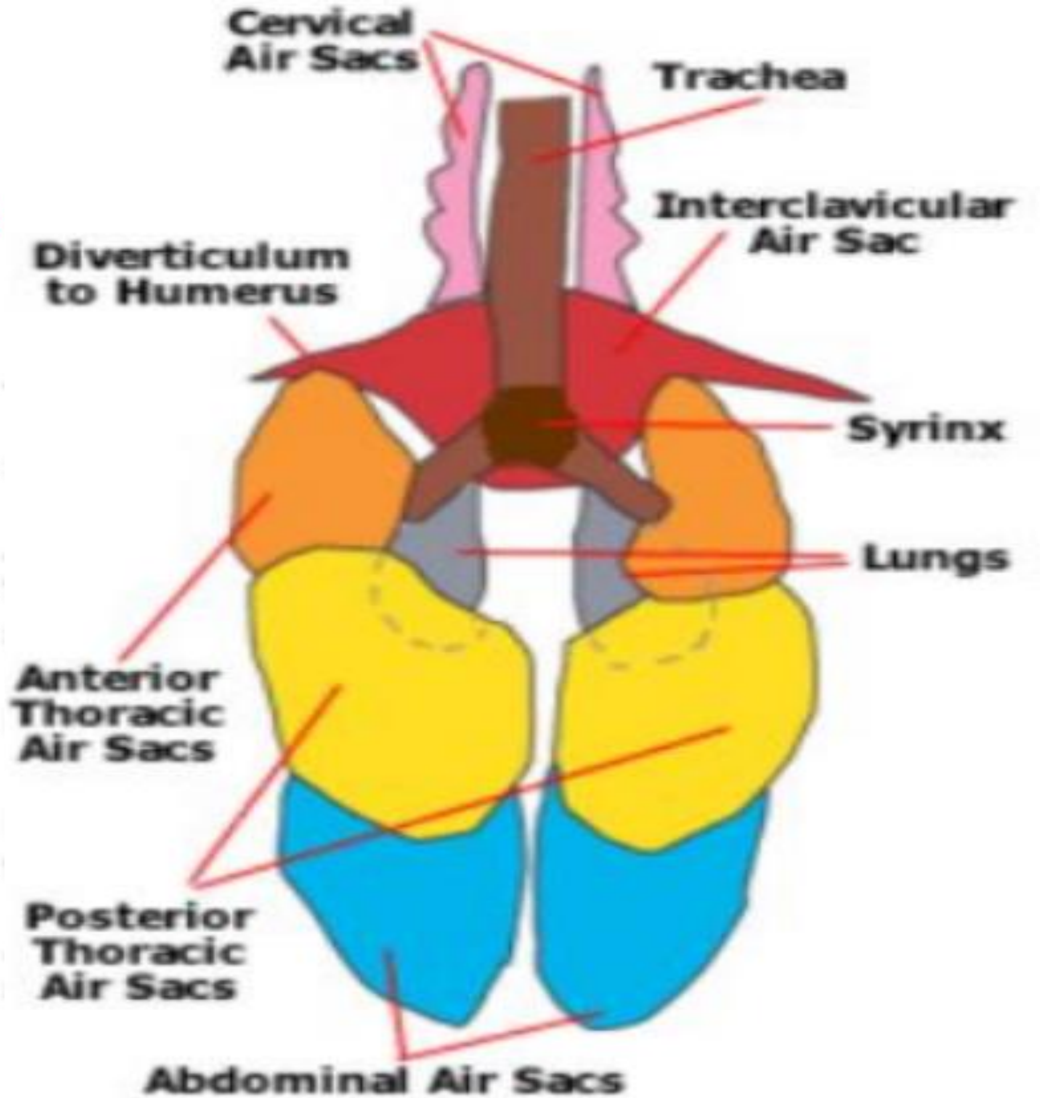




Air sacs have very thin walls with **few blood vessels**.
So, they do not play a direct role in gas exchange.
Rather, they act as a **'bellows' to ventilate the lungs**.

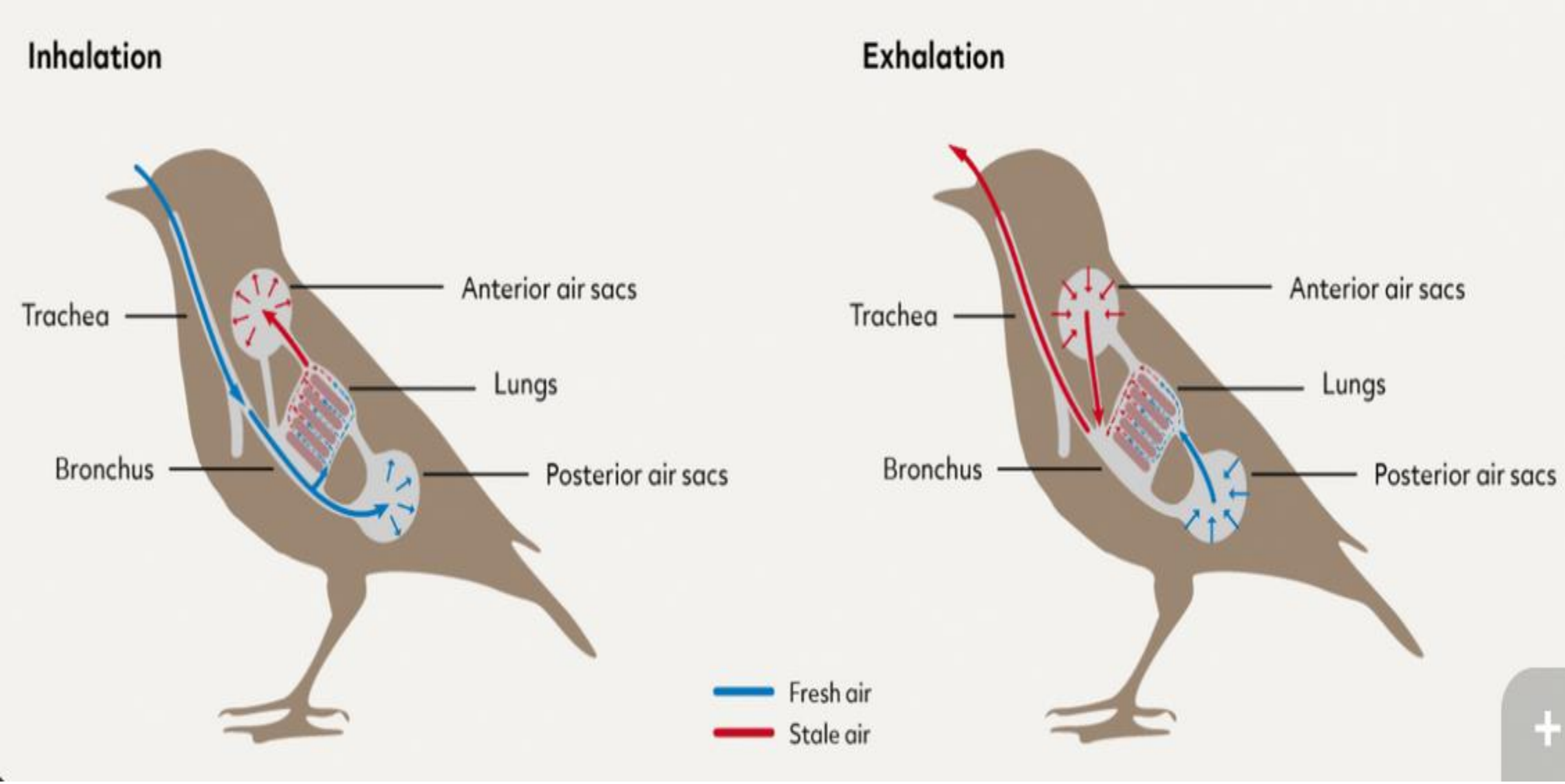
Air sacs





Air sacs are expandable sacs of serous membranes, **connected** to the **bronchi**. They are **1 interclavicular**, **2 cervical**, **2 anterior thoracic**, **2 posterior thoracic** **&2 abdominal air sacs.**

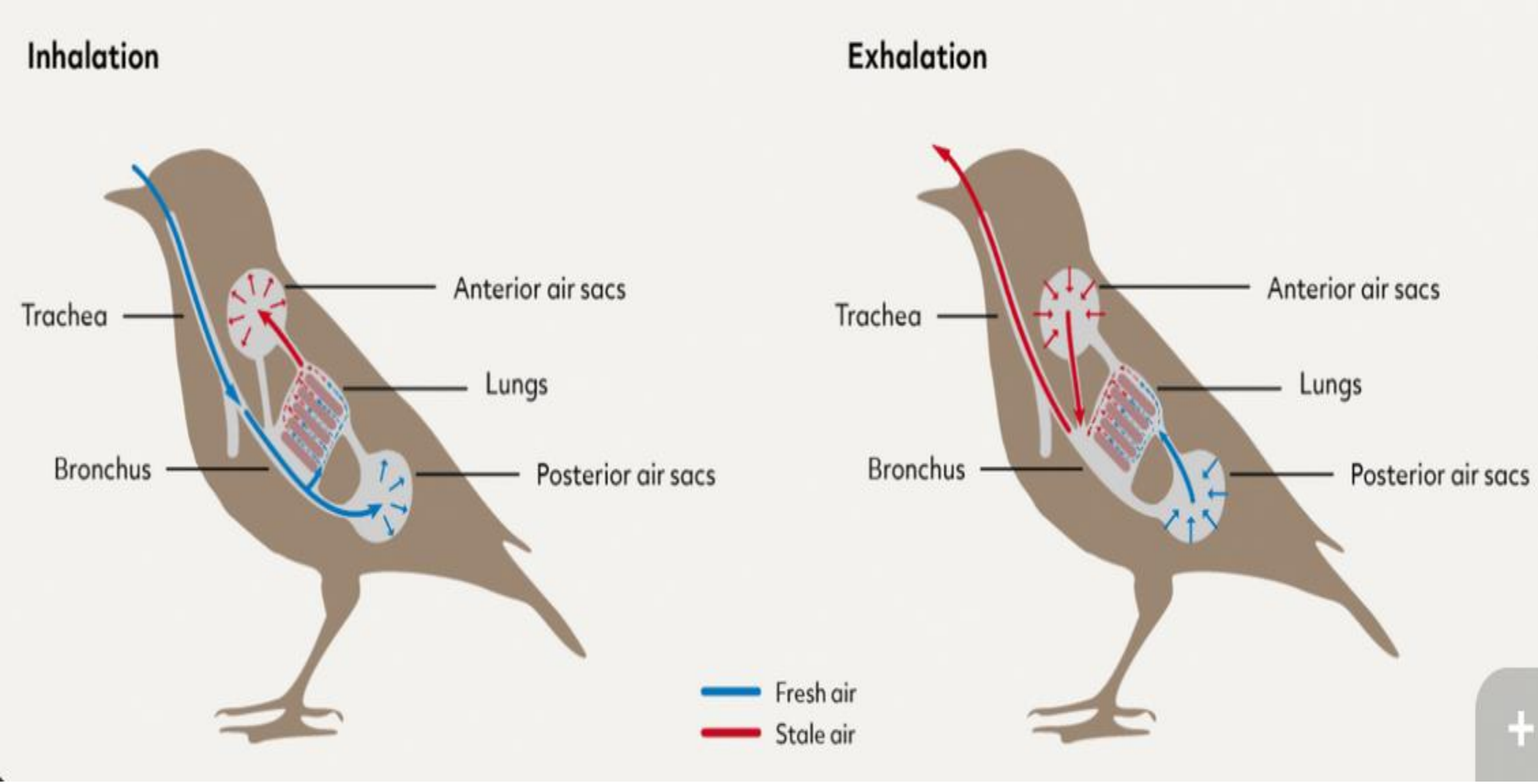
Double avian respiratory Cycle



When a **bird inhales**, fresh air (blue) enters through trachea &bronchus&flows into lungs &posterior air sacs.

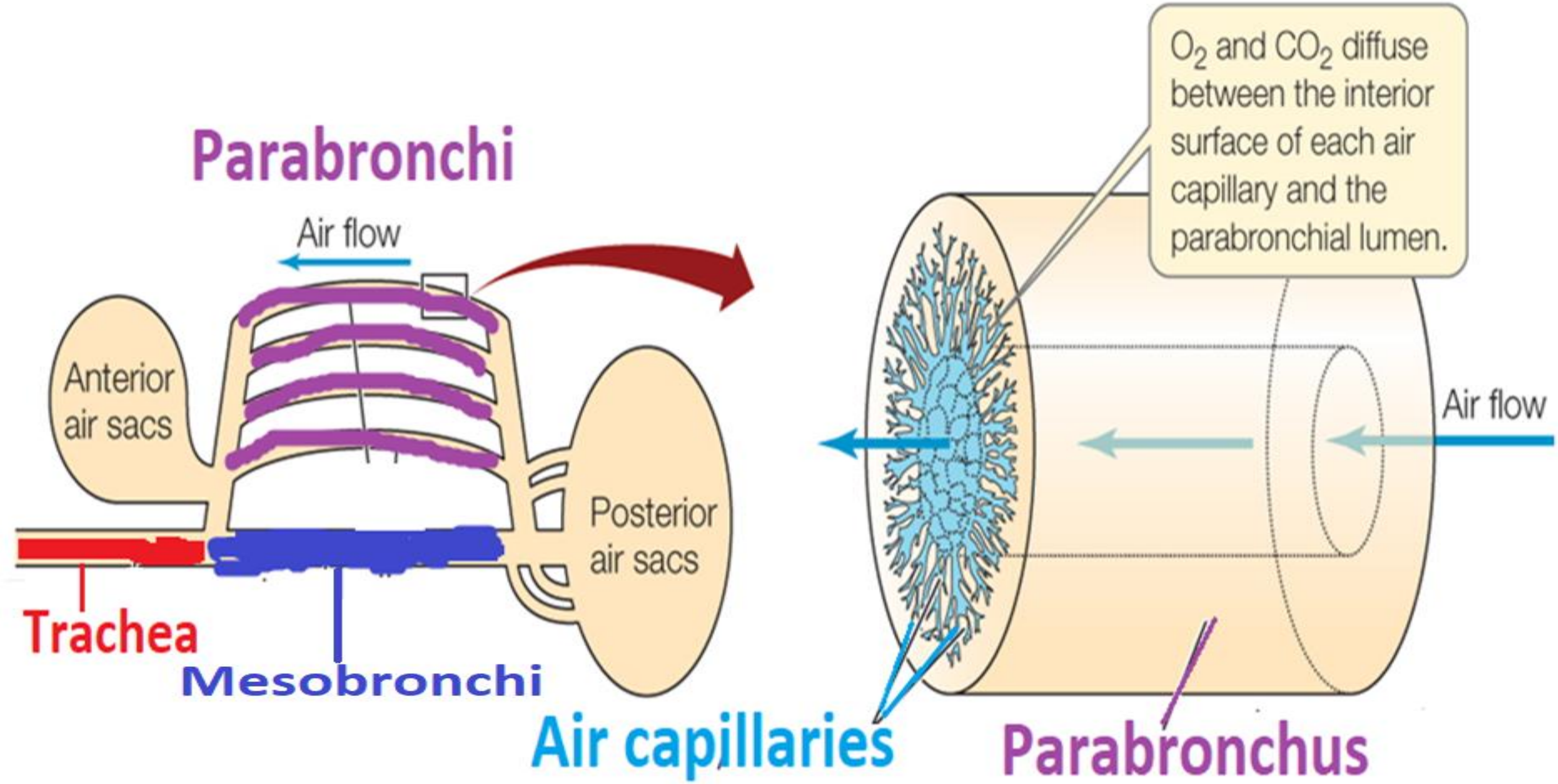
The fresh air moving into lungs displaces stale air (red) from previous breath, moving it into anterior air sacs.

Double avian respiratory Cycle



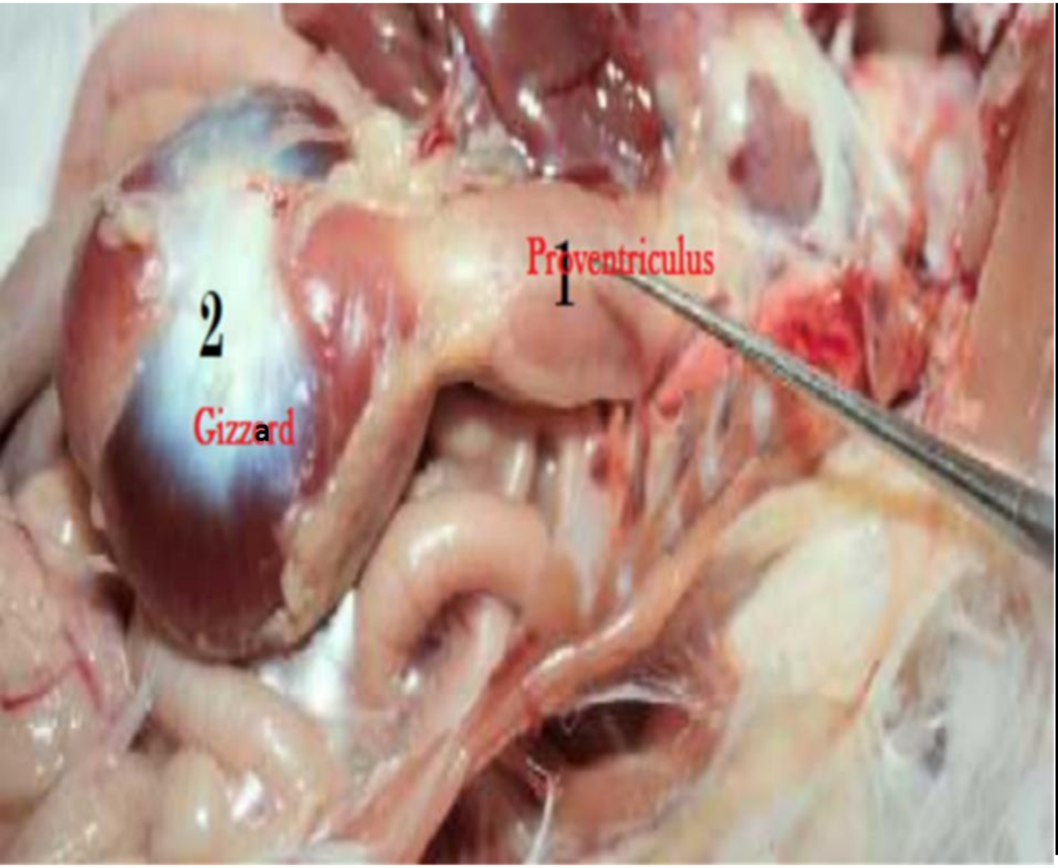
During exhalation, fresh air from posterior air sac moves into lungs, while stale air from anterior air sacs is expelled through bronchus &trachea.

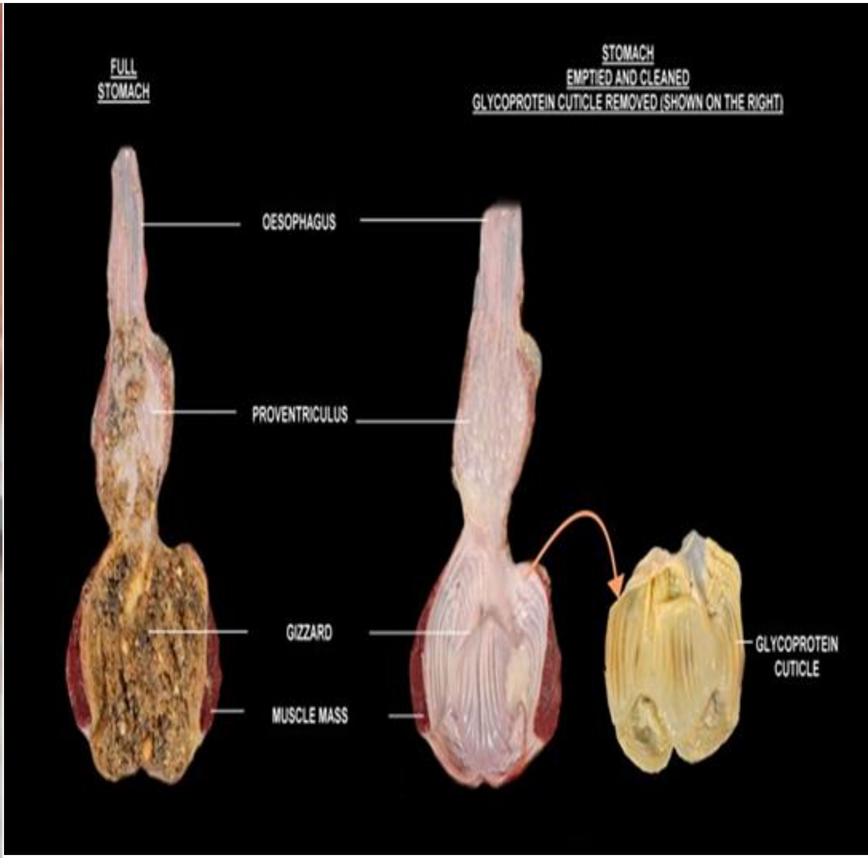
Air capillaries



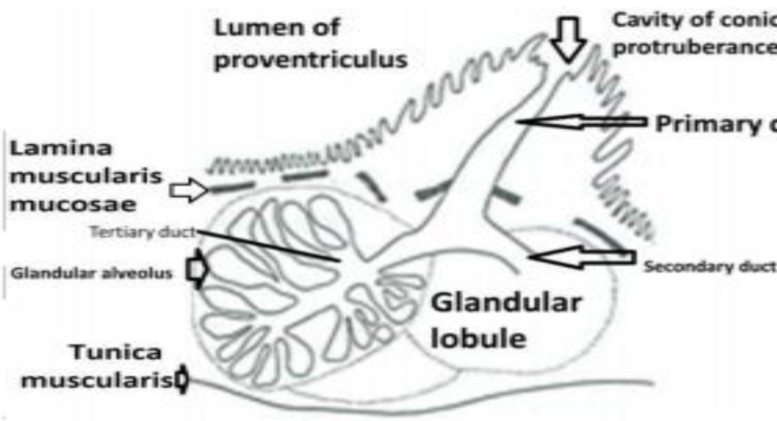
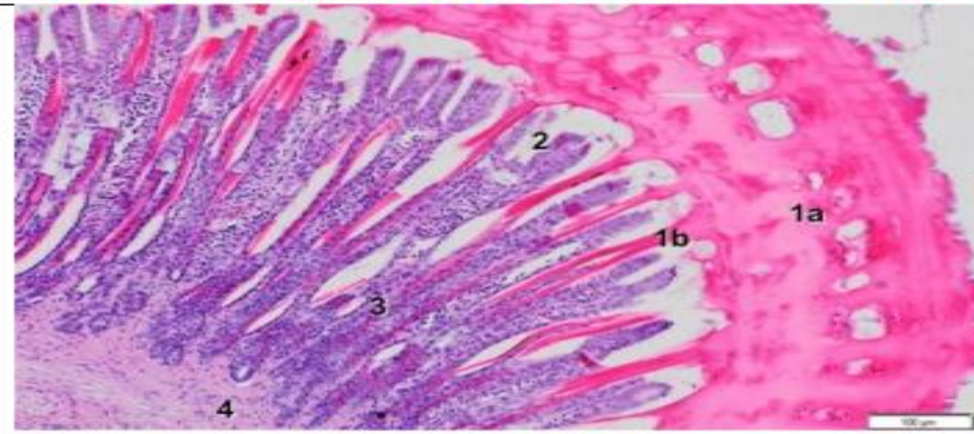
Air capillaries(pneumo-capillaries)have walls *similar* to those of **mammalian *alveoli***.

The stomach

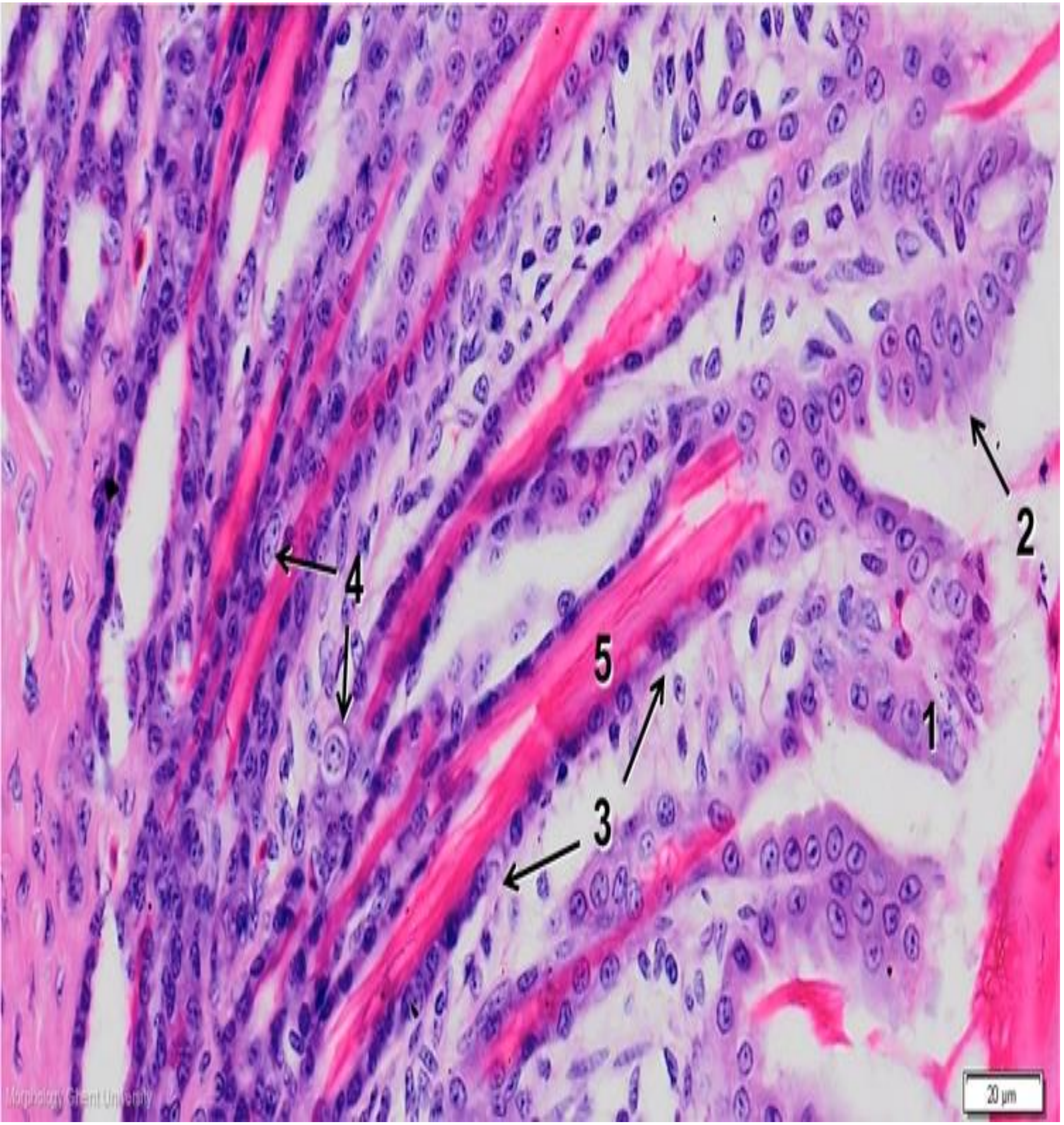




In birds, generally, the *stomach* consists of
1-Proventriculus (cranial *glandular* part)
2-Gizzard (caudal *muscular* part).

Proventriculus (*glandular*)	Gizzard (caudal *muscular*)
	
Submucosal glands	***Tunica cuticula*** is **secretory product** of **gizzard glands**, is a **thick , hard &*keratin-like*** layer covering **epithelium**.
Lining epithelium is columnar *mucous secreting* cells.	**Simple columnar *surface epithelium*** continues into **branched tubular glands**
Tunica muscularis consists of **thick, *inner circular* layer** & thinner, *outer longitudinal* layer of smooth muscles.	***Tunica muscularis*** ***single* thick *circular* & *mixed* (smooth & striated muscles).**  ***No muscle* fibers** at the **center of the *aponeurosis* (*dense regular***

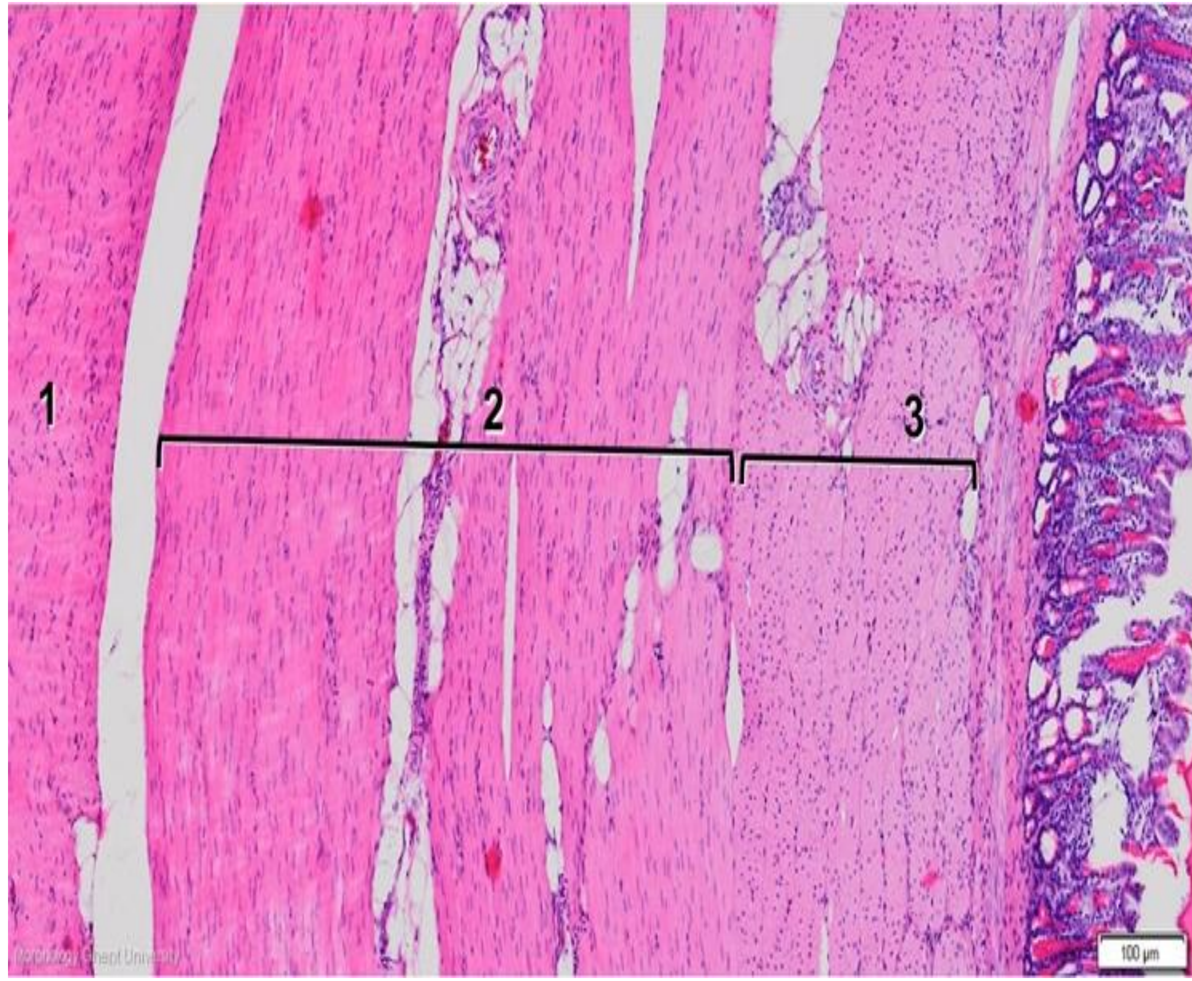
Gizzard



1-single cuboidal epithelium
2-apical secretion
3-chief cells
4-basal cells
5-vertical layer of the koilin

a. The **simple columnar** ***surface epithelium*** continues into **pits** **&into simple** or **branched tubular glands** that **open** into the **pits**.

The muscular part (ventriculus, gizzard)



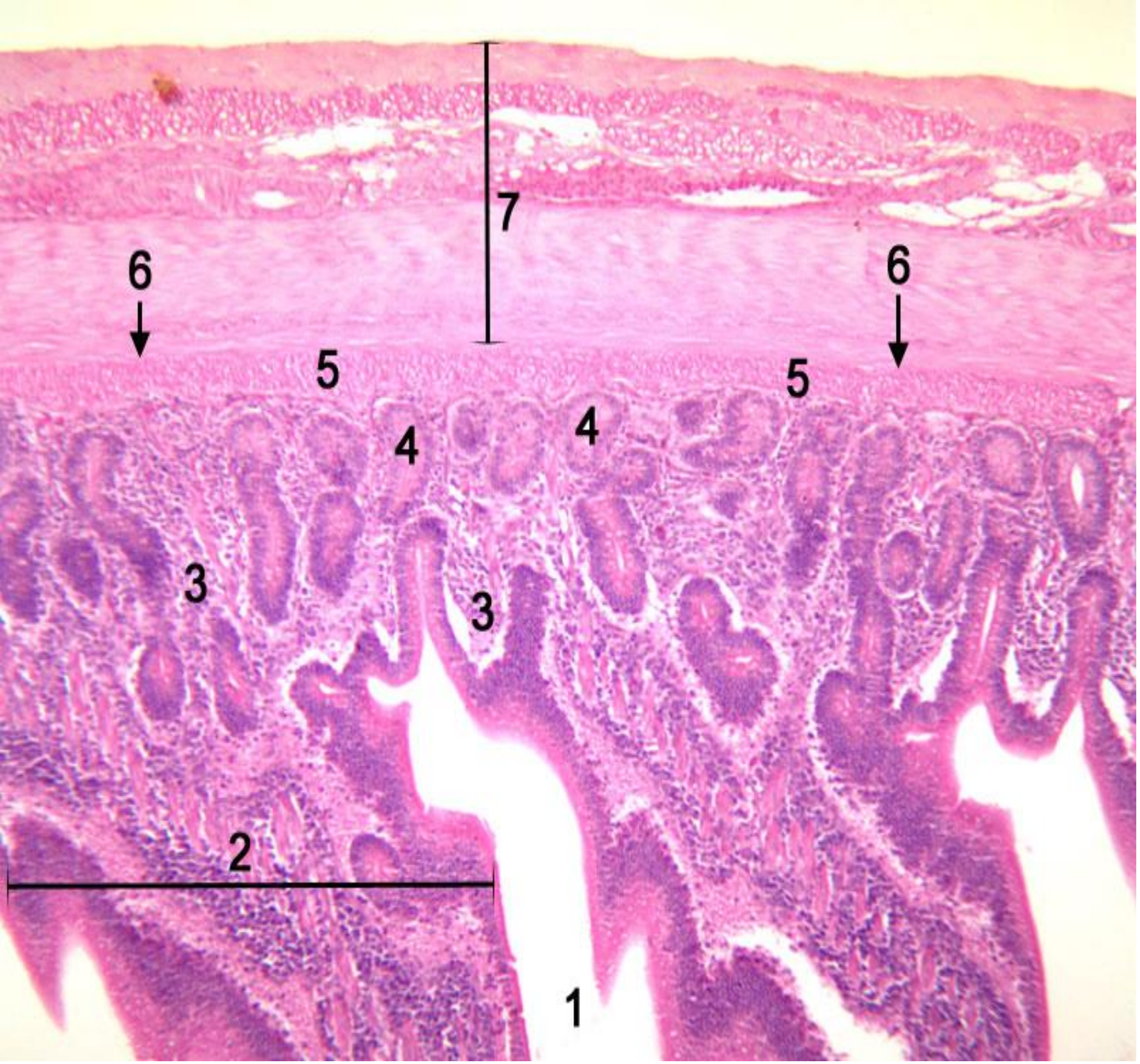
The *tunica muscularis* is a *single* thick *circular* & *mixed* (smooth & striated muscles).

The muscular part (ventriculus, gizzard)

There are ***no muscle* fibers** at the **center of the *aponeurosis (dense regular*** *C.T.)*, which **spreads out from 2 areas in the center of each side** of the gizzard.

The muscular part (ventriculus, gizzard)

Tunica serosa, the gizzard is covered externally by a *peritoneal* coat.



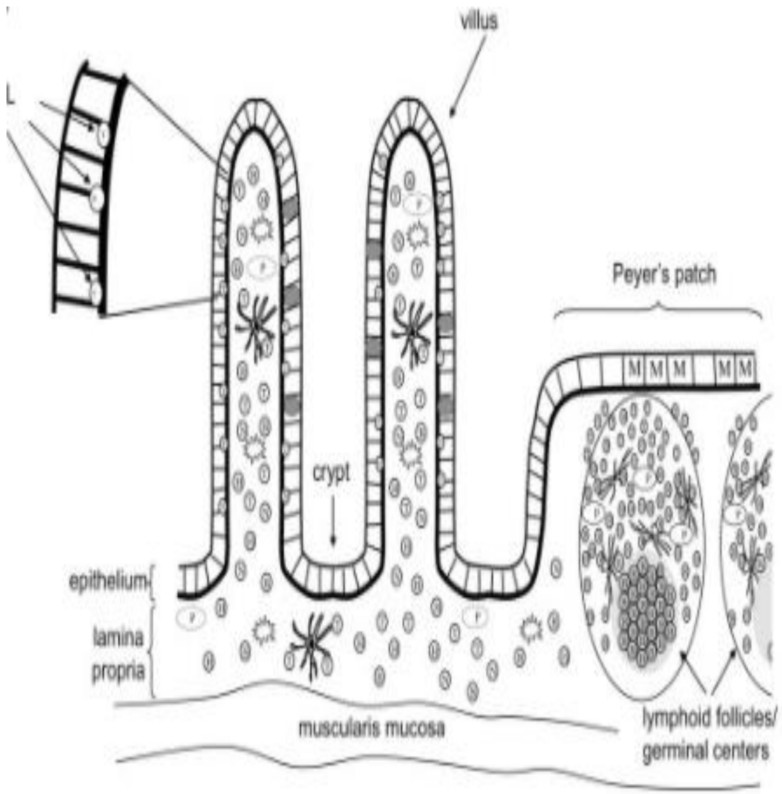
Small intestine

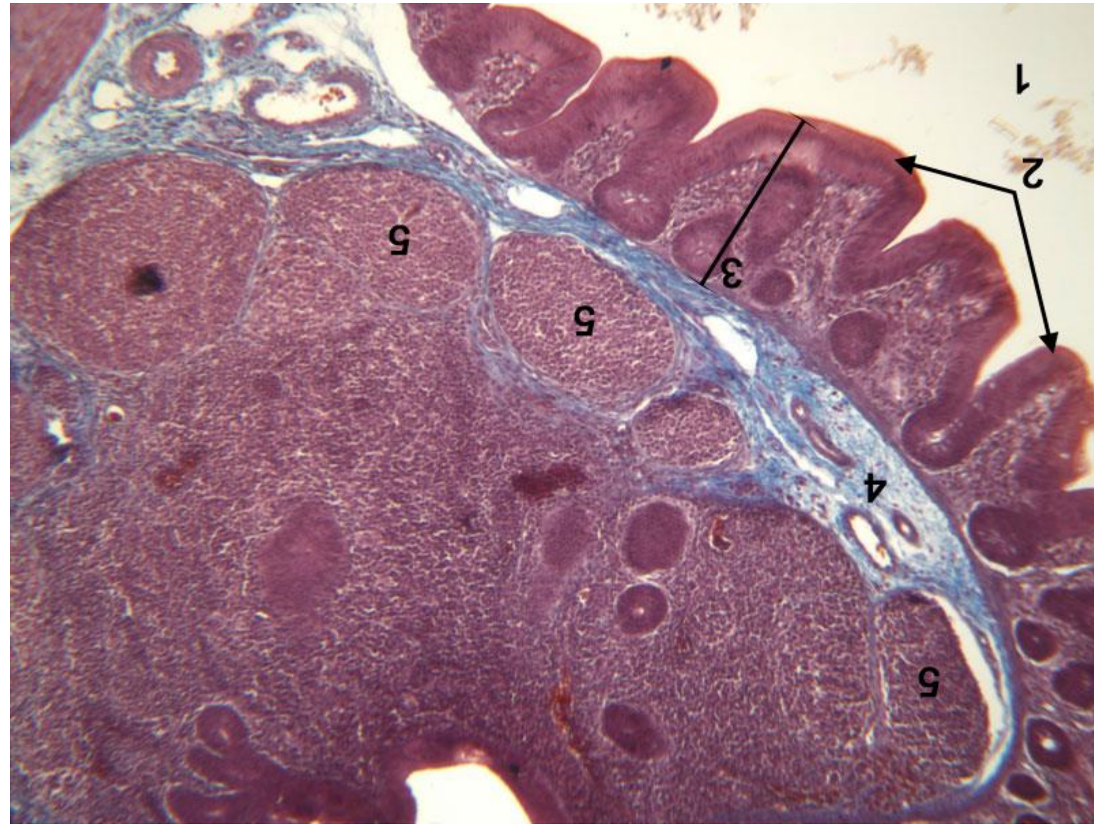
1. Lumen
2. Villi
3. Lamina propria
4. Crypts of Lieberkühn
5. Muscularis mucosae
6. Submucosa
7. Muscularis externa

Small intestine histologically differs from that of the domestic mammals in the following:

1-Intestinal *villi* are *long* & *leaf* shaped.

Large intestine



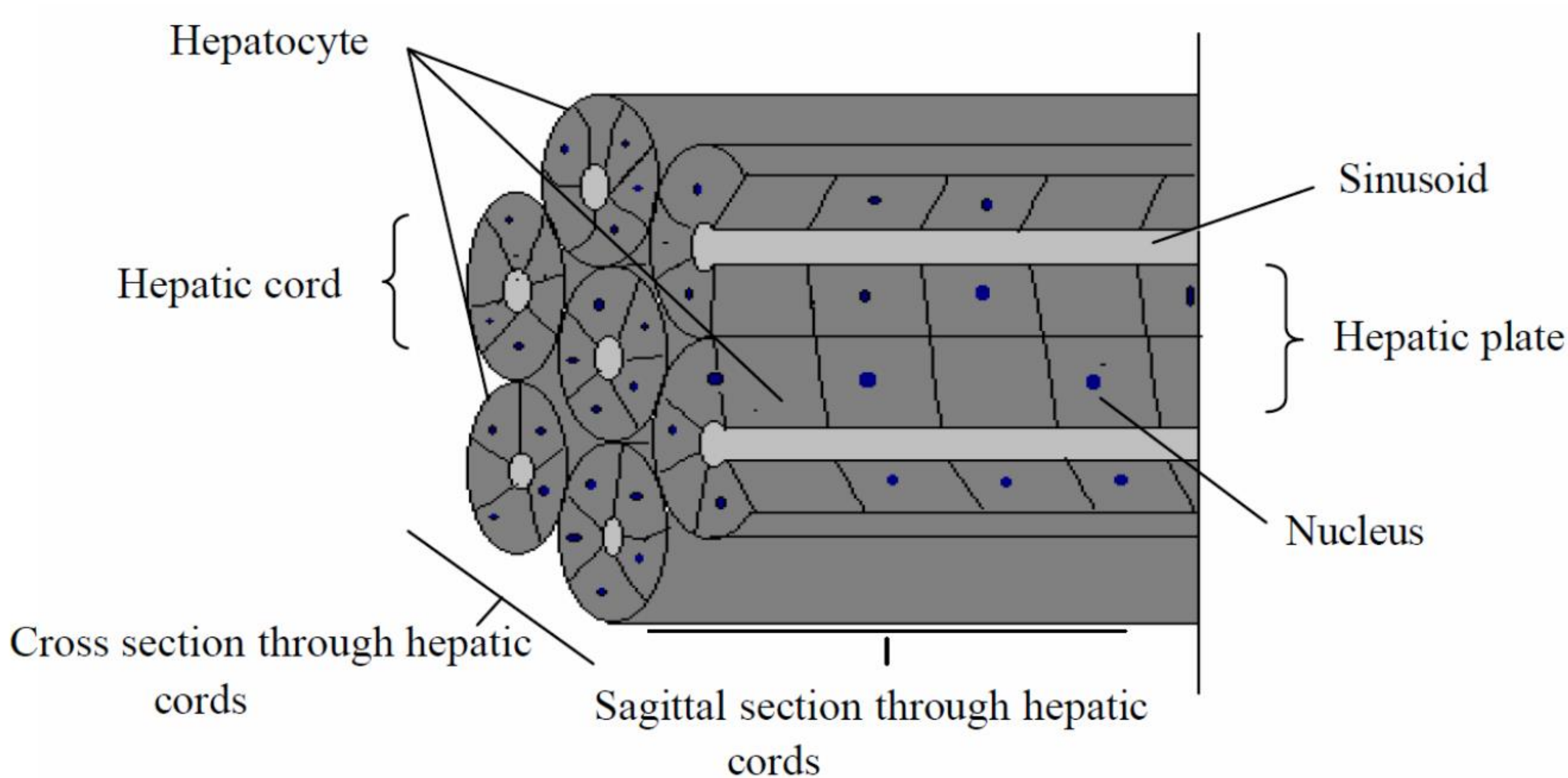


It differs from mammalian pattern in:

*1-**Villi*** are *present* in ***ceca &rectum.***

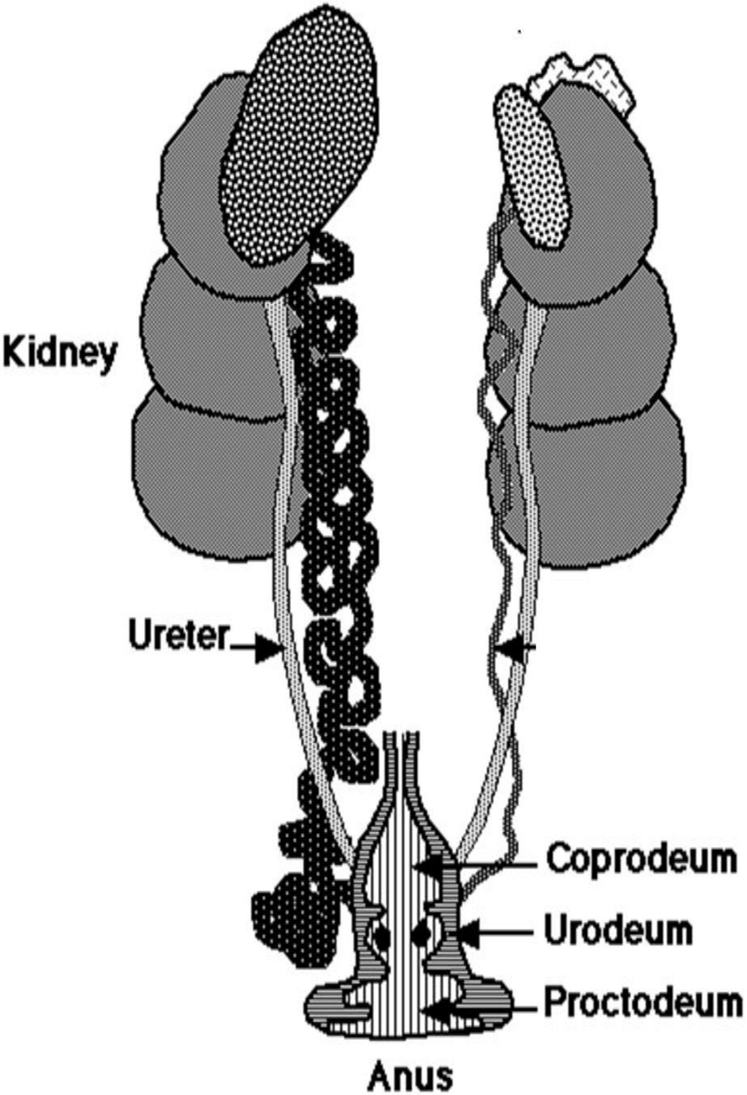
2-Solitary &aggregated *lymphatic* tissue are more *plentiful* in *propria* &tunica *submucosa*.

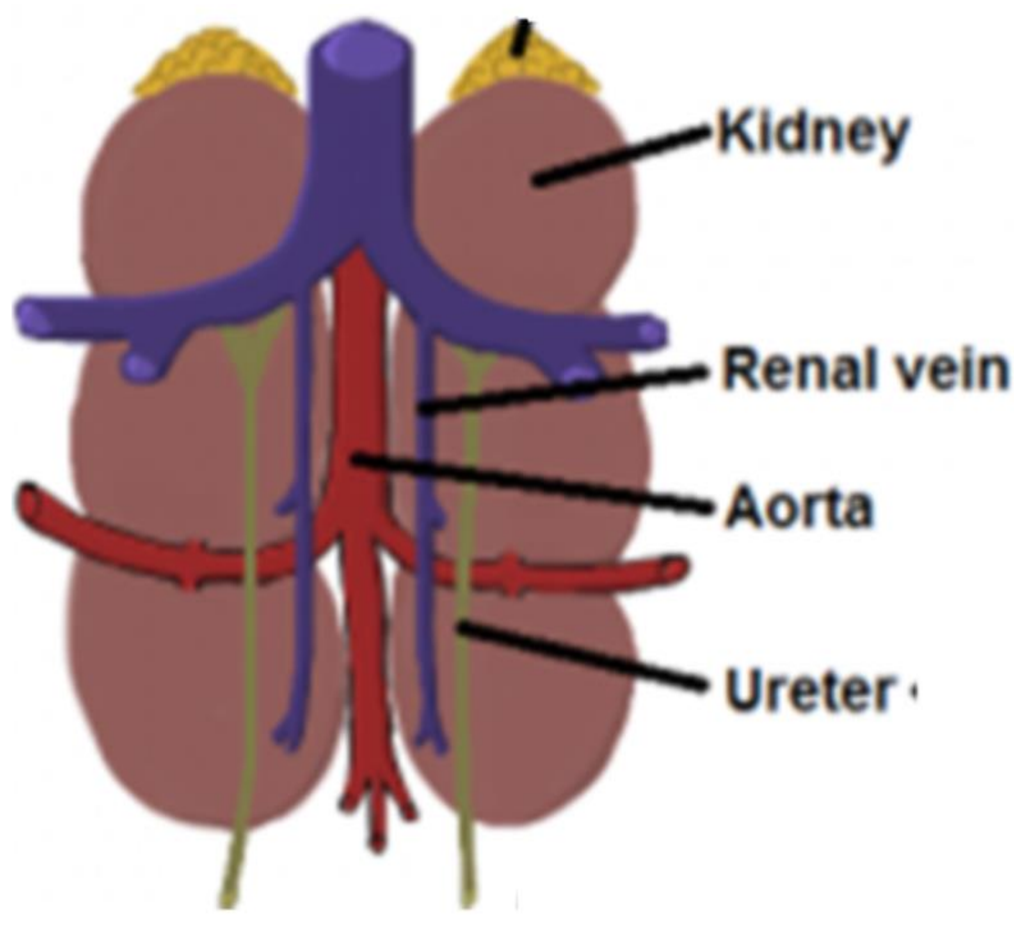
The **liver**



- The radiating **plates of hepatocytes** in each lobule are **2 cells wide** in the **chicken**. Although, *mammalian models* are **1 cell wide**.

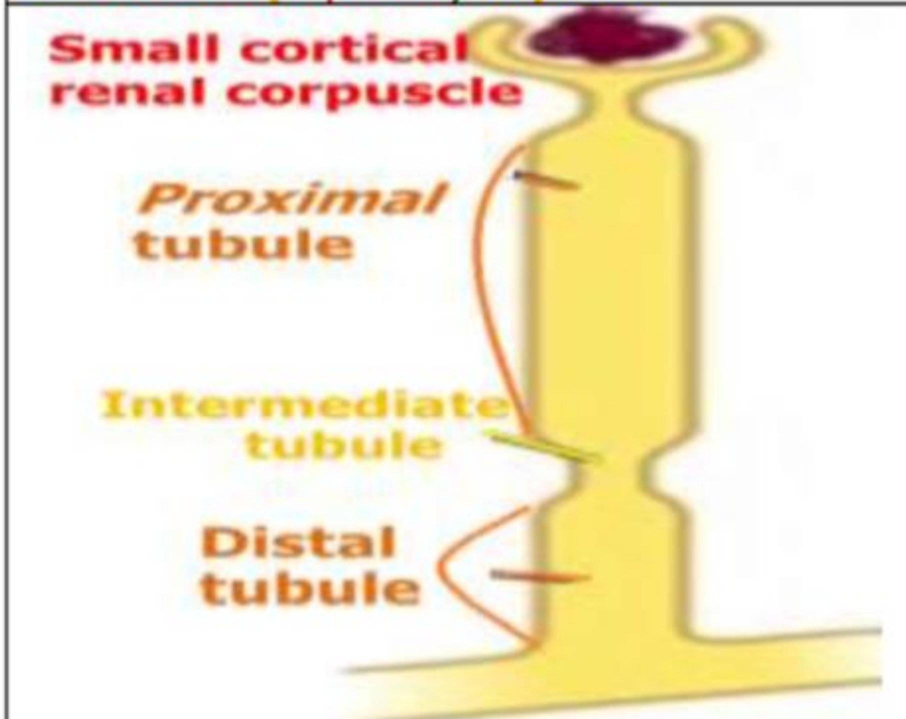
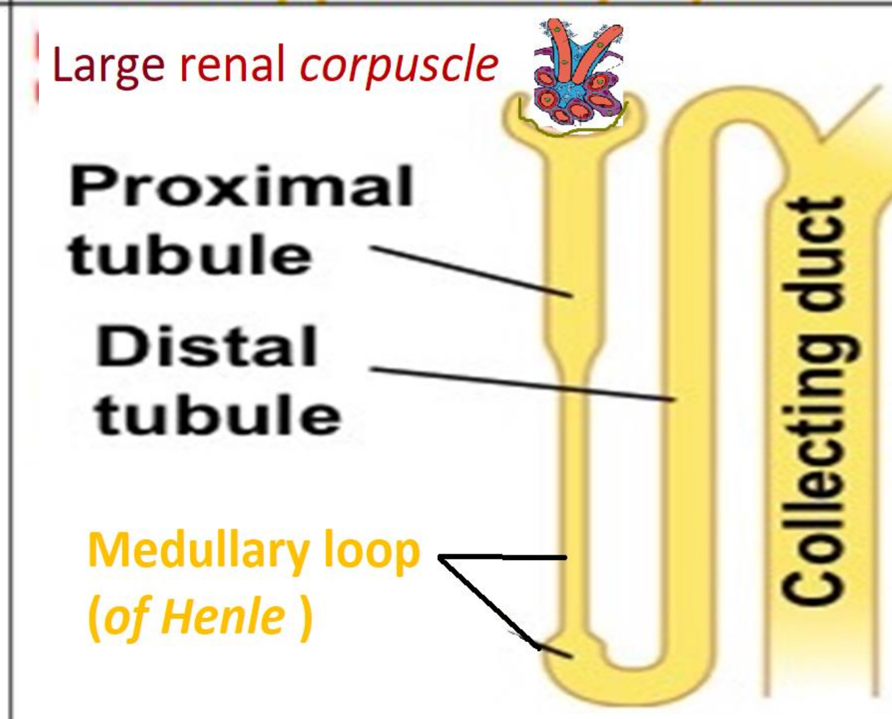
Urinary system



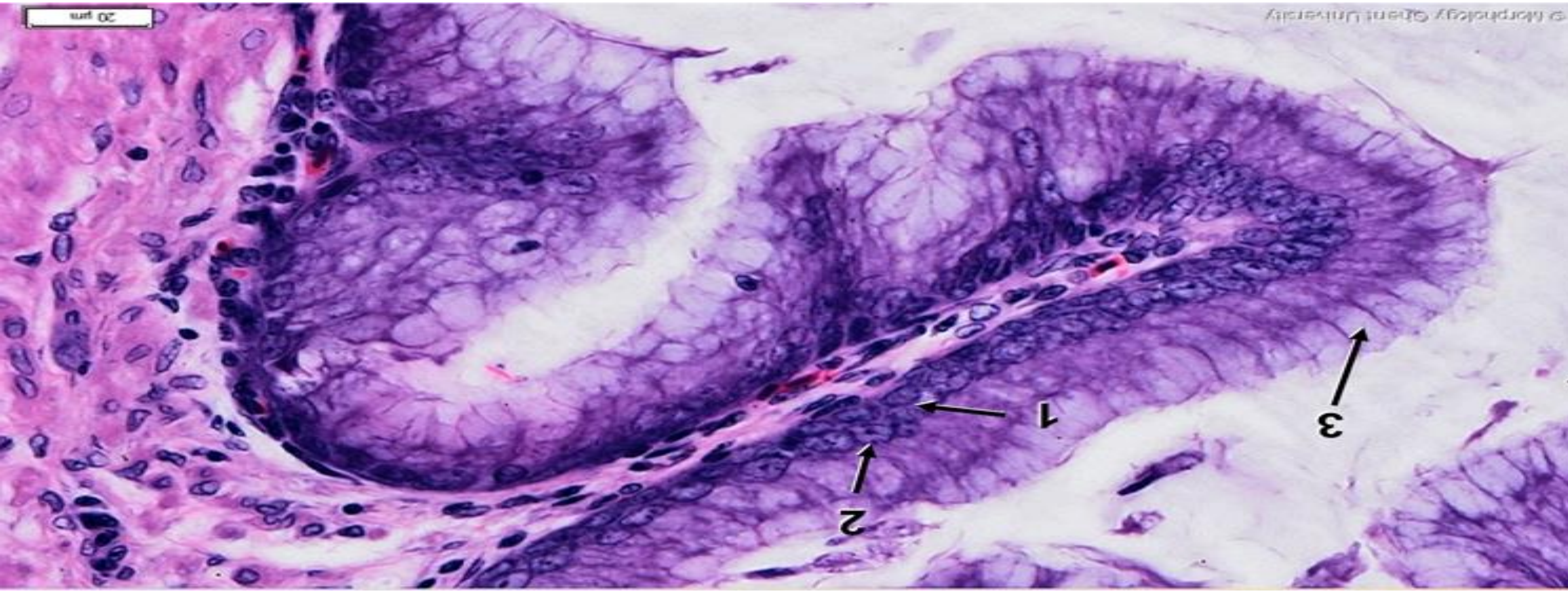


The urinary system of the chicken consists of large, elongated, paired *kidneys*.

Main histological difference between 2 types of nephrons in avian kidney

1-*Cortical* (reptilian) *nephron*	2-*Medullary (mammalian) nephron*
	
Small cortical renal corpuscle.	**Large medullary** renal corpuscle.
No loops of Henle. **short *intermediate tubule***	**Have long or short *medullary* (Henle's) *loop***
75-85%	**15-25%**

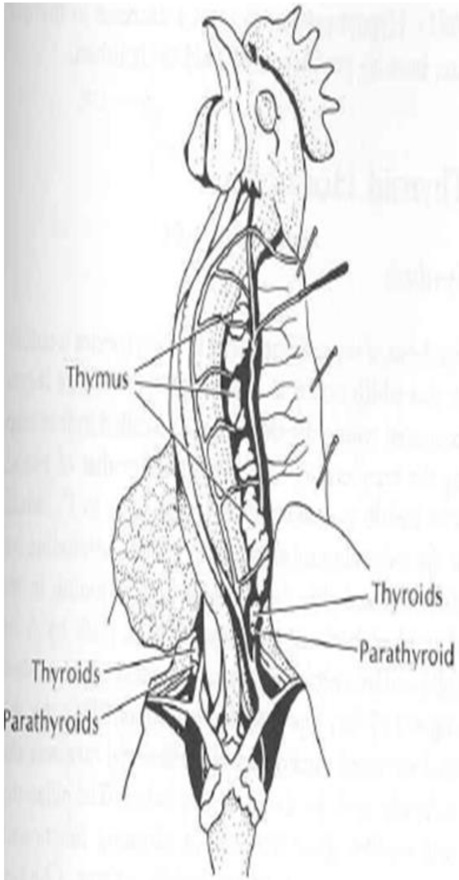
Avian ureter

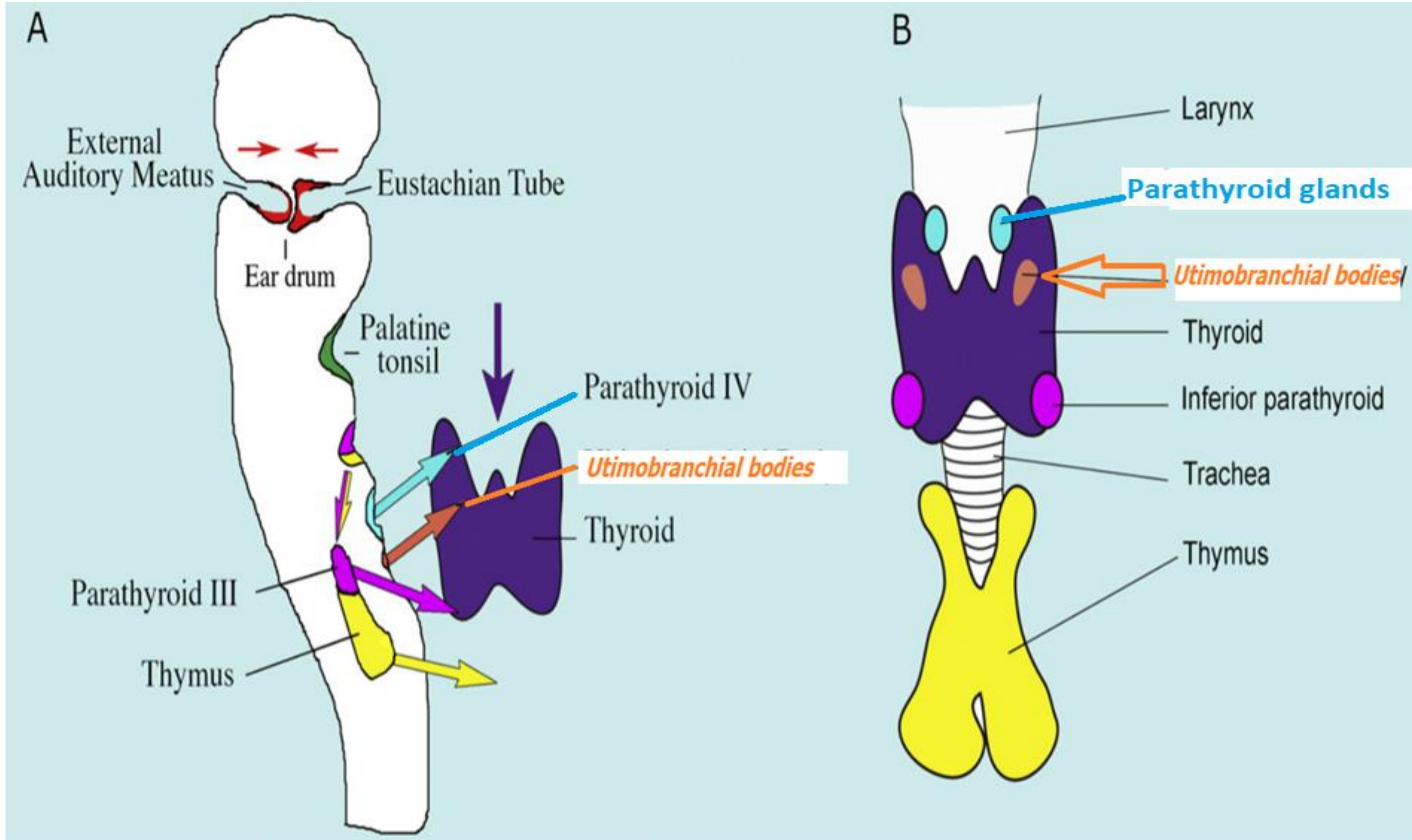


Ureter is lined by a ***pseud.str.*** **columnar** epithelium.

Vacuoles filled with **mucus** play a role in **binding** (precipitated uric acid), that birds produce **instead of urea in mammals.**

Avian parafollicular cells present in *Ultimobranchial bodies*

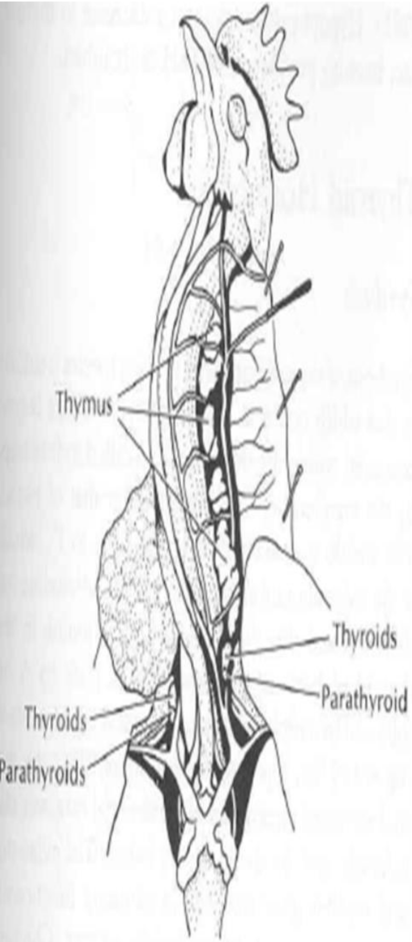


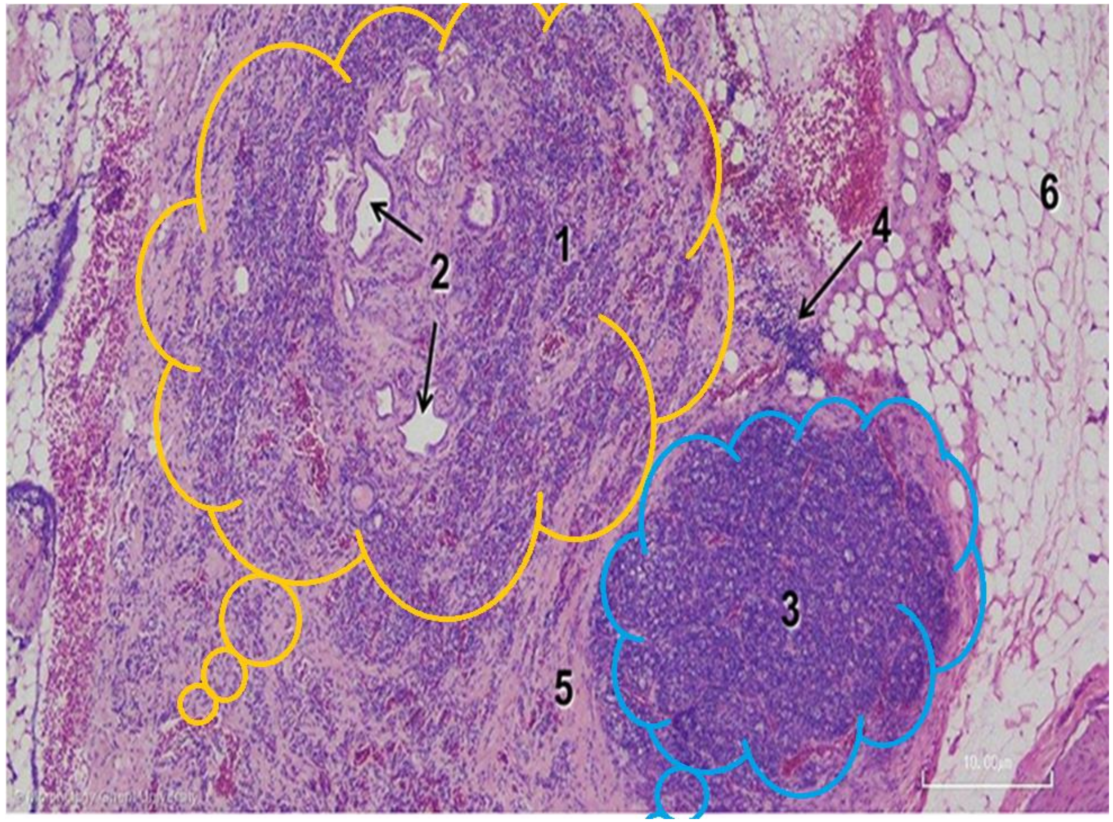


Migration of the caudal glands and final position in mammals. (A) Migration of the caudal glands. (B) Final position in midline.

Ultimobranchial bodies(=glands=organs), are paired 'organs' that lie closely associated to **parathyroid glands.**

Left ultimobranchial body of a laying hen



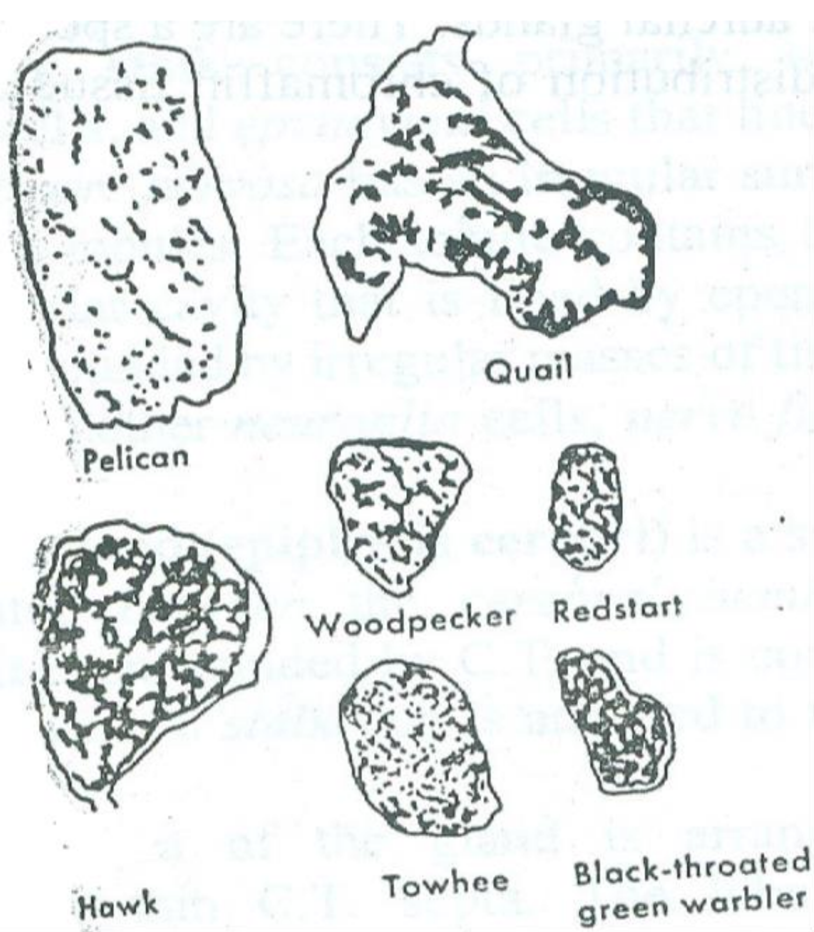


-Calcitonin of the ultimobranchial bodies

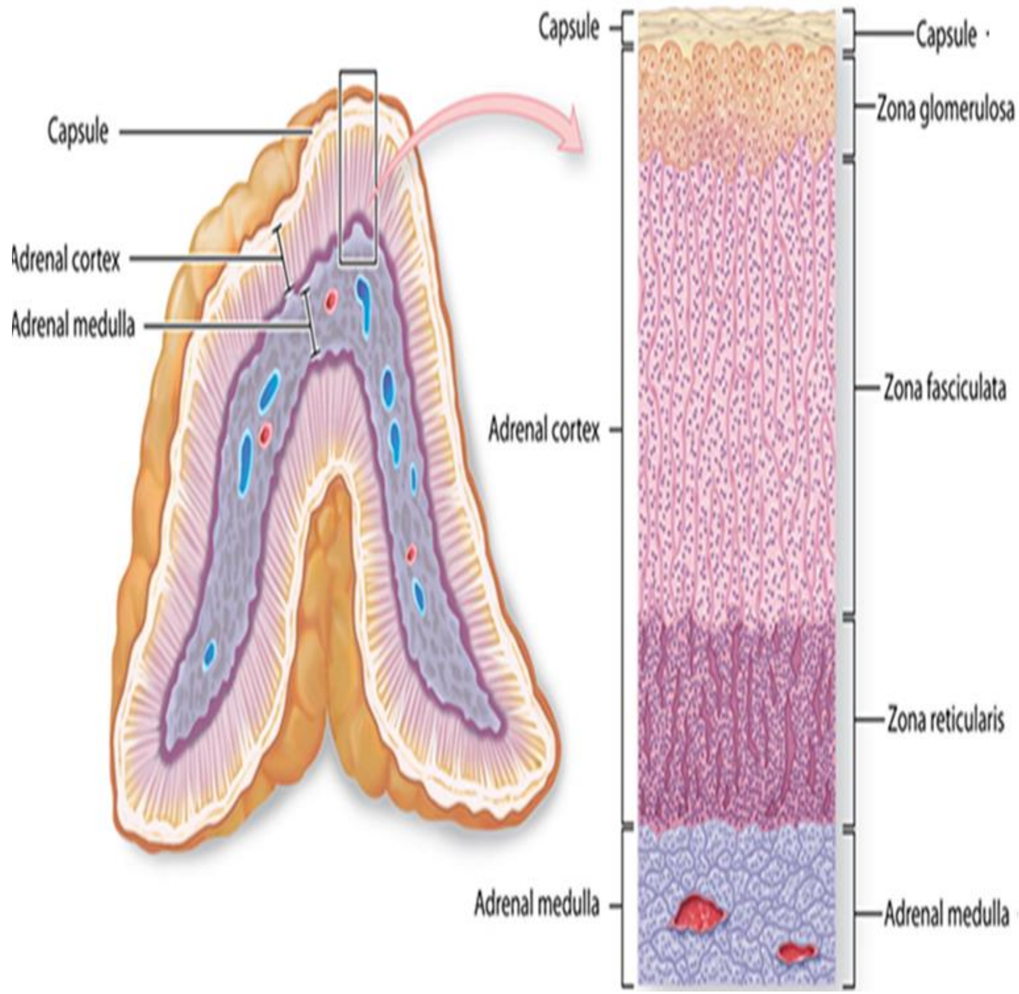
-Parathormone of parathyroids

- Must be in balance if calcium levels in blood are to be in balance to requirements.

Avian adrenal glands

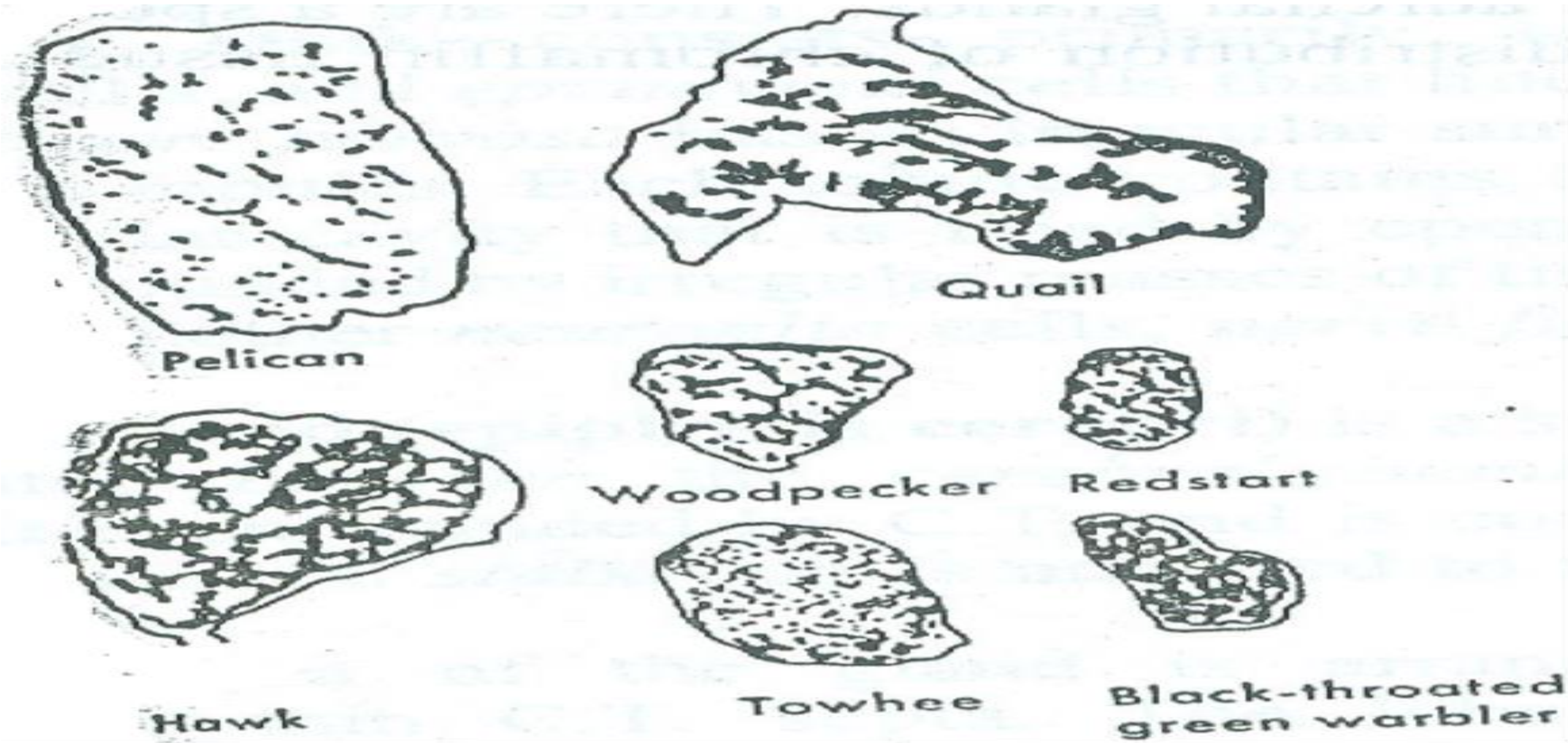


Mammalian adrenal gland



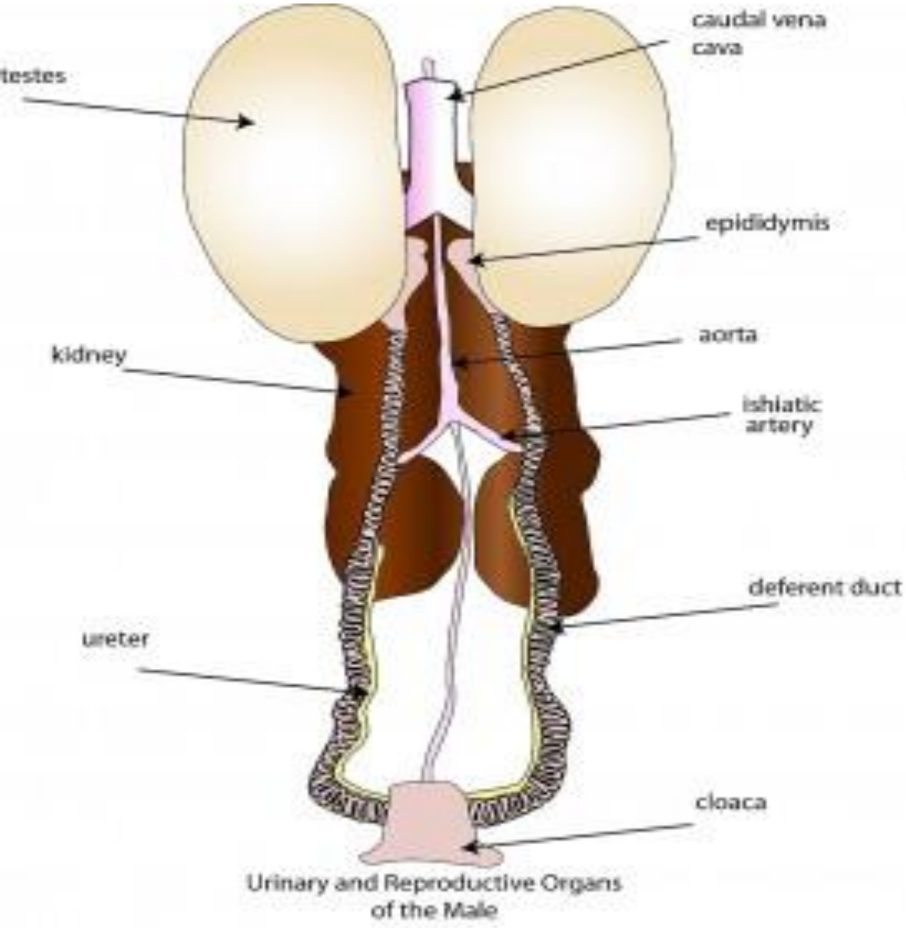
Unlike mammals, the **parenchyma** is not organized into a distinct **cortex** &medulla.

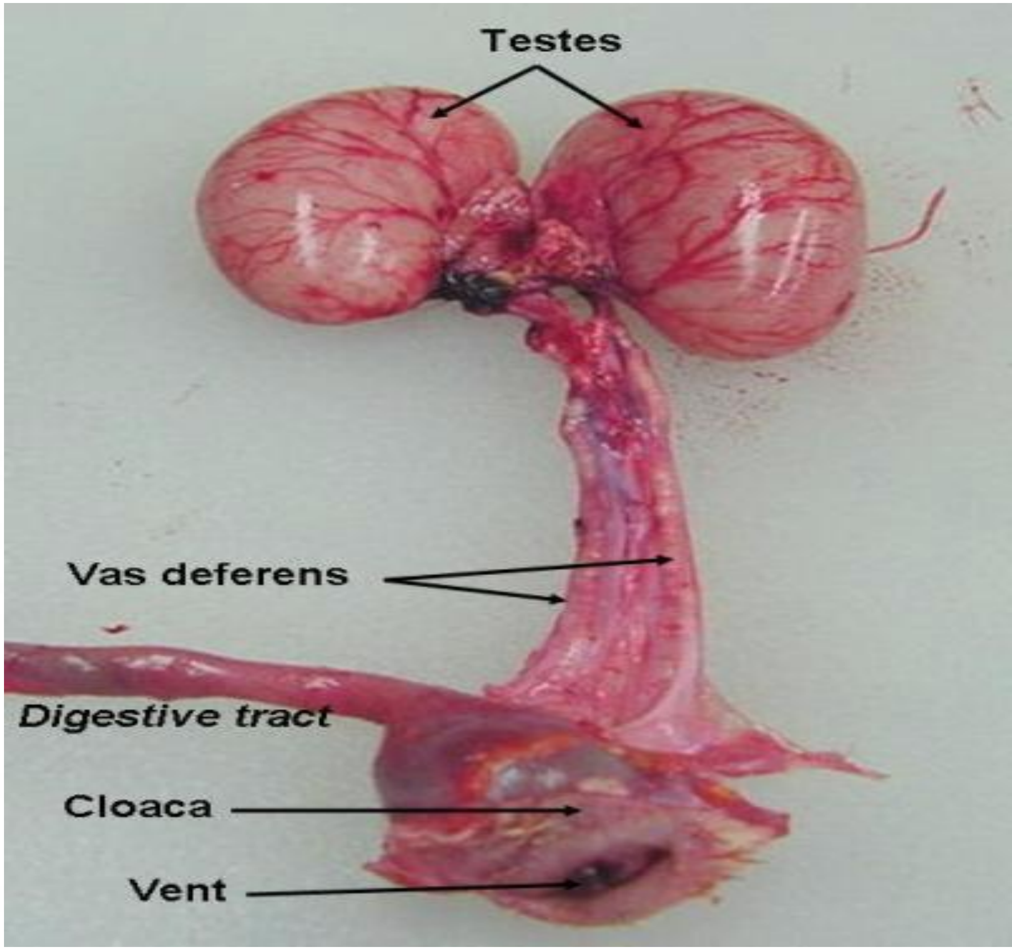
Avian adrenal glands



It is composed of ***intermingled cortical*** =interrenal tissue white & **medullary (chromaffin tissue= (black)) tissue.**

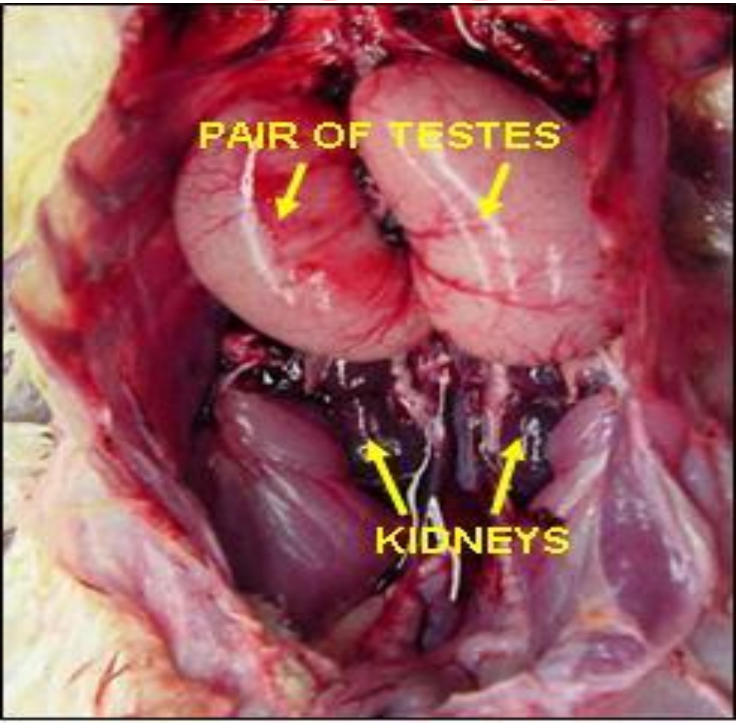
The male reproductive system

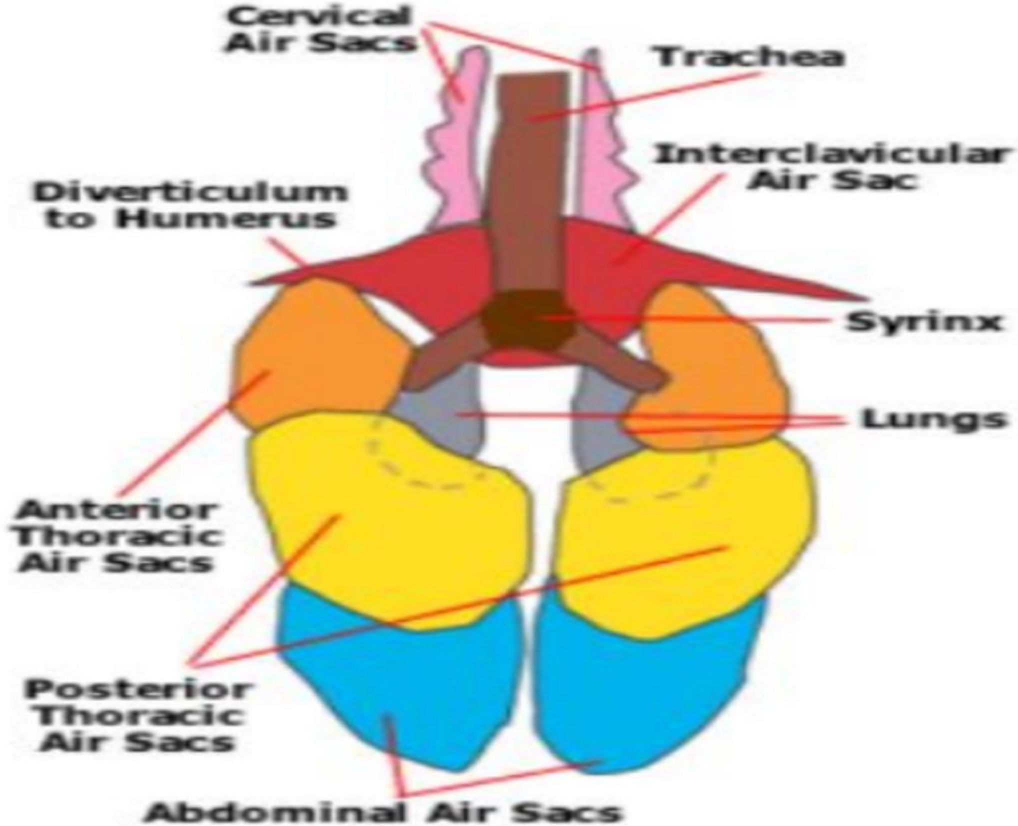




It is composed of **paired testis**, **small epididymides**, **long coiled ducti deferentes** opening into **urodeum** of the cloaca via ejaculatory ducts &certain erectile organs within the cloaca.

Testes



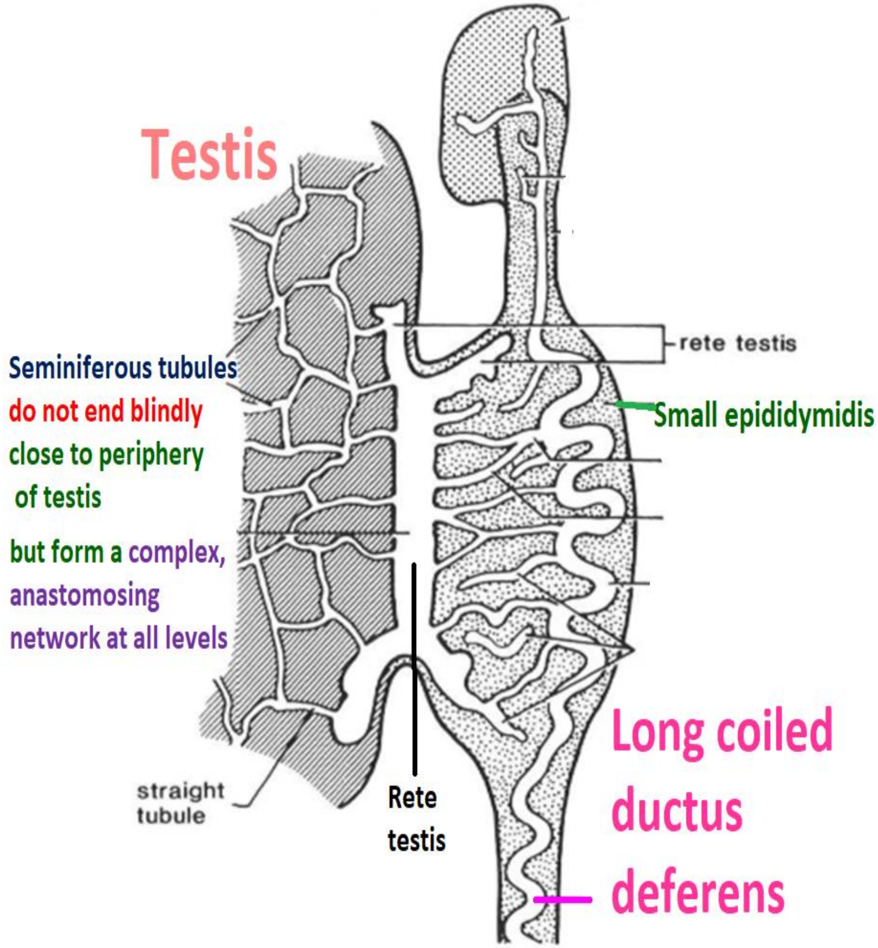


-In birds, **testes** must be in **abdomen** because of **aerodynamic** issue in **evolution.**

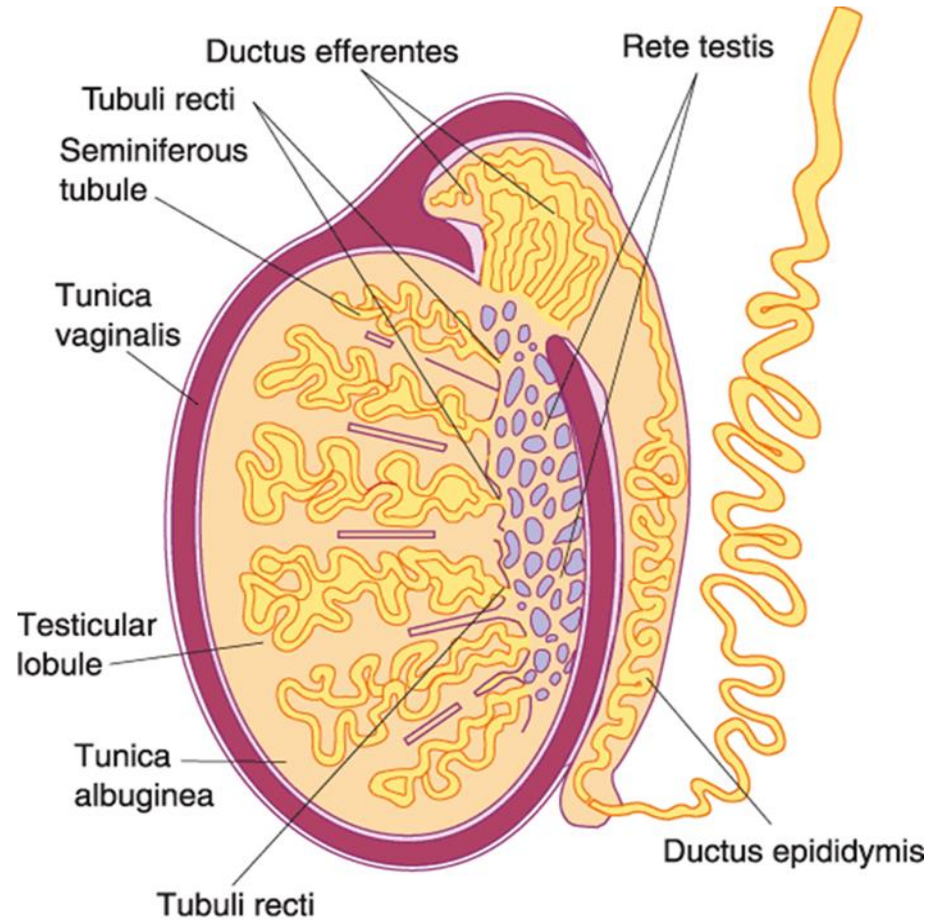
Thermoregulation is arranged by **air sacs .**

-Testes in **birds** is near **air sac** in **abdomen** so it is **not need scrotal cooling**.

Avian testis



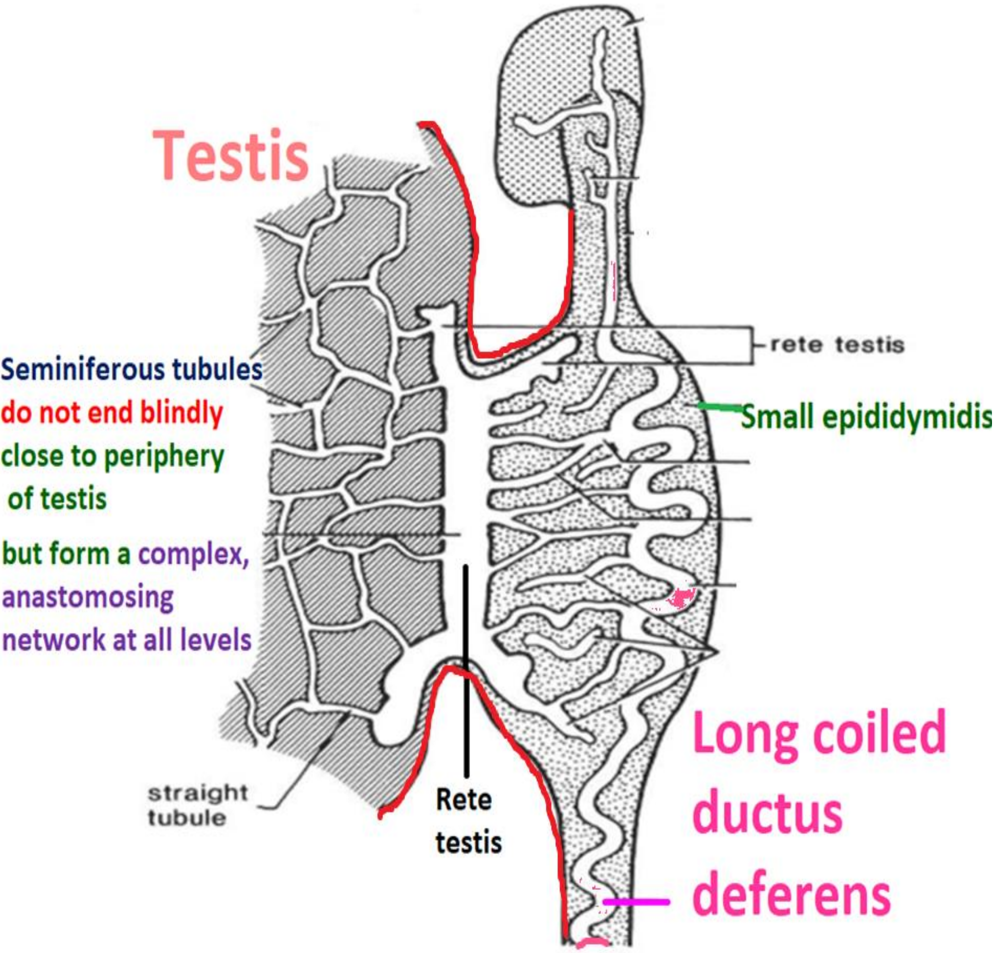
Mammalian testis

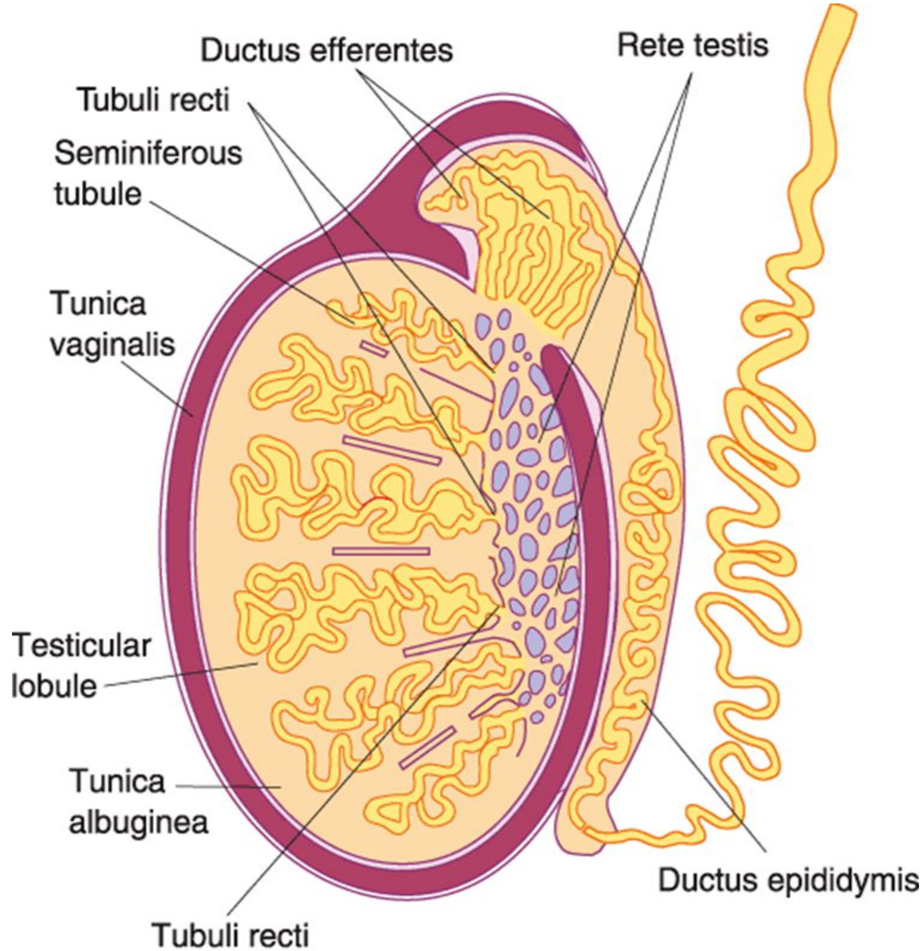


In cock, **testis** capsule is very thin &it also does not give off septa to divide testis into separate lobules.

Avian testis

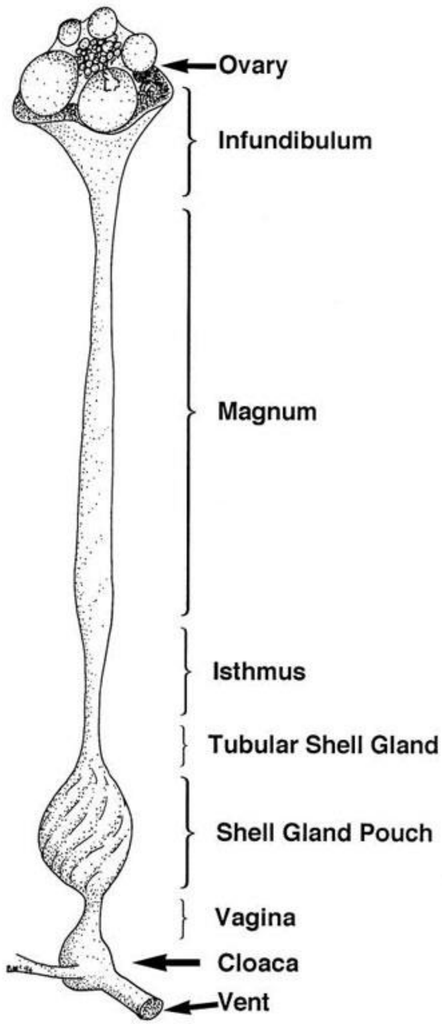
Mammalian testis

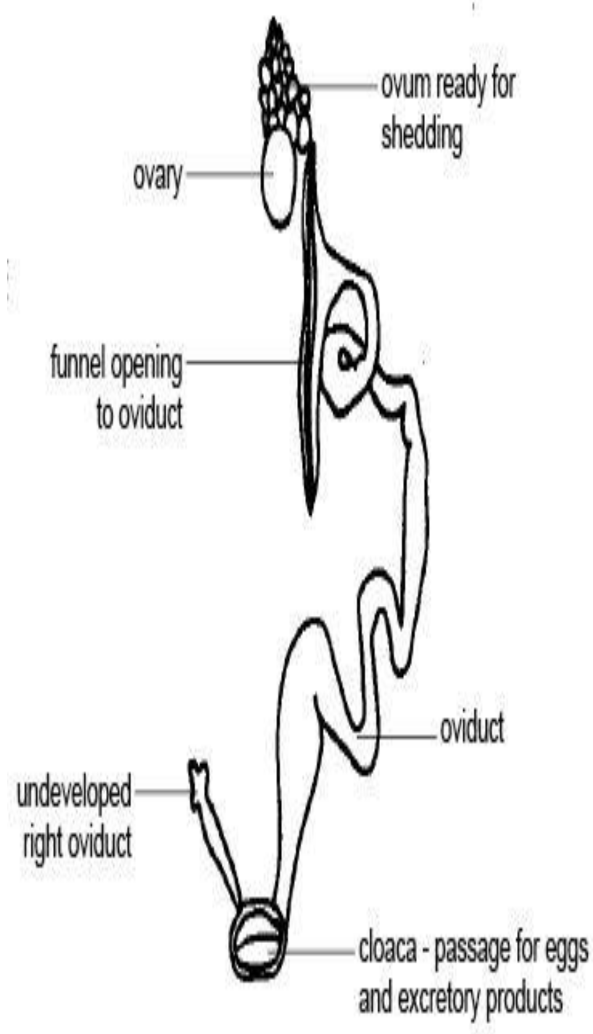




The seminiferous tubules of cock, do not end blindly close to periphery of the organ, (as in mammals), but form a complex, anastomosing network at all levels.

Avian female reproductive system

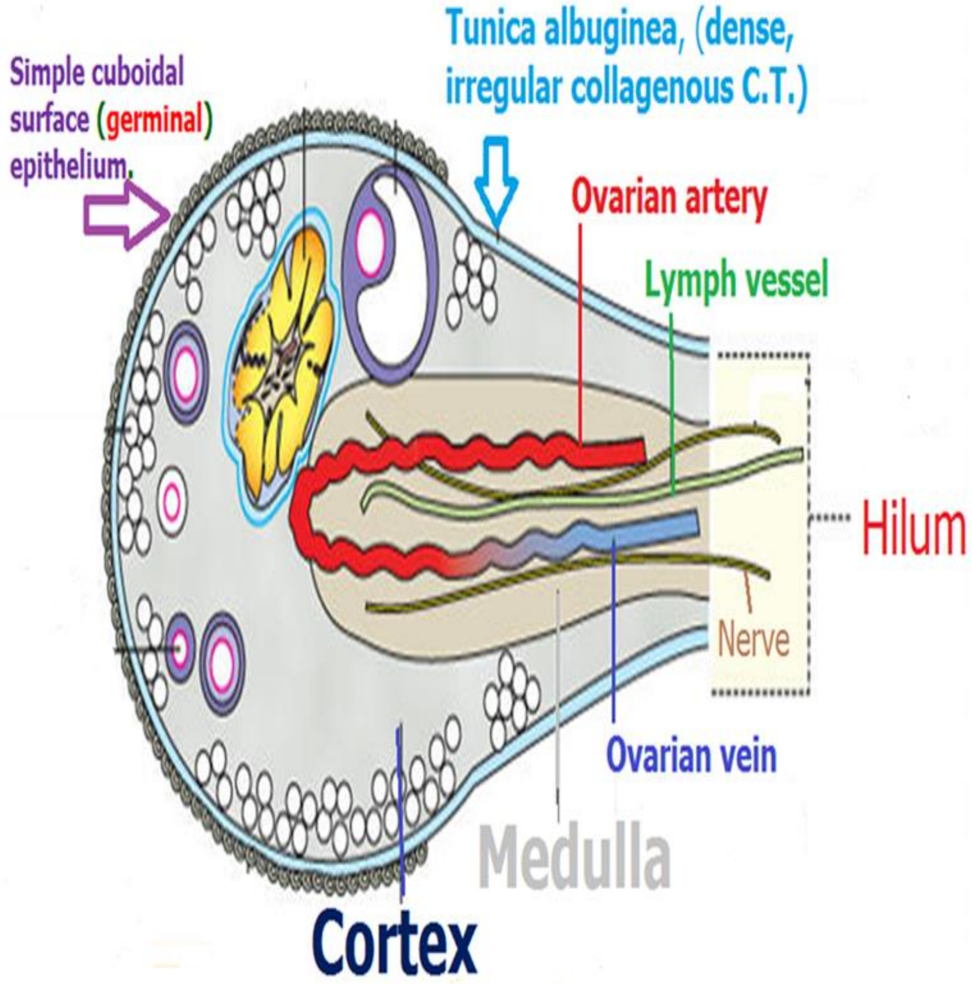




A.Ovary: ***Only*** **the** ***left*** **ovary develops, except owls &*hawaks* have *right &left* ovaries.**

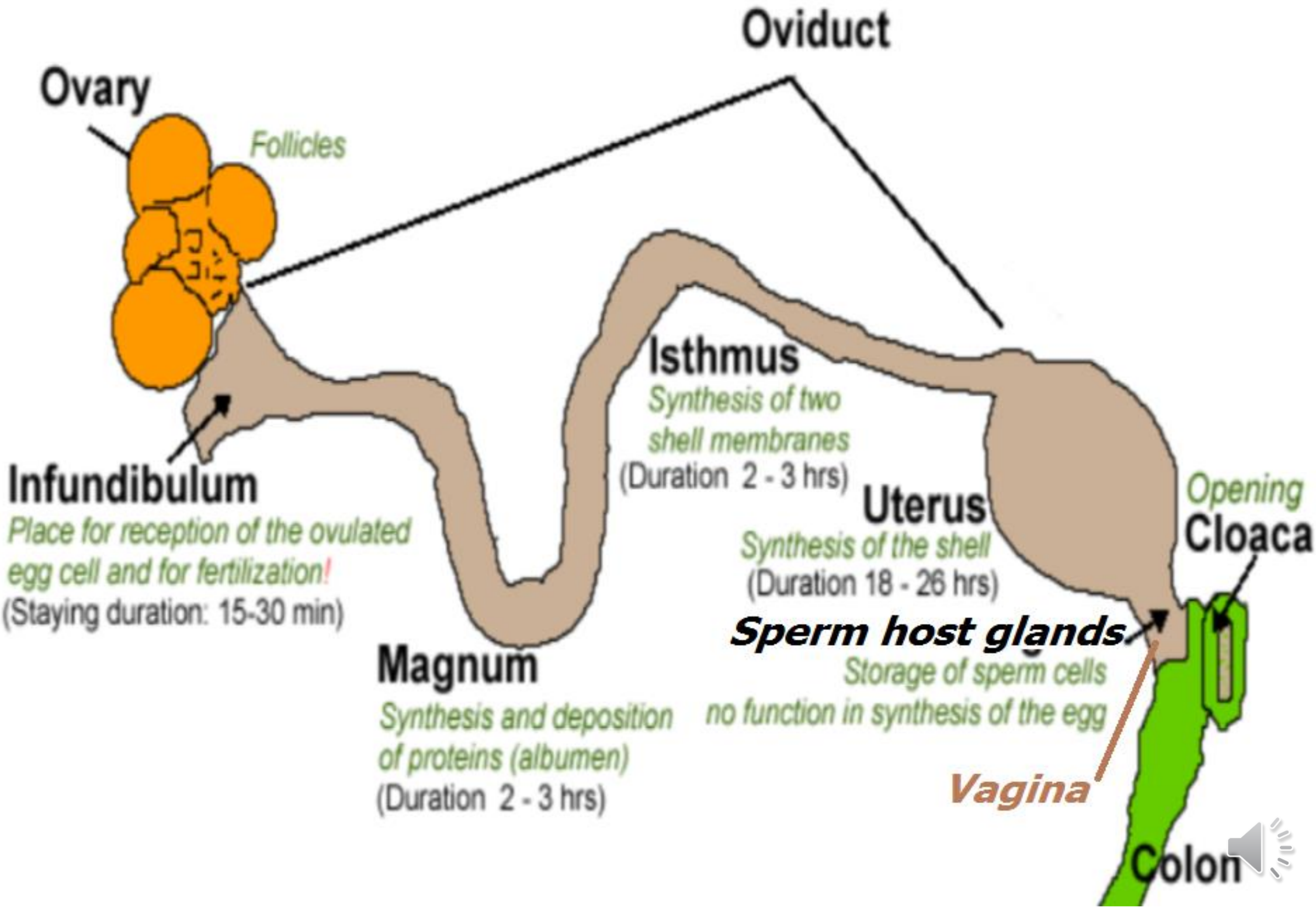
Avian ovary

mammalian ovary

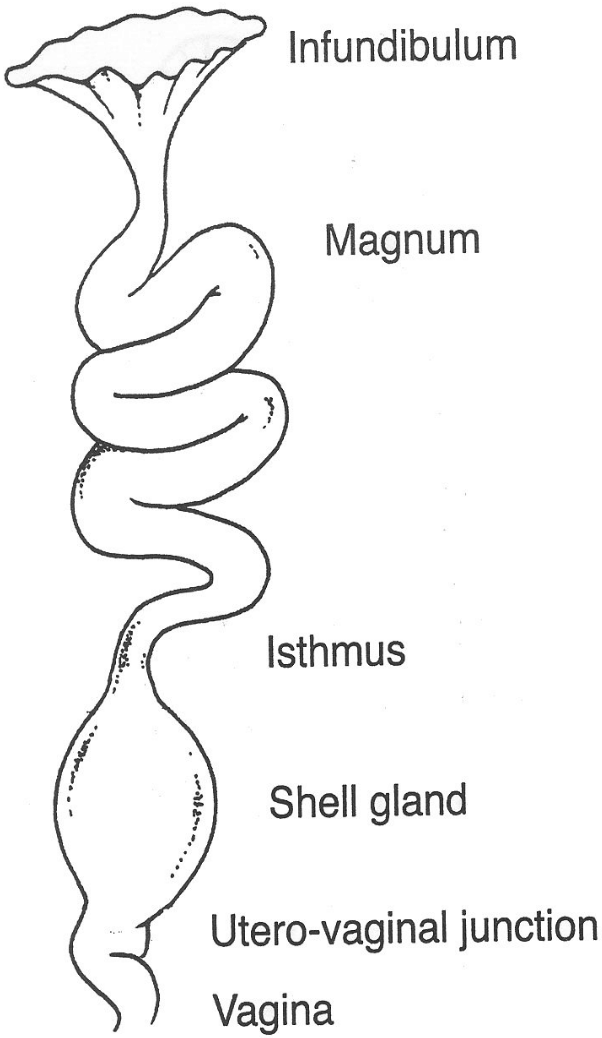


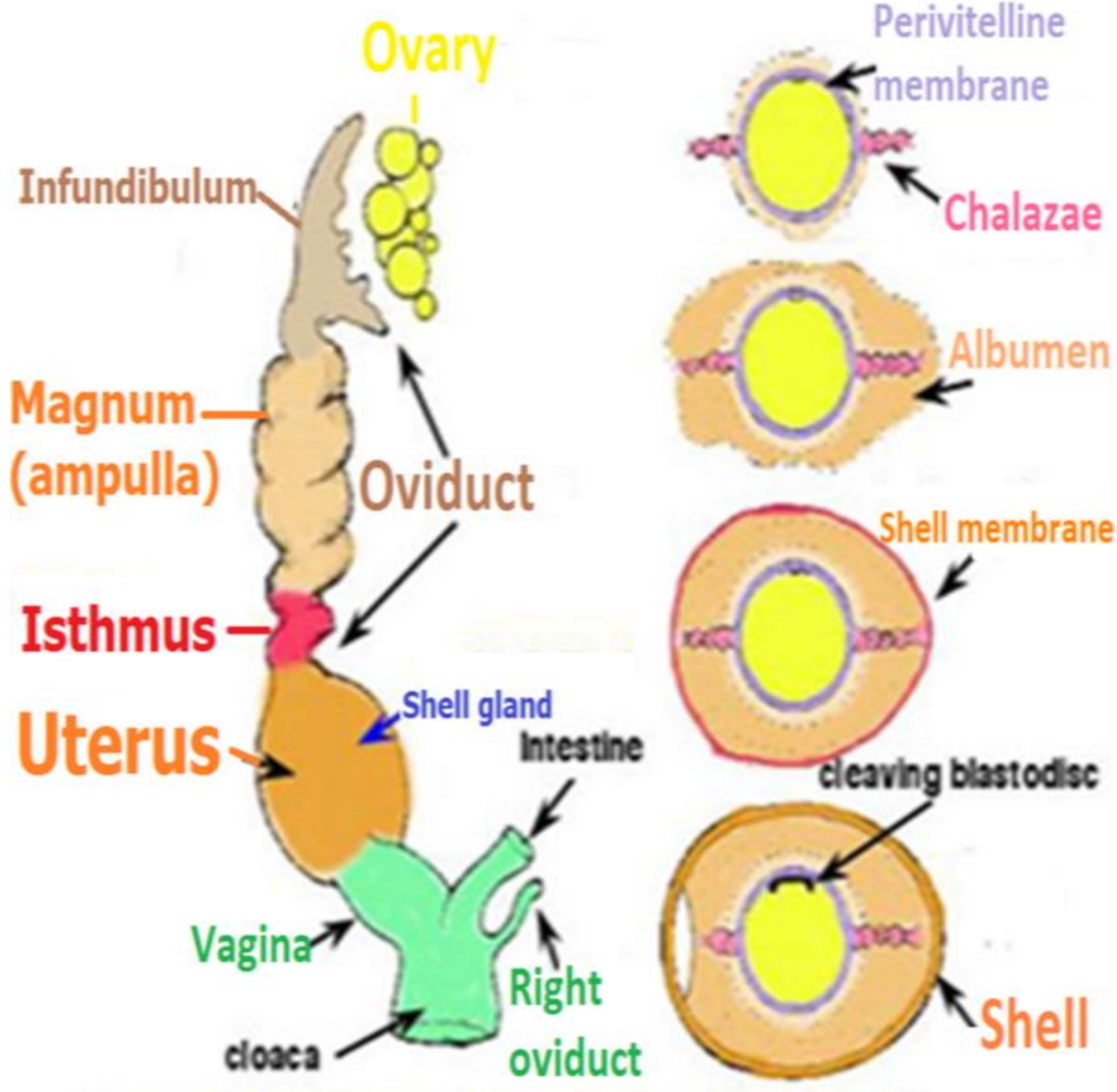
Ovarian cortex and medulla are not clearly defined as **mammalian** pattern. ***Ovarian cortex*** is solid, stalked ovarian follicles that ***never ever develop an antrum***.

Avian female reproductive system



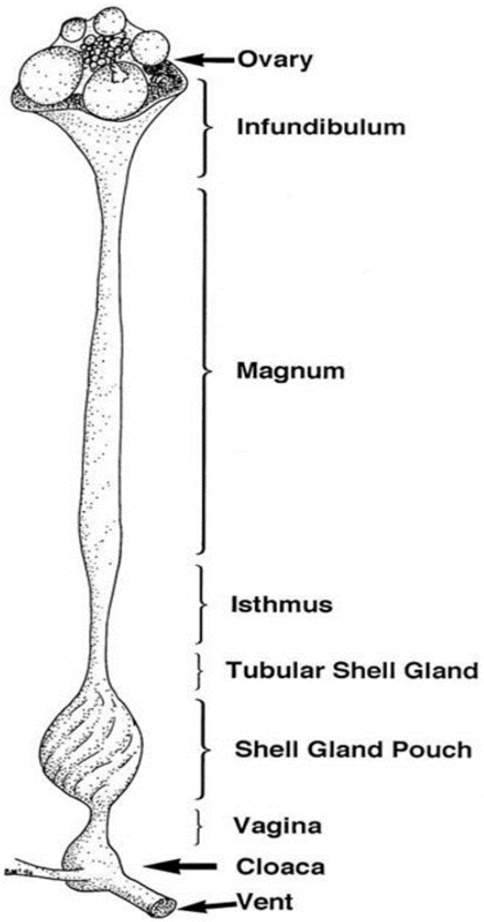
Oviduct

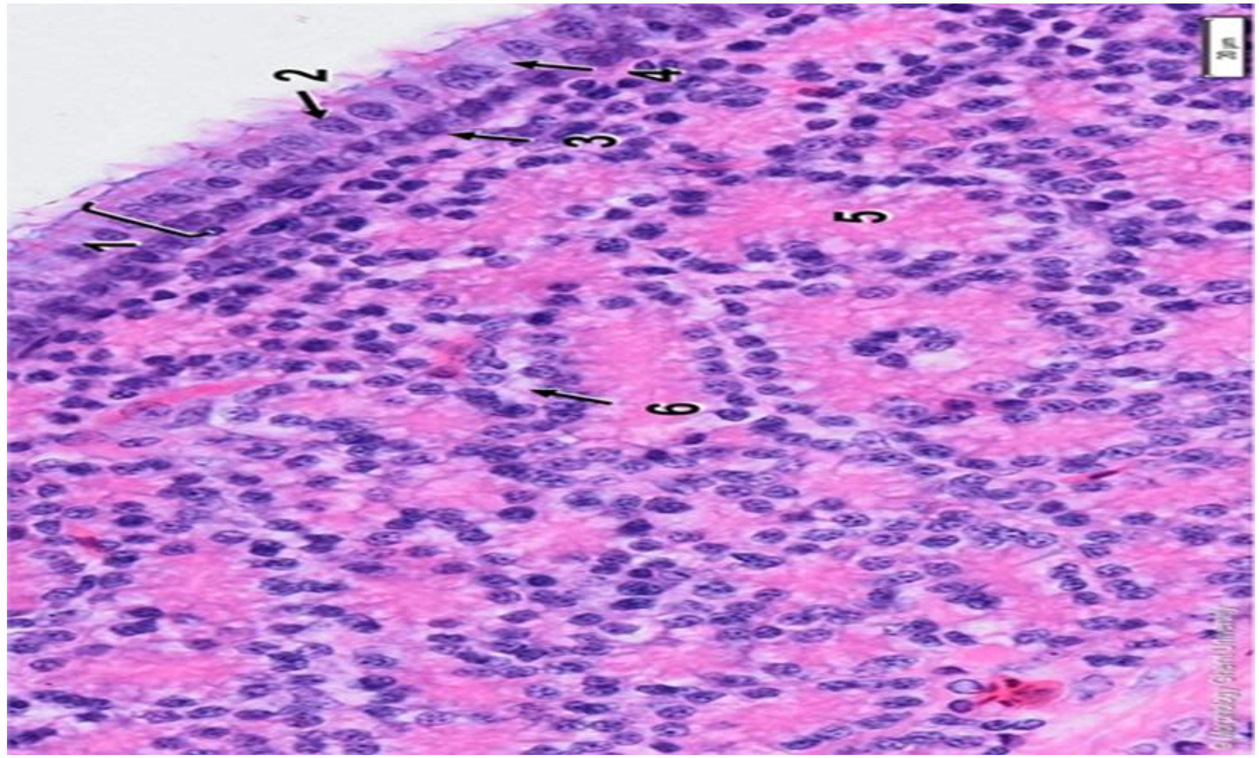




Oviduct secretes *albumin*, shell membranes & shell.

Infundibulum

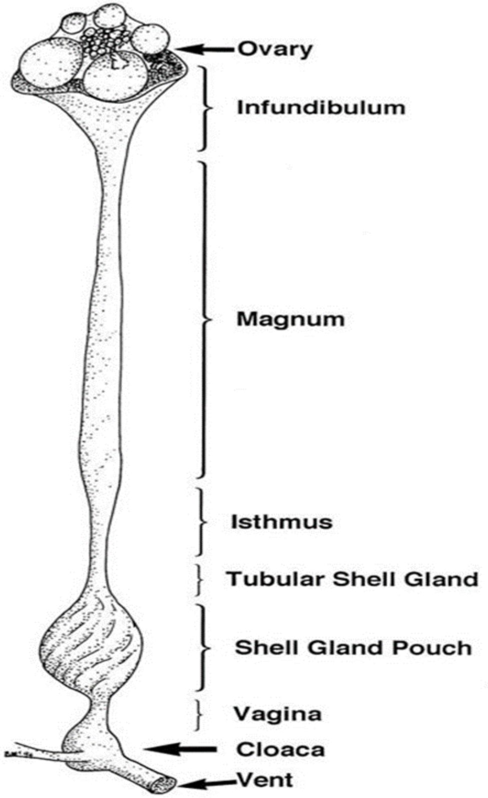


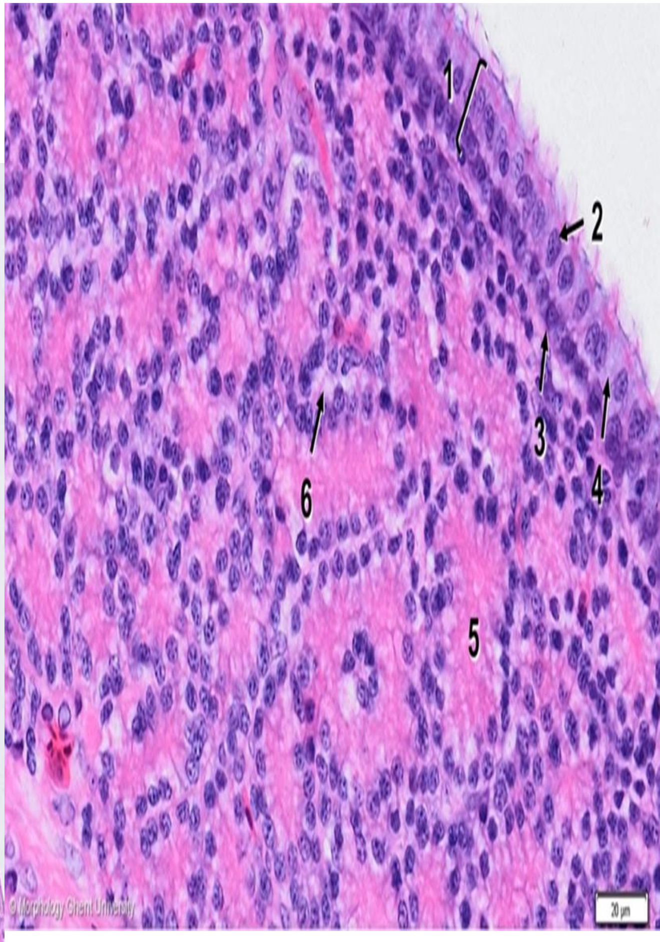


The *epithelial* lining ranges from *s.* col. to *pseud.str.* col.*partially ciliated* & partially secretory (old name by zoologists *goblet* cells).

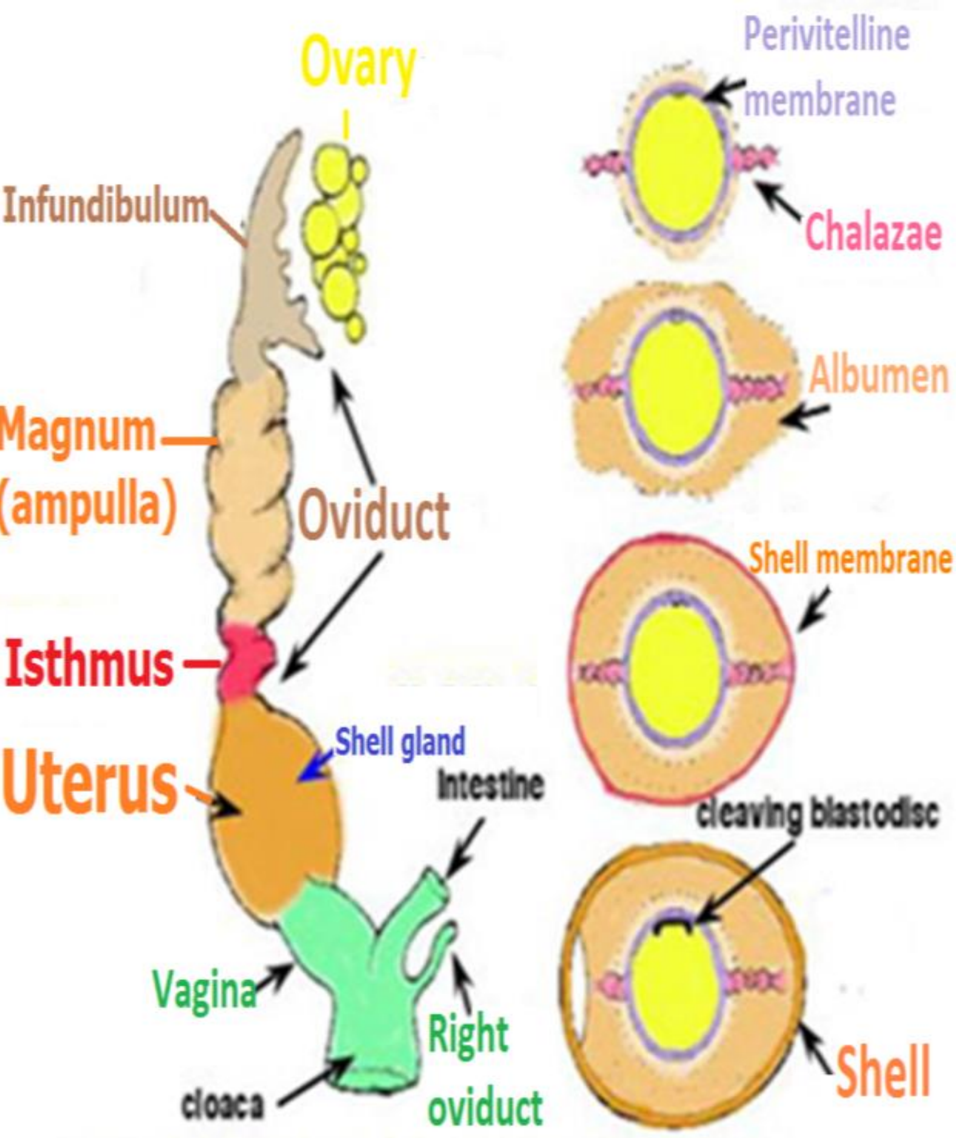
Simple tubular *glands* are present in *propria-submucosa*.

Magnum(ampulla)

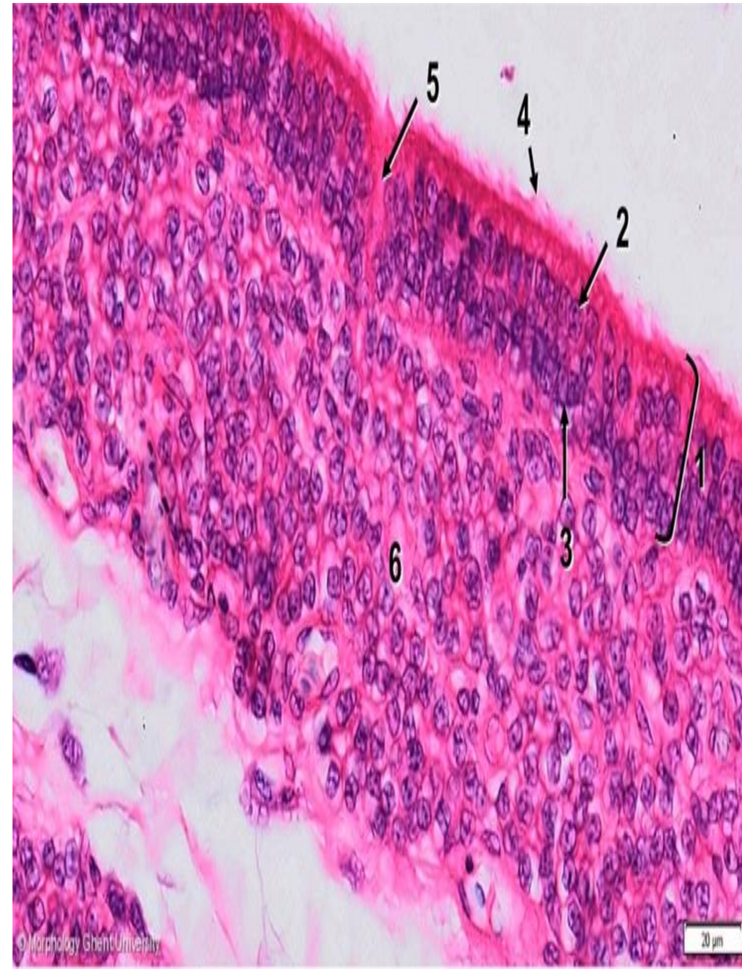




Magnum produces ***albumin*** for ***oocyte*** (or **zygote**). Branched, tubular coiled *glands* in *propria-submucosa* are lined with ***ovalbumin*** &***lysozyme*-secreting cells.**

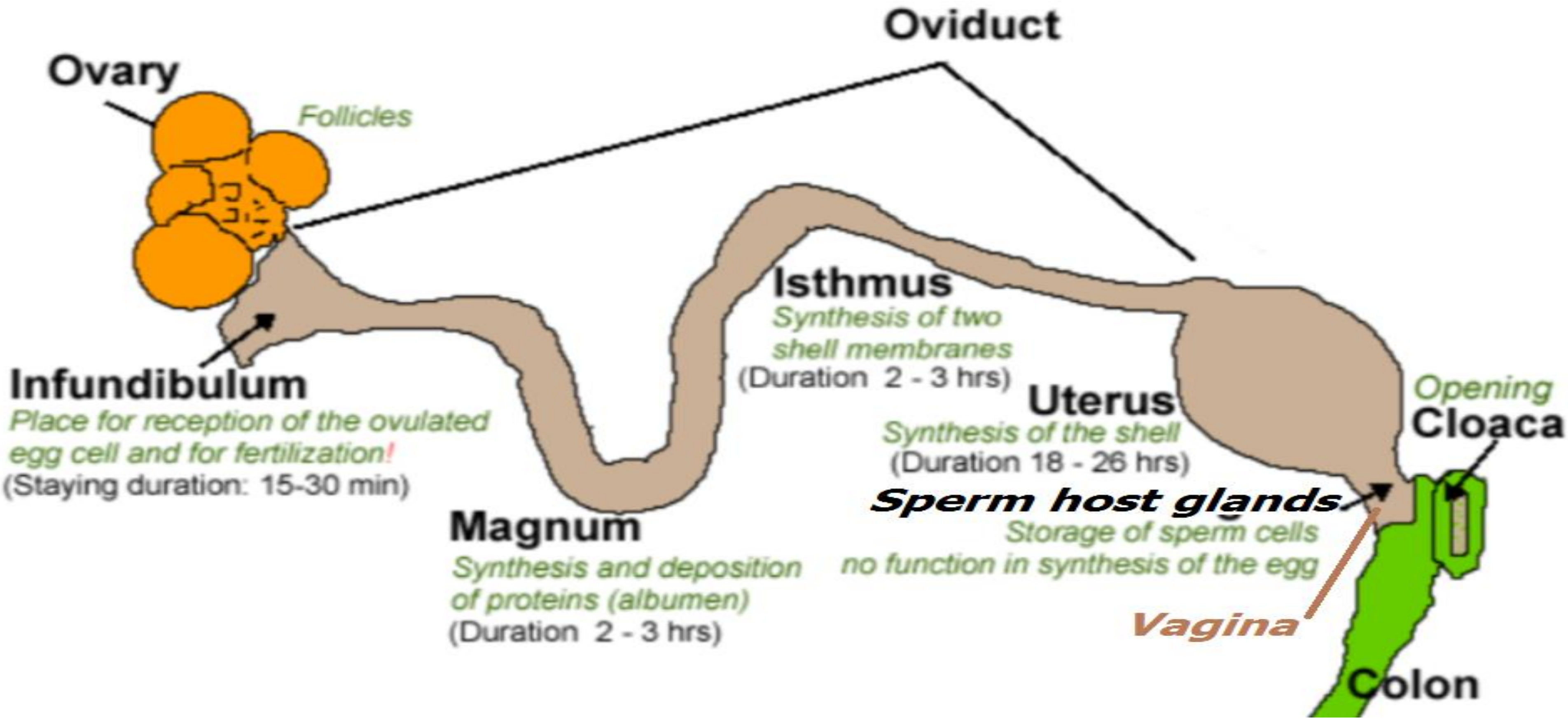


Uterus (shell gland)



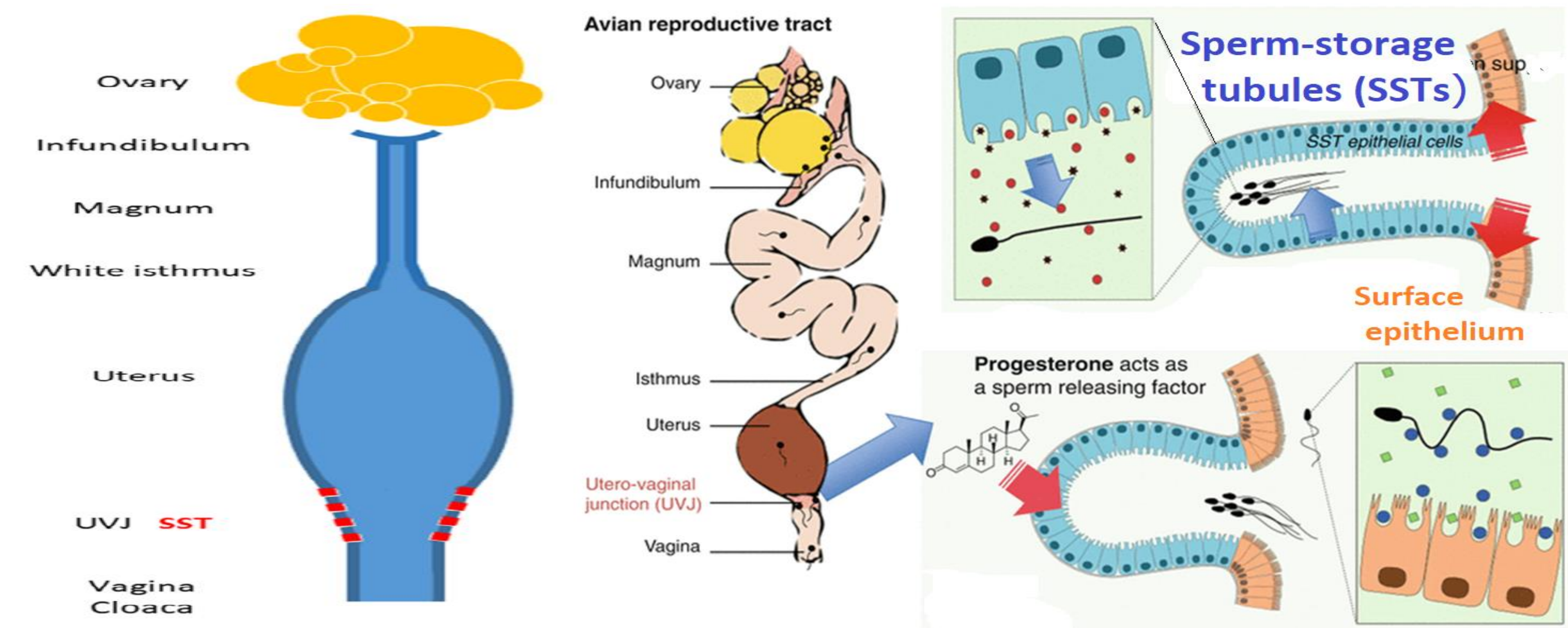
Simple tubular coiled *glands* in propria-submucosa secrete ***inorganic*** part of ***egg*** **shell**.

Sperm host glands =**sperm-storage tubules (SSTs)**

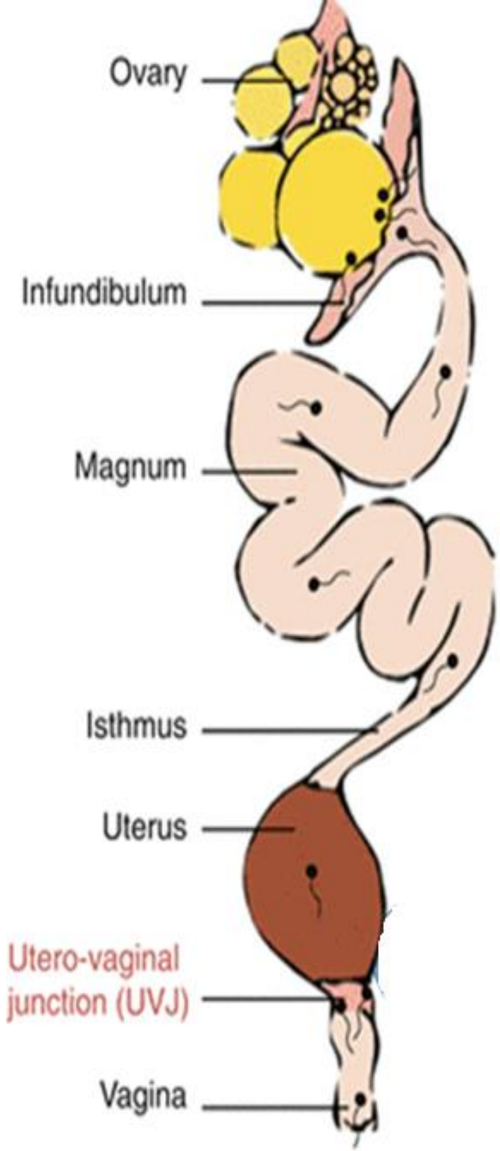


Simple tubular *sperm host glands,* in which spermatozoa may be stored for up to **21 days** are present at **uterovaginal** junction.

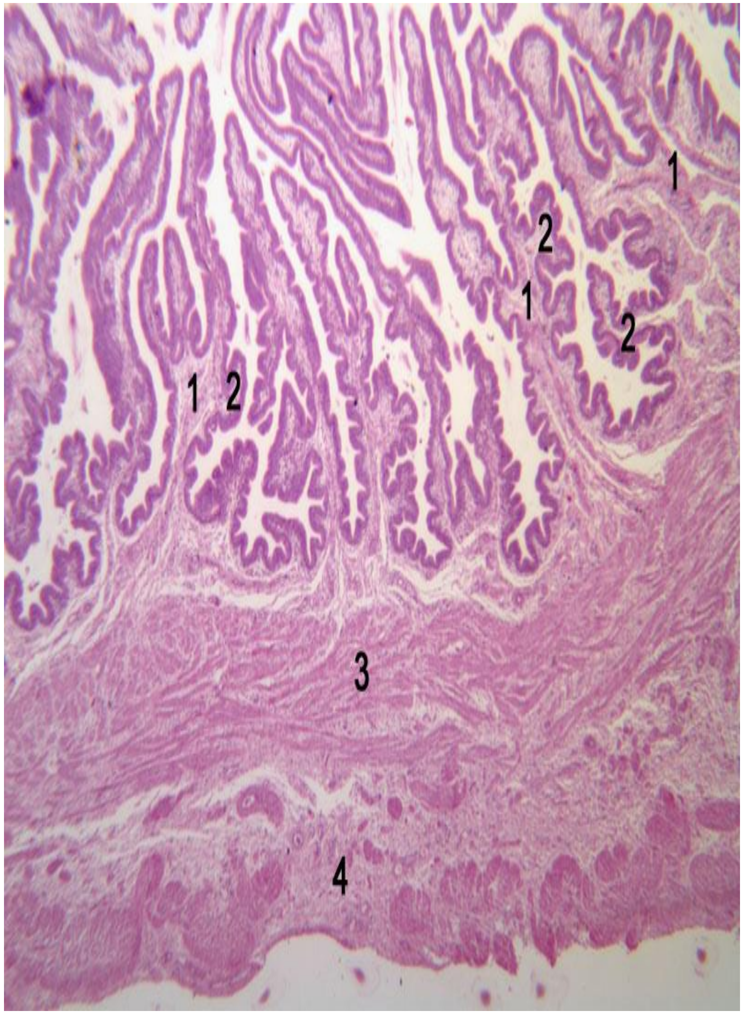
Sperm host glands = Sperm-storage tubules (SSTs)



The arrow indicates transition between **pseudo-stratified columnar ciliated epithelium** of uterovaginal junction & **simple columnar epithelium of SST** that is characterized by supra-nuclear vacuole.



Avian Vagina



1. **Primary fold**
2. **Secondary fold**
3. **Muscularis**
4. **Serosa**

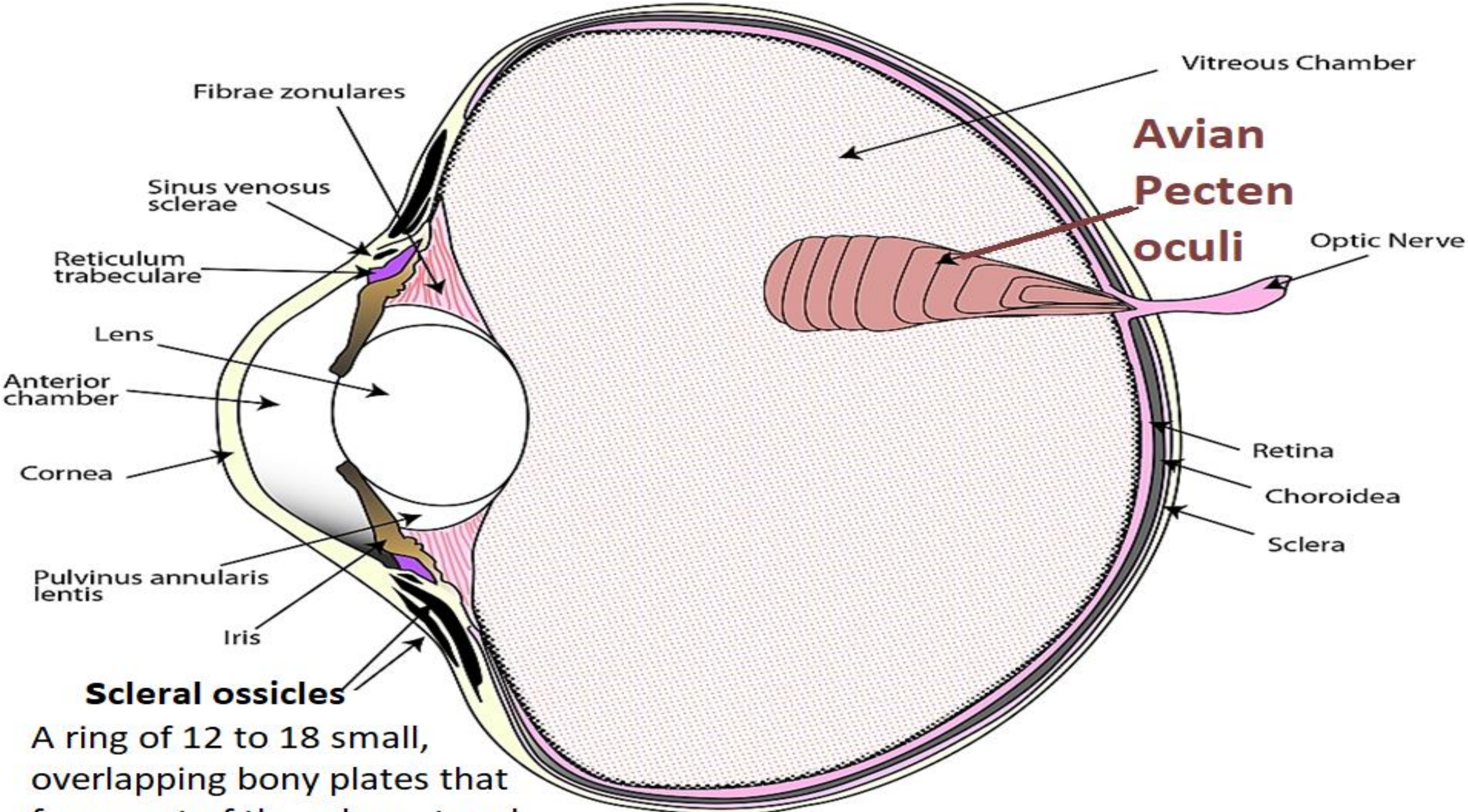
Lining is ***pseud.str.*** **col.** Epith.*partially ciliated*.

جامعة الإسكندرية
ALEXANDRIA
UNIVERSITY

3-Main points of applied avian histology
3rd part

Prof. Dr. Mohamed Ibrahim Abdel Aziz
Digital Veterinary Histology

Avian eye



Scleral ossicles

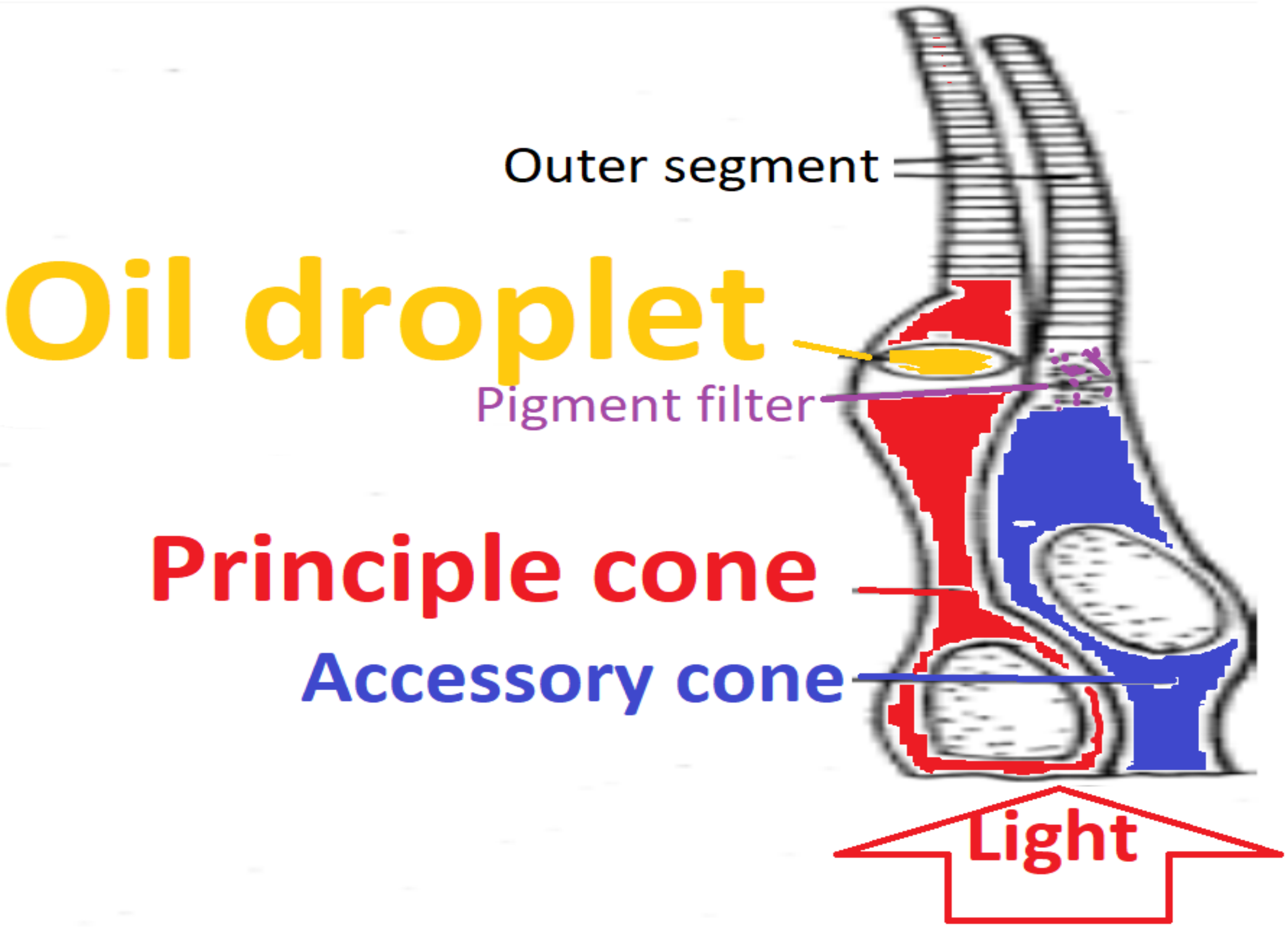
A ring of 12 to 18 small, overlapping bony plates that form part of the sclera, tough outer layer of the eye.

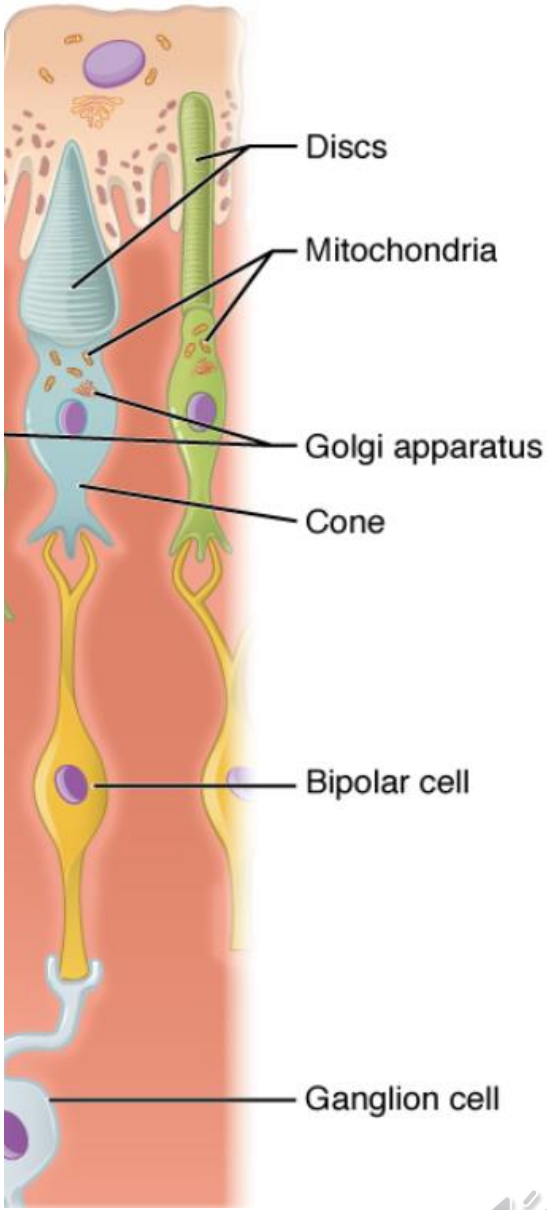
Avian cones

Mammalian cones

Double cones

Lipid droplet in principle





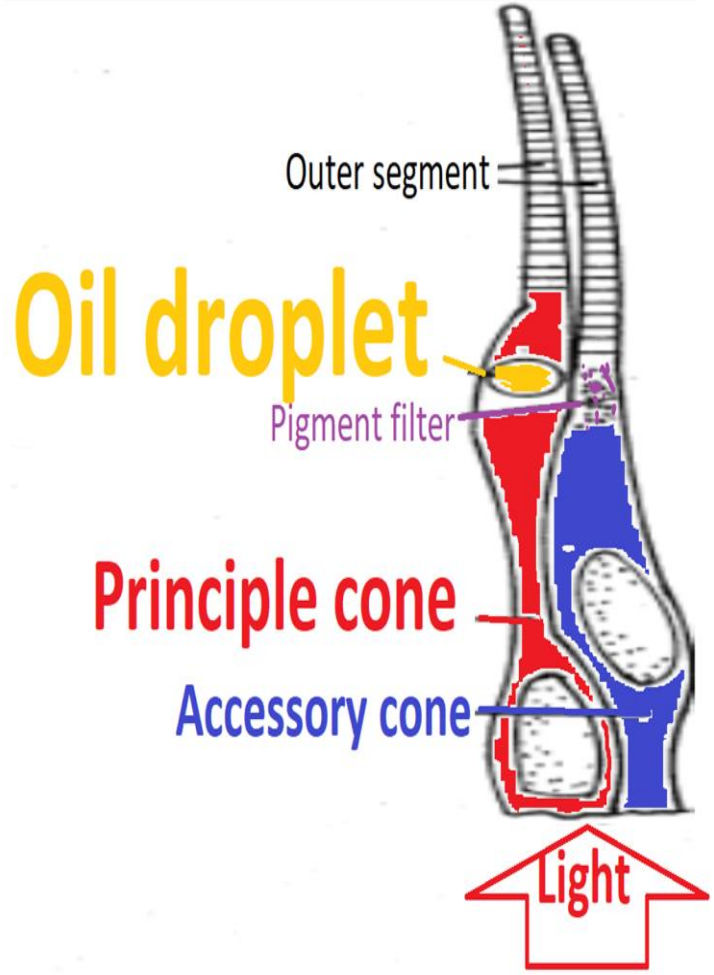
Lipid droplets

Have 2 main functions:

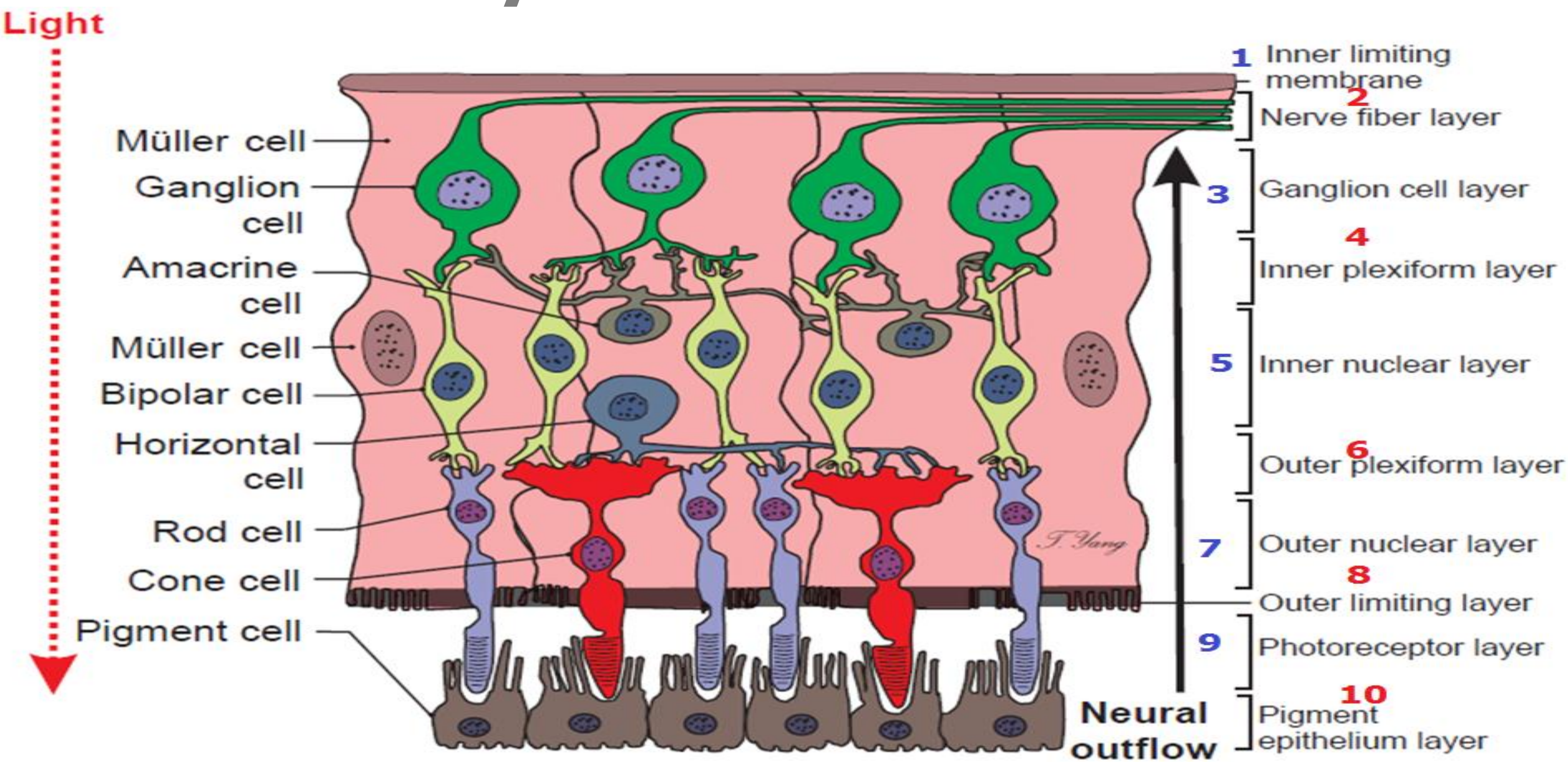
1-They act as **intracellular microlenses** that enhance light delivery to outer segment

2-They **filter spectrum of light** reaching outer segment, thereby **improving color** .

Lipid droplet in principle CONE

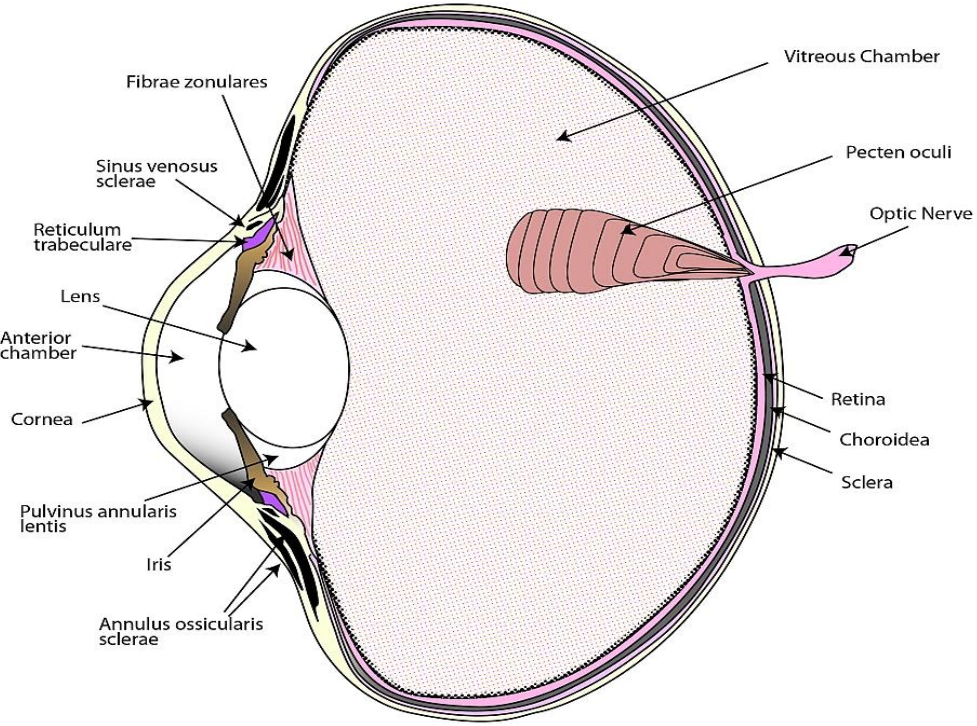


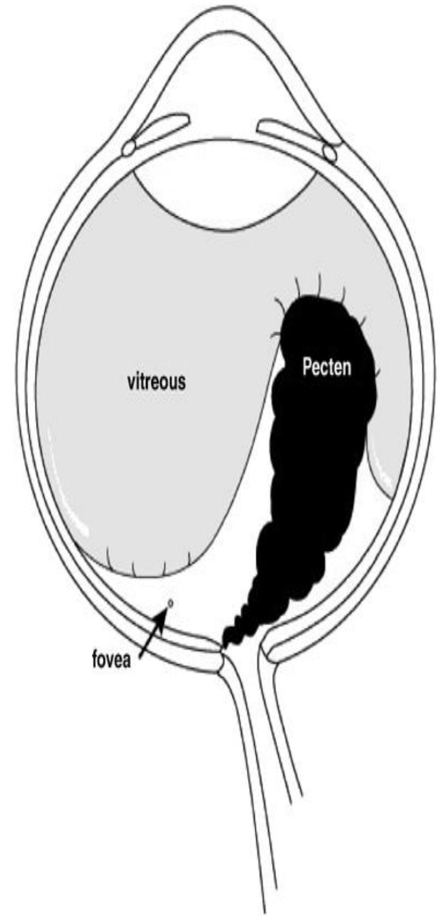
Avian eye **Photosensitive retina**

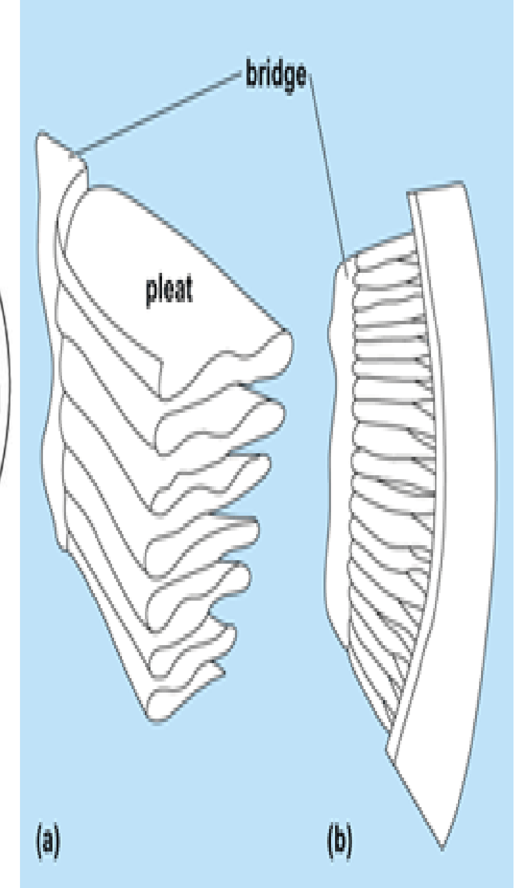


The photosensitive retina of the chicken is composed of *10 layers*, as in *mammals*, but *unlike* that in *mammals* is a *vascular*.

Avian pecten oculi



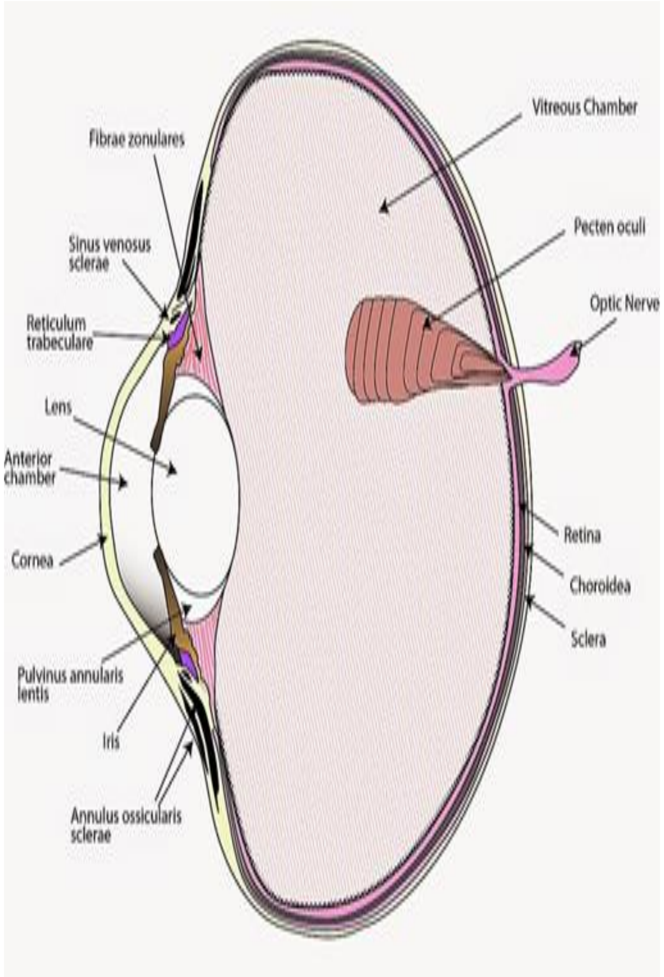


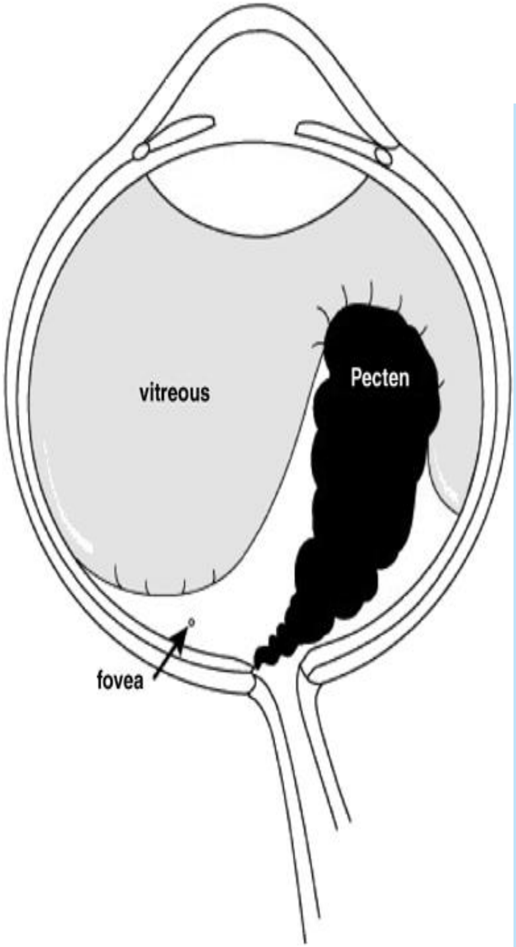


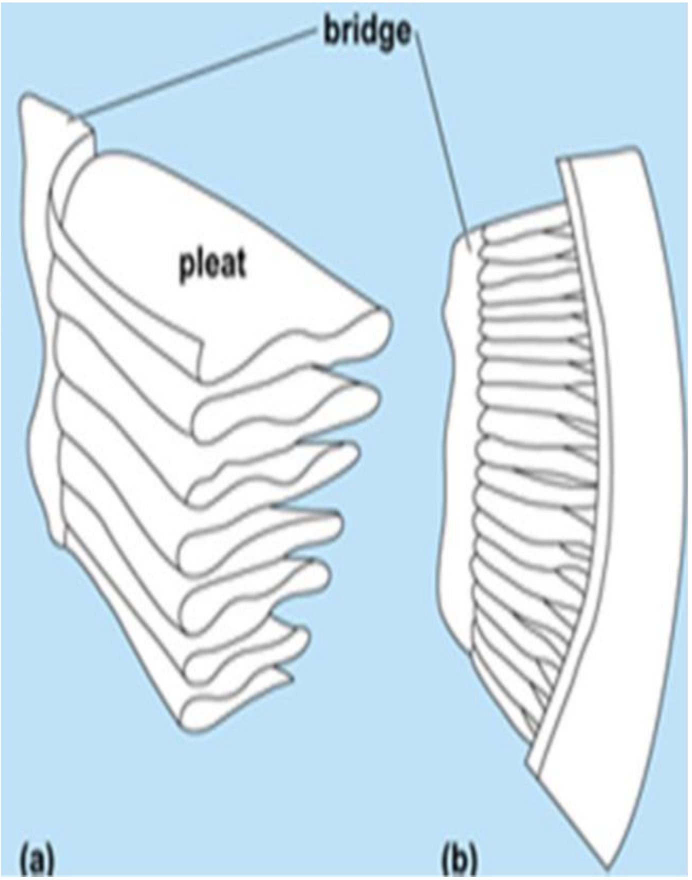
Found in both birds located with base positioned over **"blind spot"**

Pecten base is secured to *optic disc*

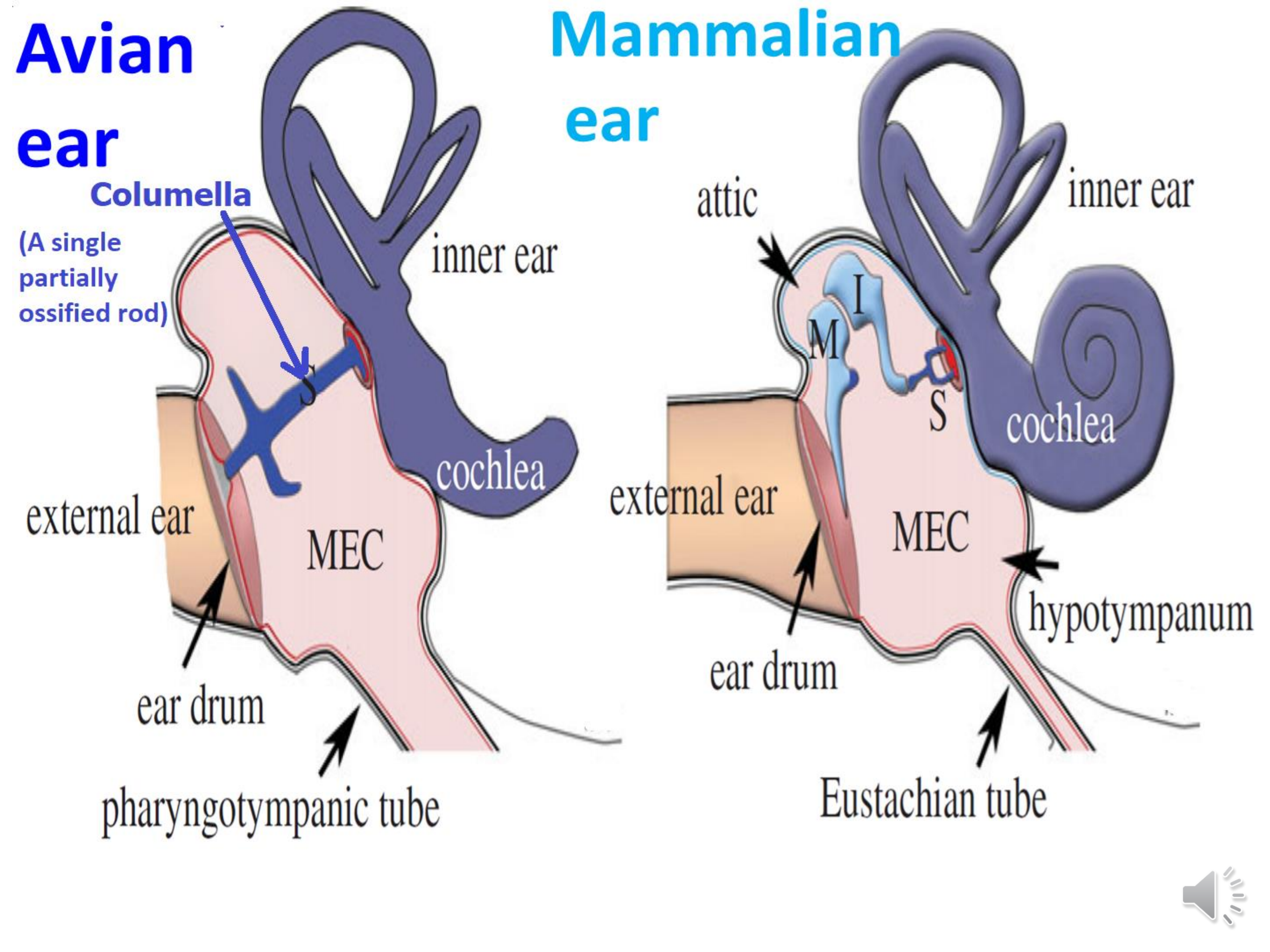
The *pecten*



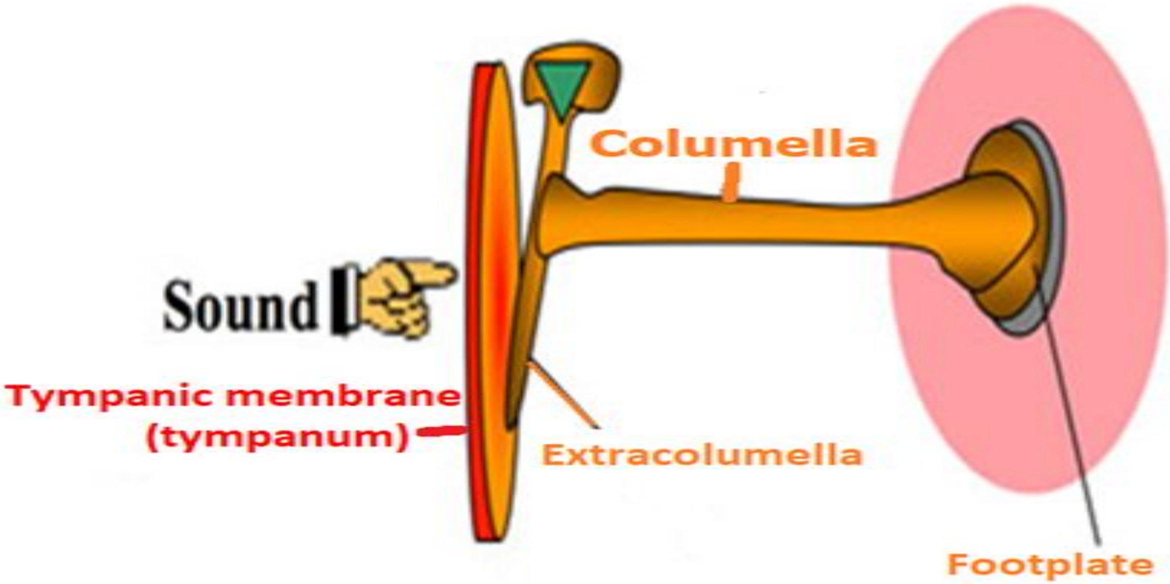


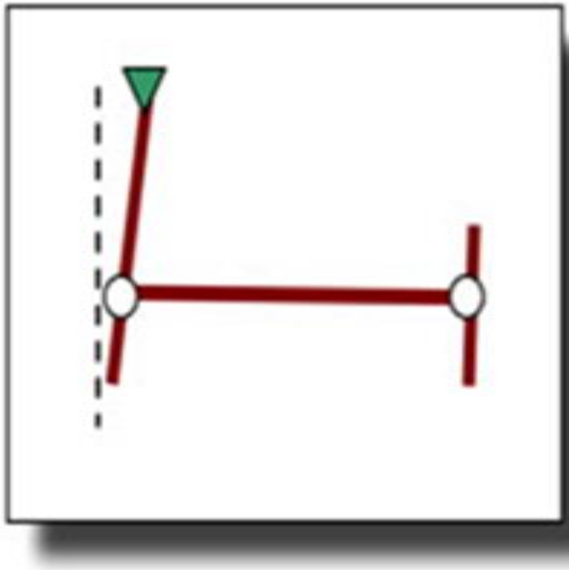


Pecten is thin, highly *vascular, pleated* membrane belonging to **choroid** ,that *protrudes* into *cavity* of viterous humor from ventral surface of eye.


Avian ear
Columella
(A single partially ossified rod)
S
inner ear
cochlea
external ear
MEC
ear drum
pharyngotympanic tube
Mammalian ear
attic
inner ear
I
M
S
cochlea
external ear
MEC
hypotympanum
ear drum
Eustachian tube

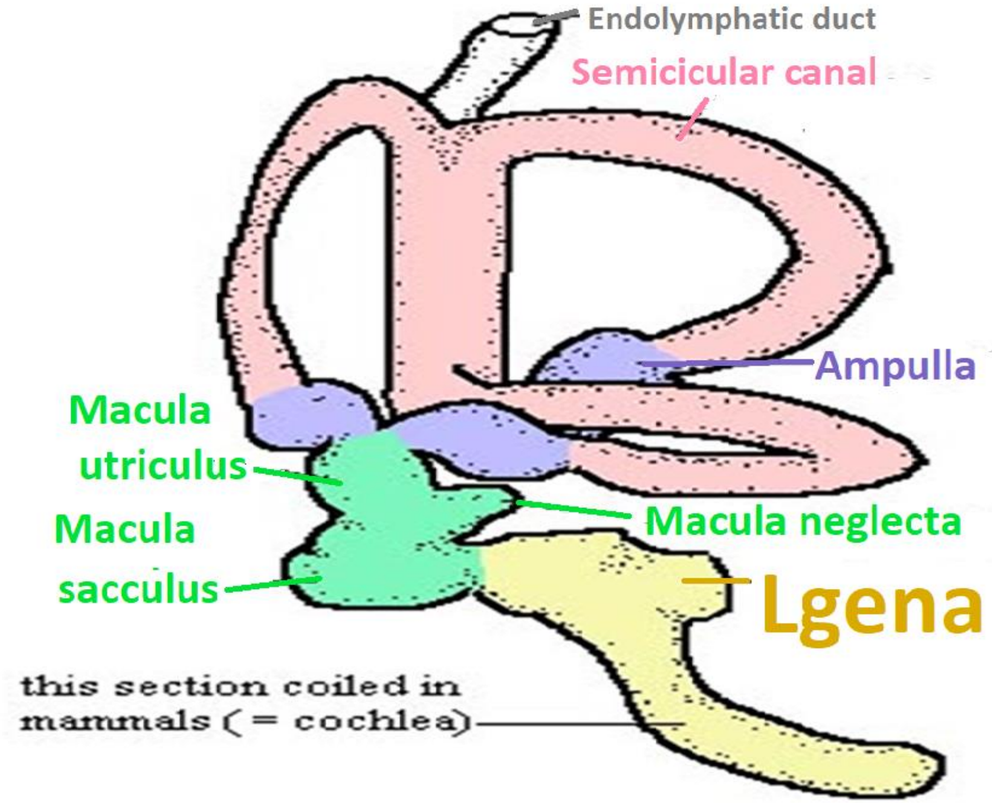
Avian Ear

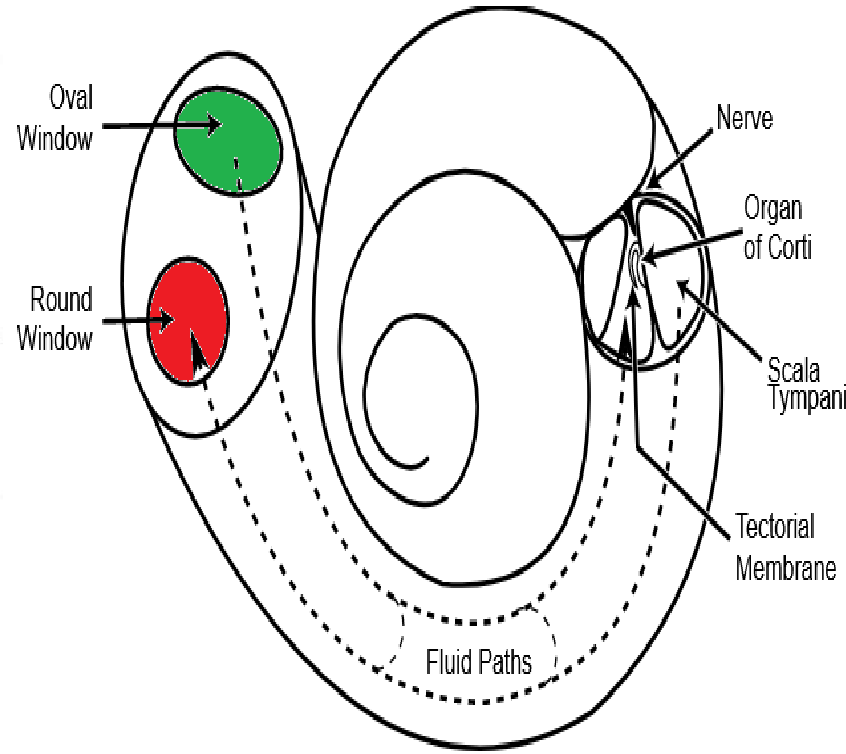




Columella is a single partially ossified rod (malleus, incus &stapes of mammals)
Extends from **tympanic membrane** to **oval window**.

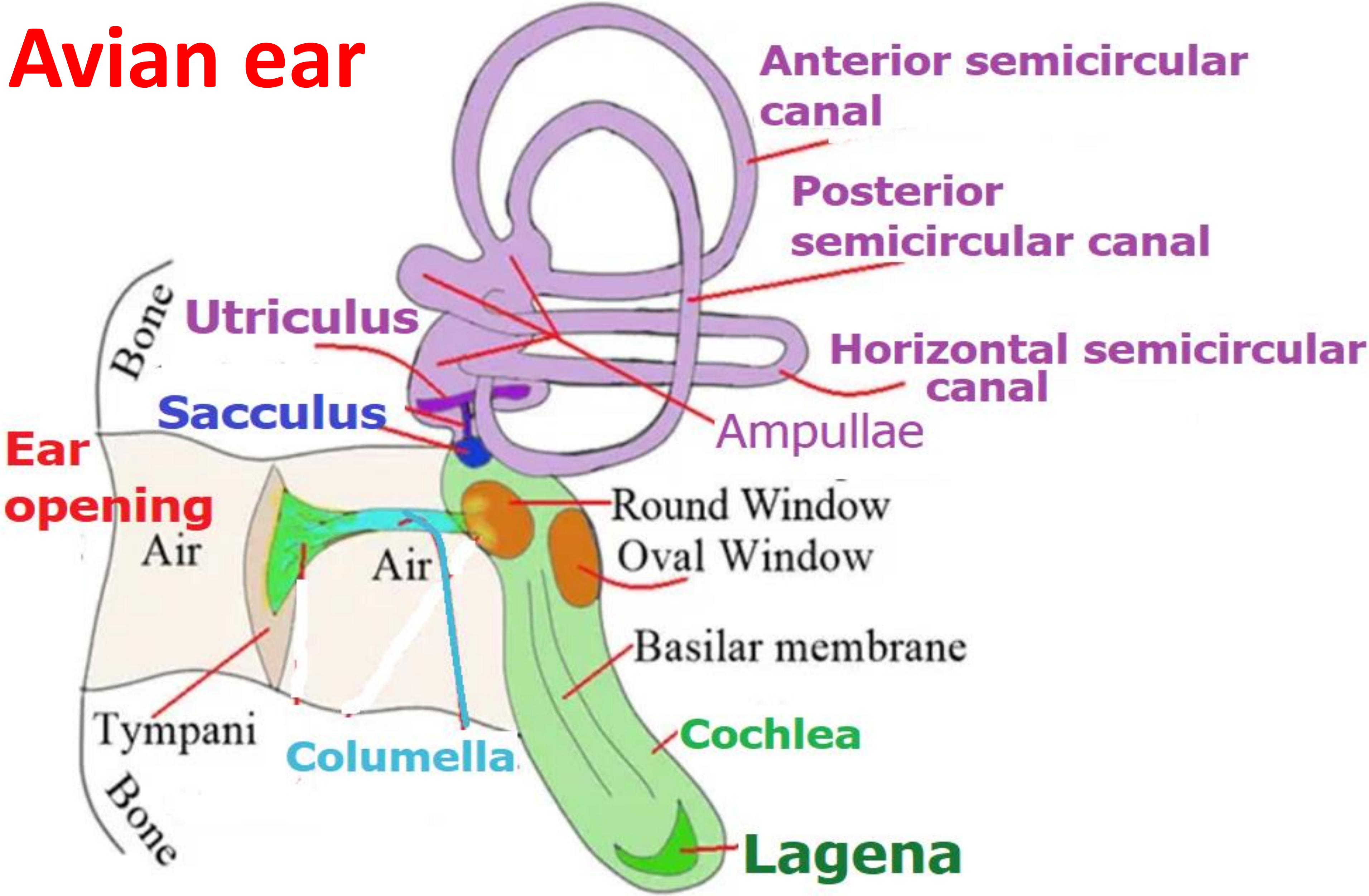
Avian vs mammalian **cochlea**





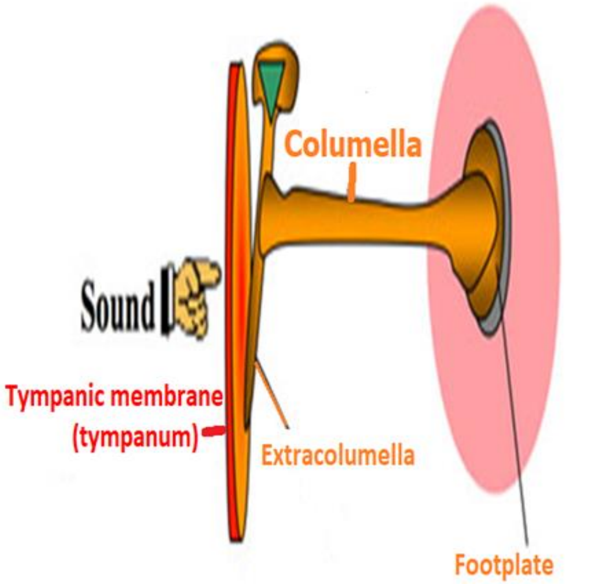
Avian cochlear duct is a short, narrow, slightly curved tube.

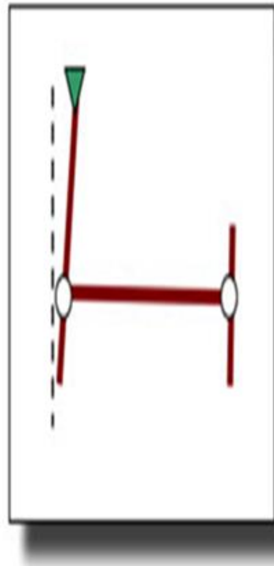
Avian ear

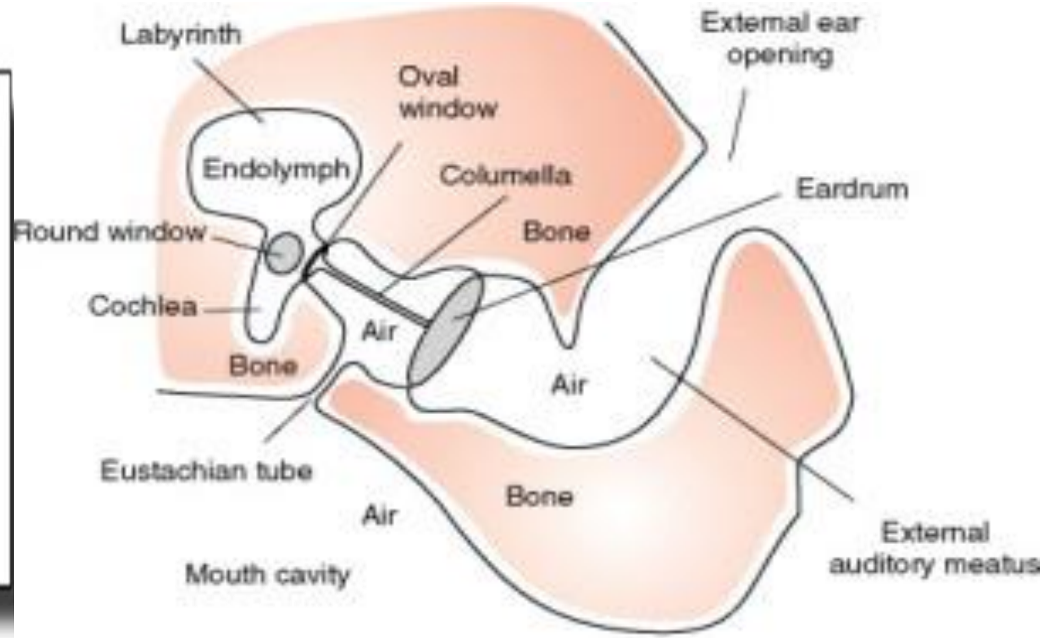


Anterior semicircular canal
Posterior semicircular canal
Utriculus
Horizontal semicircular canal
Sacculus
Ampullae
Ear opening
Bone
Air
Air
Round Window
Oval Window
Basilar membrane
Tympani
Columella
Cochlea
Bone
Lagena

The **Avian Ear** & Hearing

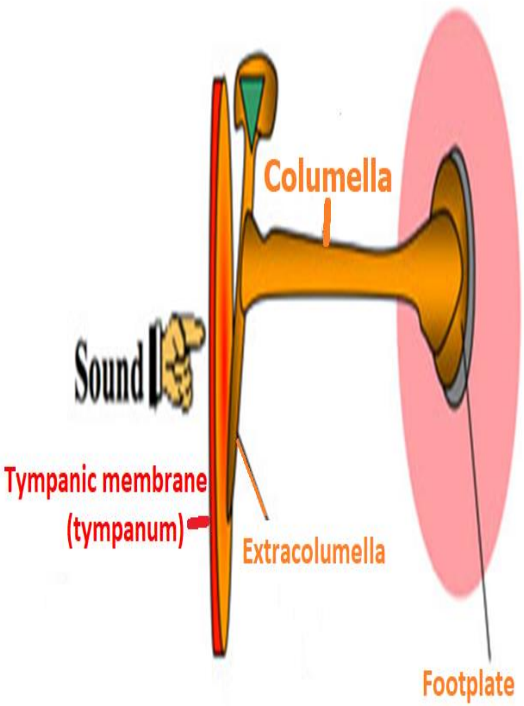


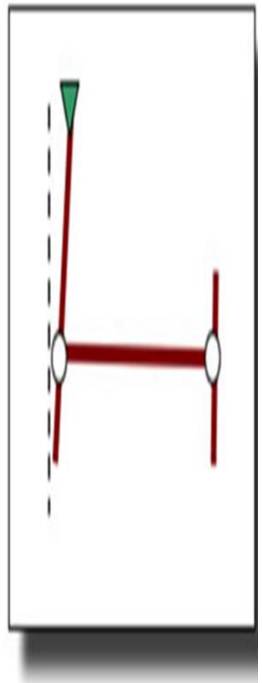


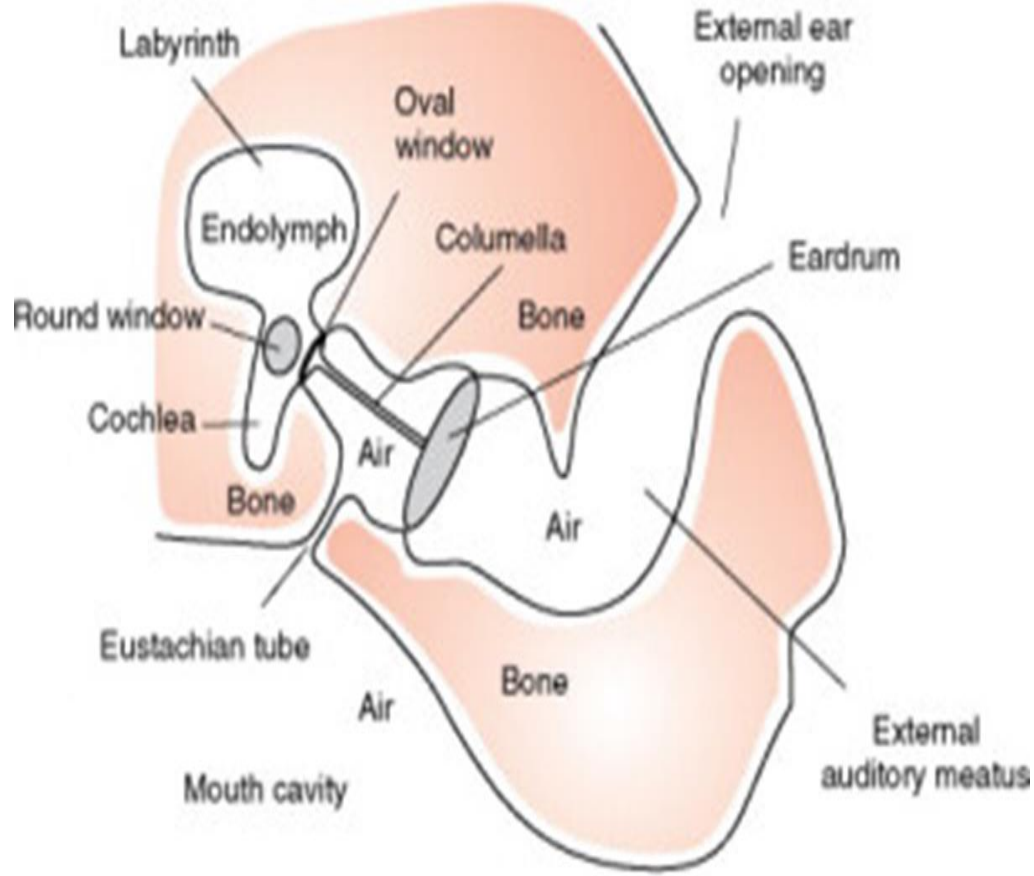


Middle ear is lined by a cuboidal epithelium that also covers **Columella (axis of a spiral shell)**, a **single partially ossified rod** that extends from **tympanic membrane** to **oval window**.

Columella

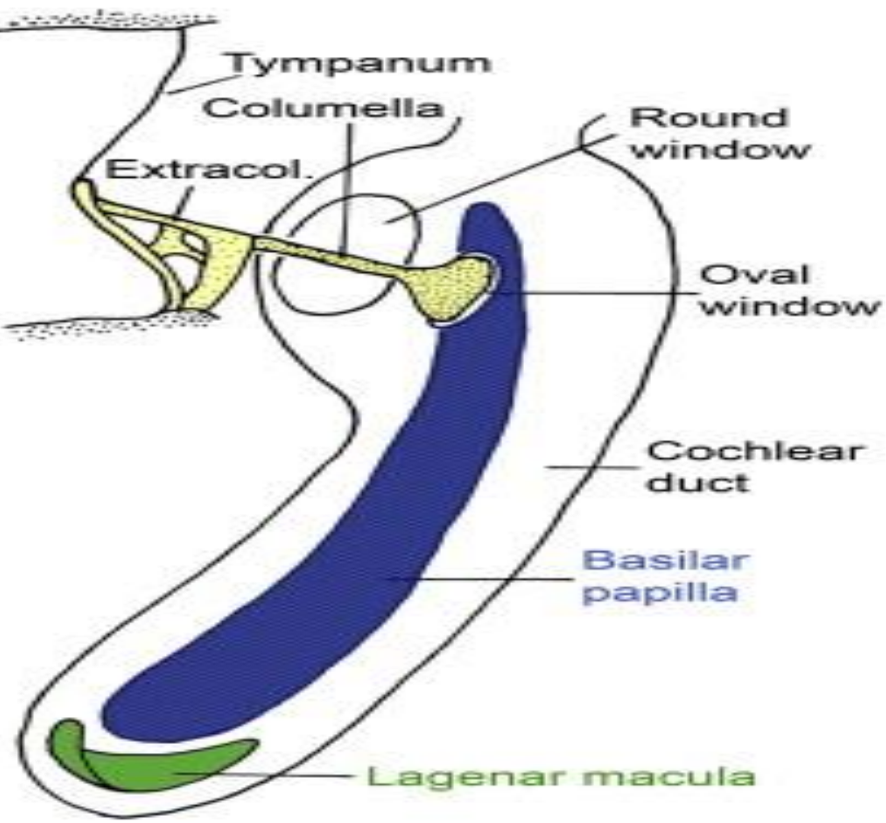




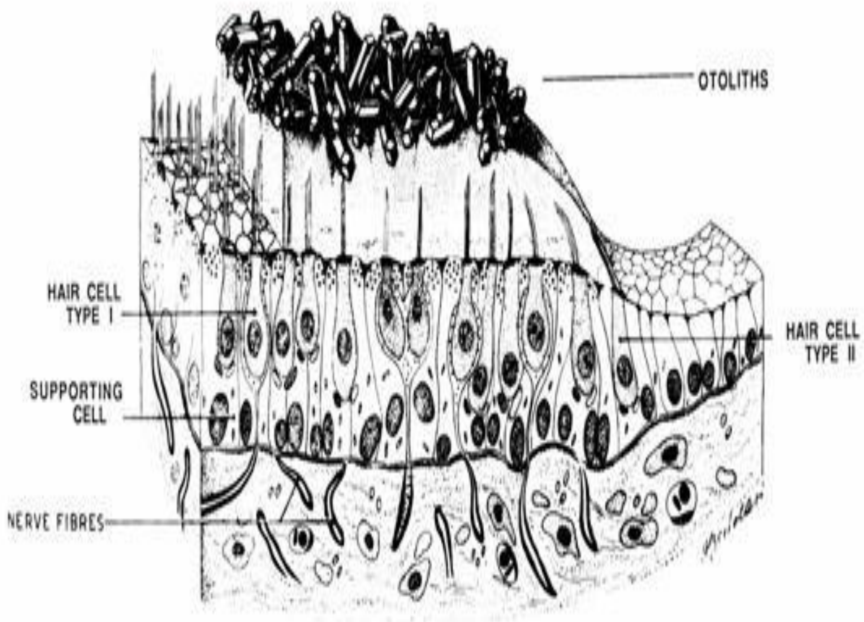


Columella transmits vibrations from tympanic membrane(tympanum) to internal ear, taking place of malleus, incus &stapes of mammals.

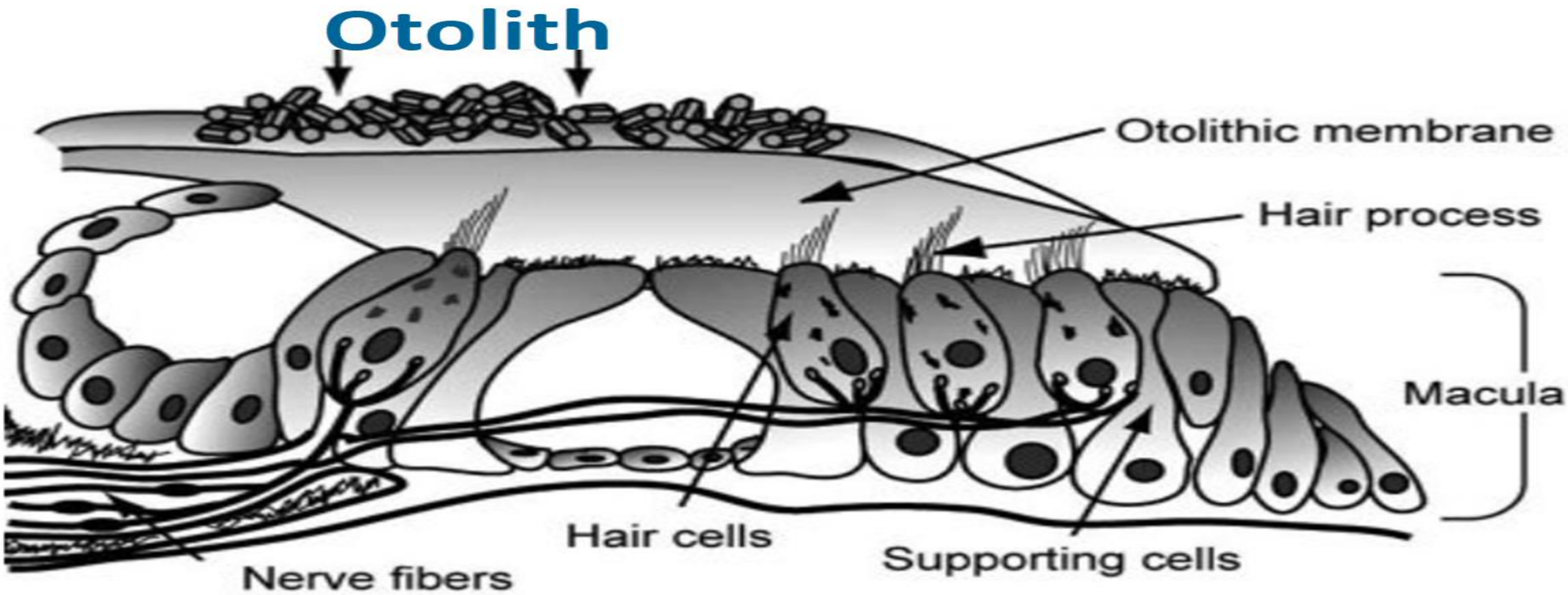
Avian cochlear duct



Mammalian otolith organs (saccule &utricle)



Possesses a terminal expansion, ***lagena (3rd otolith end organ in vertebrates)***, a structure **unique *to birds***.



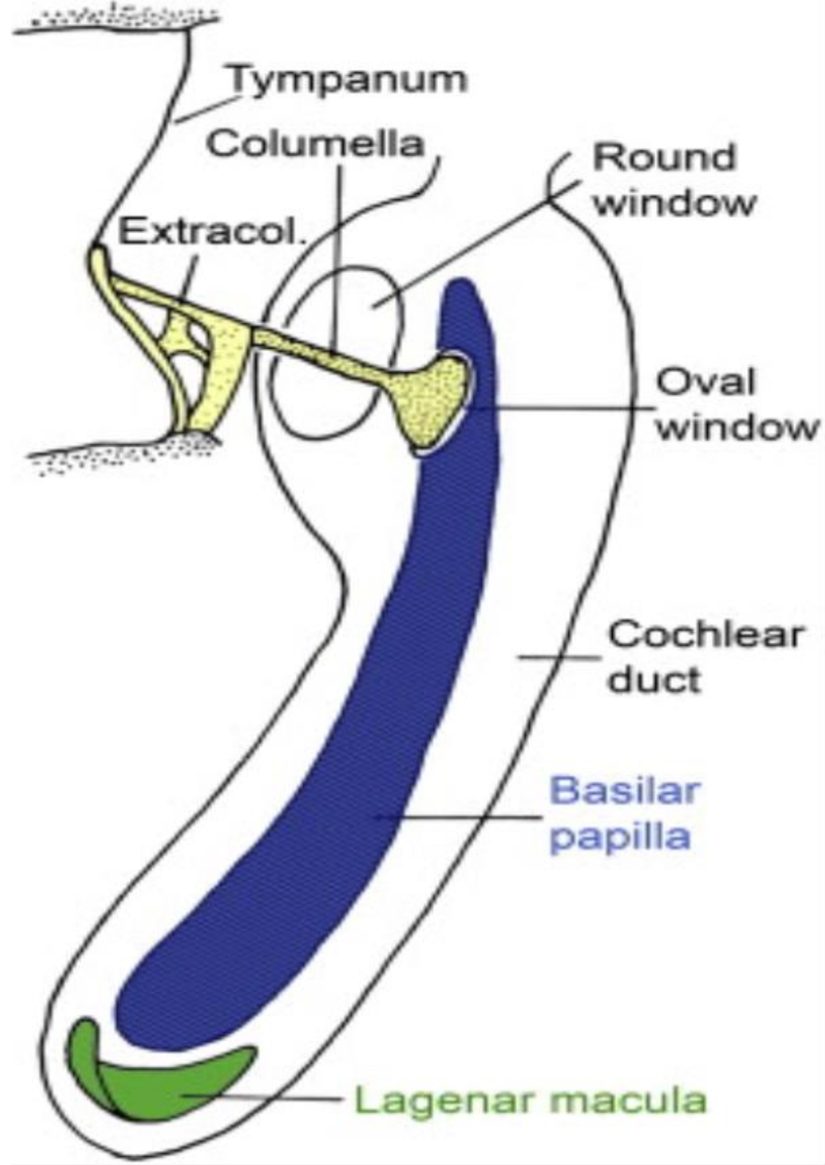
Otolith“ear stones” =**otoconia** are **calcium carbonate crystals**.

Otoconia cause **membrane to be heavier** than structures &fluid around it.

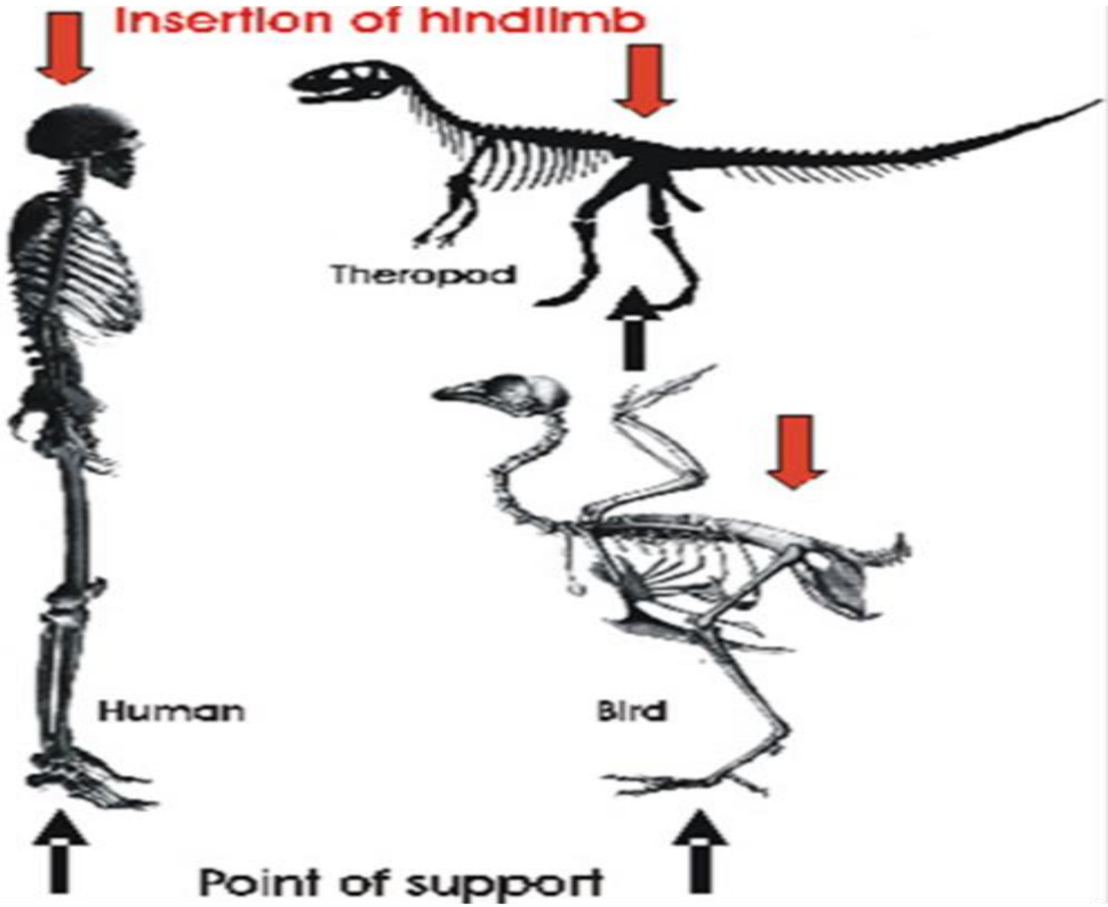
During head movement, gravity &weight of otoconia cause membrane to move, which bends hair cells, which sends signals about motion &head position to brain via vestibular nerve.

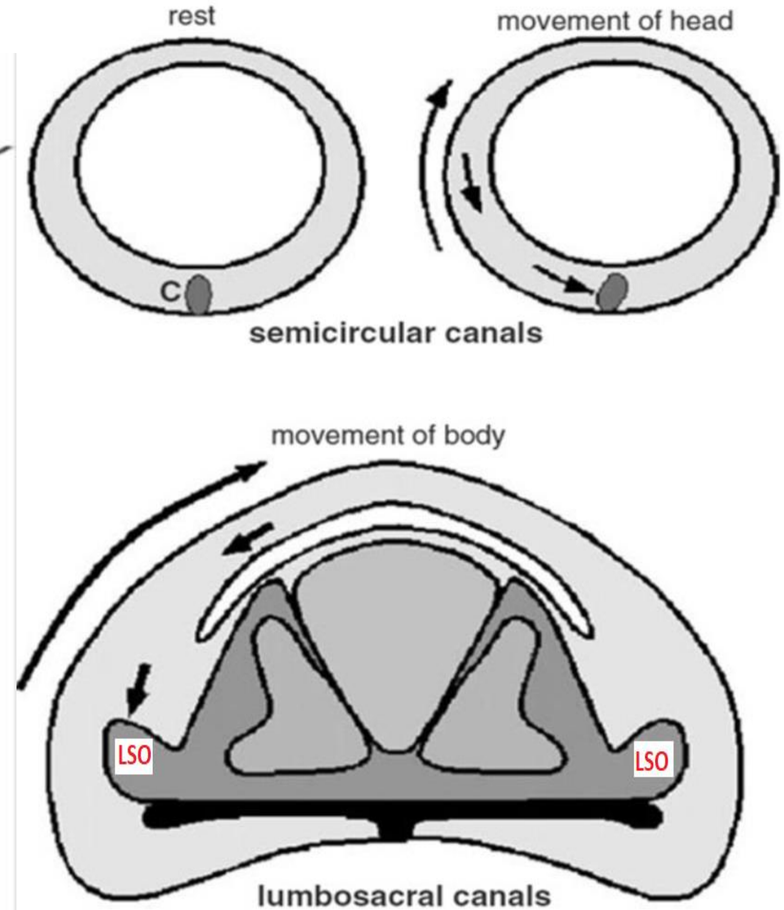
Lagena is related to their navigation abilities (orientation within magnetic field of Earth due to magnetic properties of **lagenar otoconia)**

-Detection of movements along vertical axis



Specializations in lumbosacral vertebral canal &spinal cord of birds





Lumbosacral organ (LSO) is a sense organ which is involved in control of walking.

Organization of lumbosacral spinal cord as seen from dorsal.

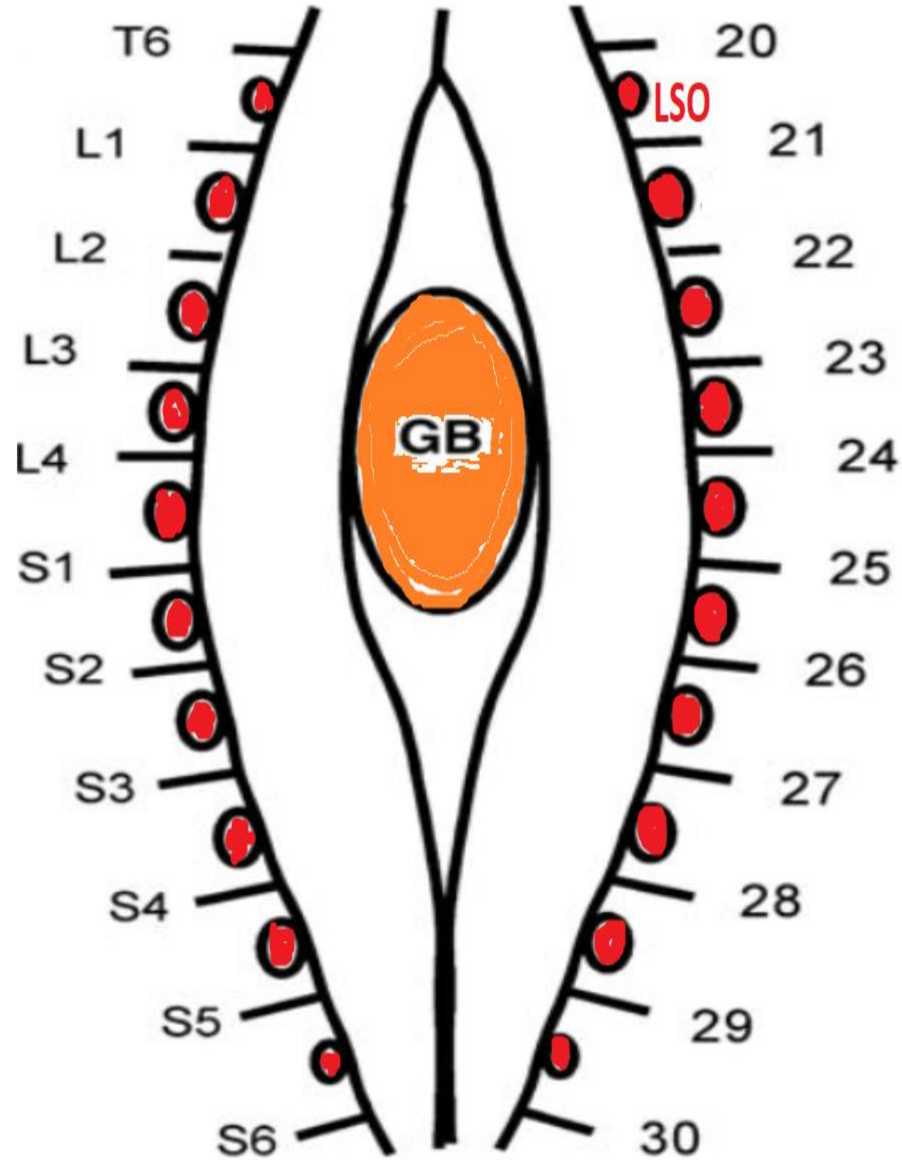
Rostrocaudal numbering of spinal segments on the right, T= thoracic, L= lumbar &

S= sacral segments on left.

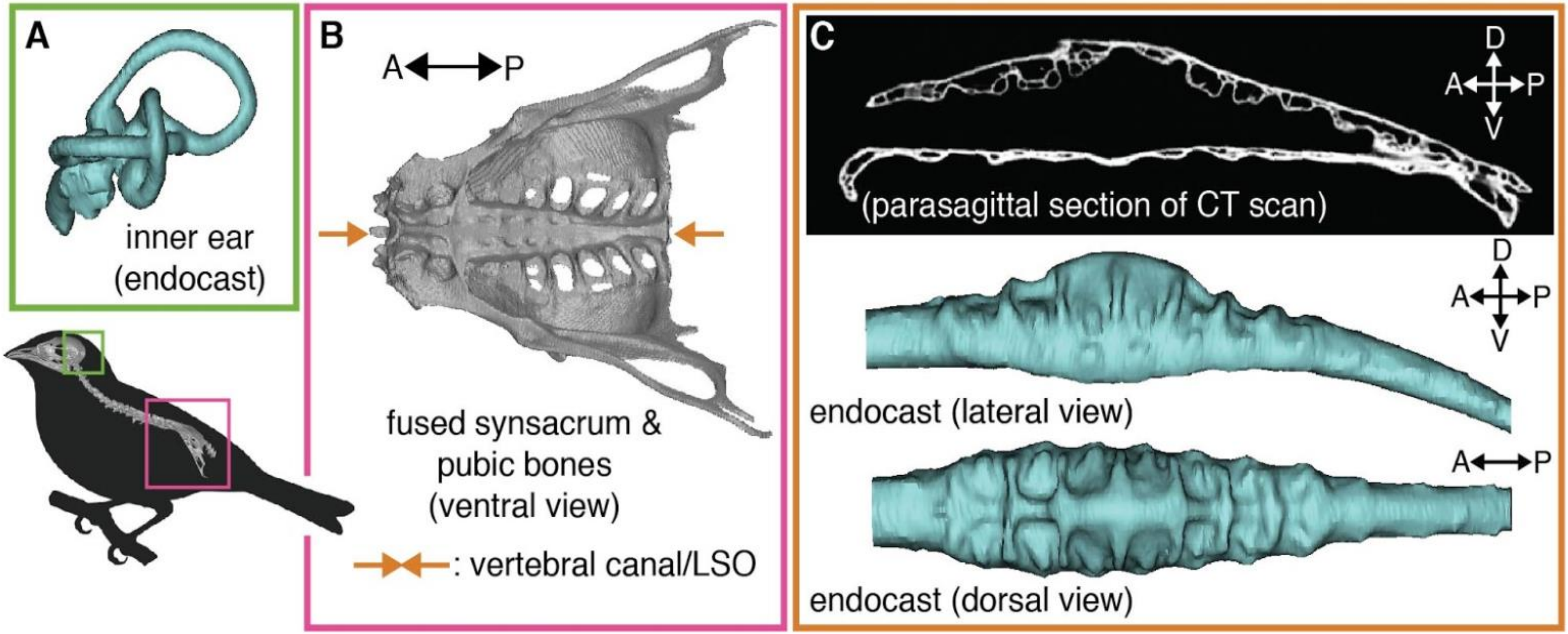
Lines indicate spinal nerves and ovals indicate

Lumbosacral organ (LSO) which are located in between the nerves;

GB =glycogen body.



Birds have2 sets of balance organs

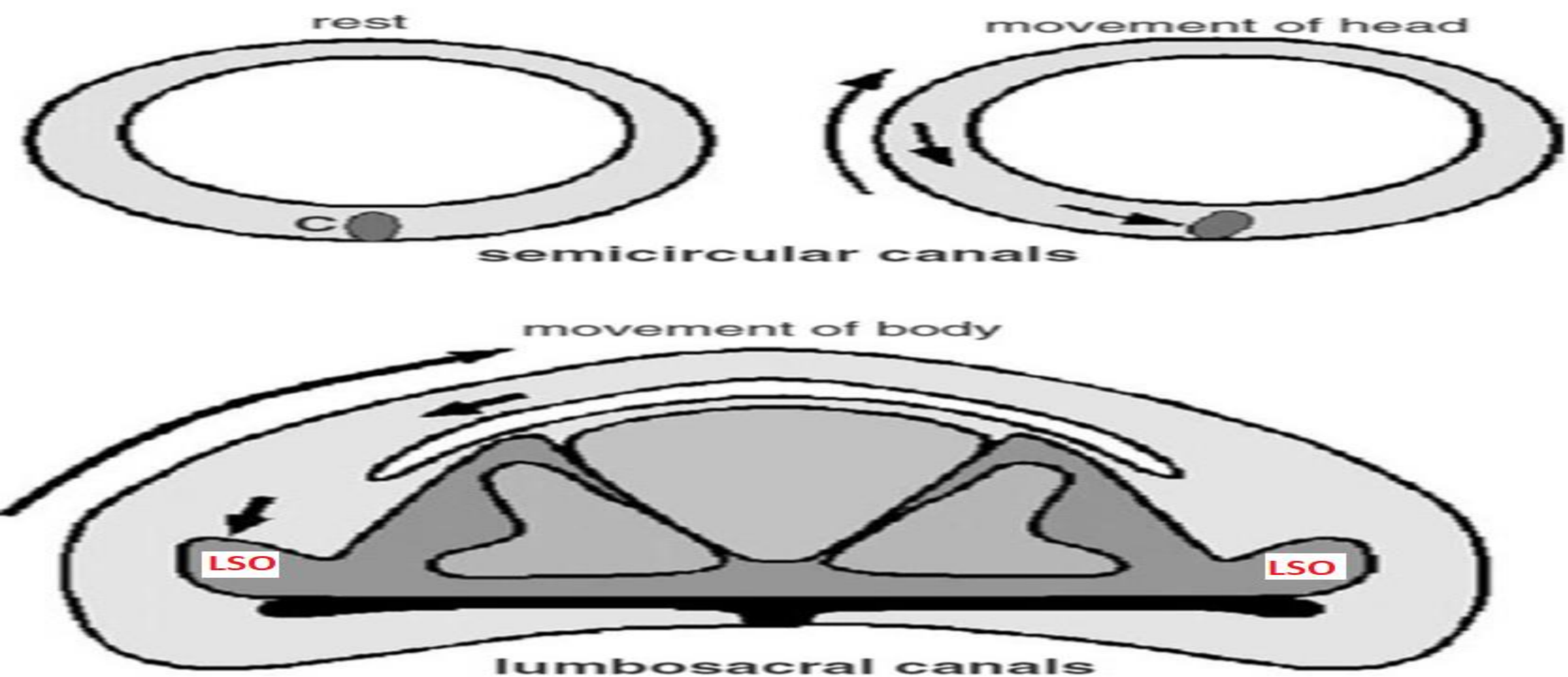


1-One in the inner ear (A) (common to terrestrial vertebrates)

2- Second within the synsacrum (B), lumbosacral organ (LSO).

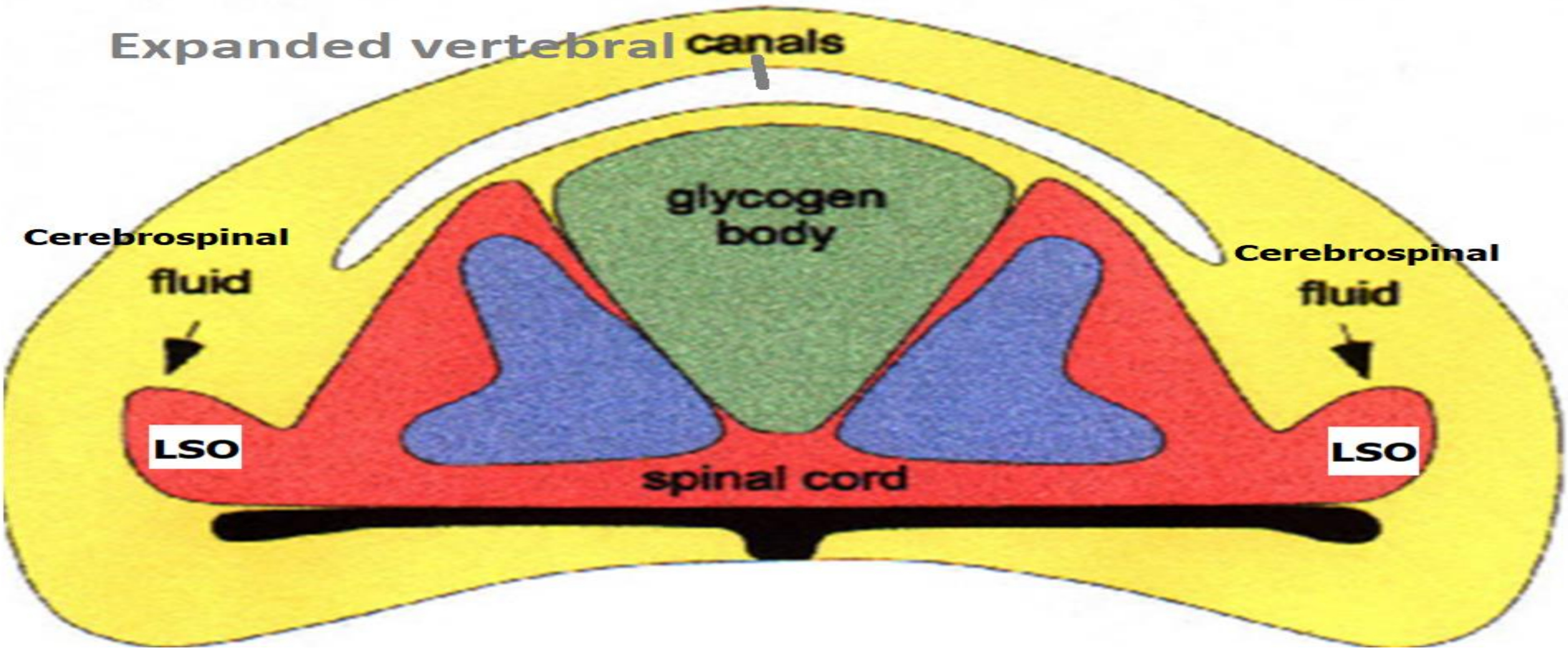
LSO is housed within an **expanded vertebral canal**, which exhibits a set of canal-like recesses (lumbosacral transverse canals, or LSTCs; C).

Lumbosacral organ (LSO)(=AccessoryLobes of Lachi=nuclei marginales)



Lumbosacral organ (LSO) is extralabyrinthine organ of **equilibrium**

Lumbosacral organ (LSO)



In lumbosacral region of birds,

segmentally organized ventrolateral protrusions of spinal cord form **Lumbosacral organ (LSO)**

Lumbosacral organ (LSO)

-Consist of

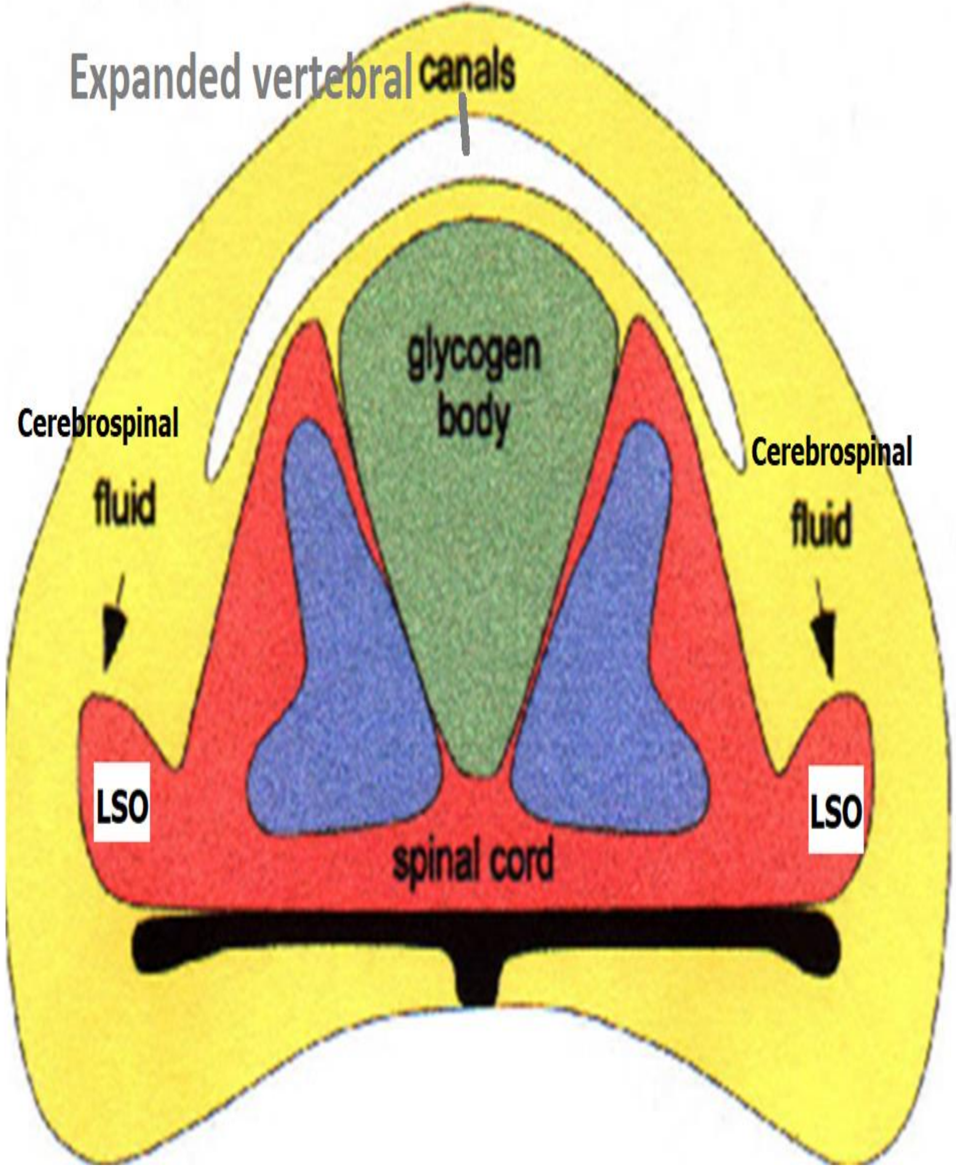
1- Sensory neurons

2-Glial cells

3-Myelinated axons

4-Unmyelinated axons

5-capillaries.

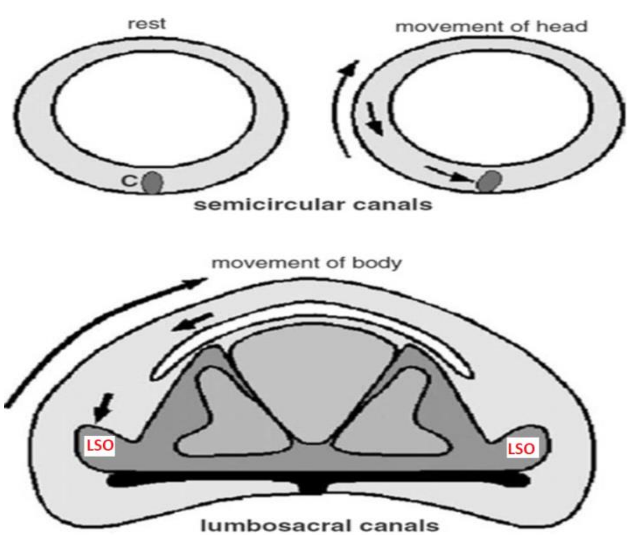


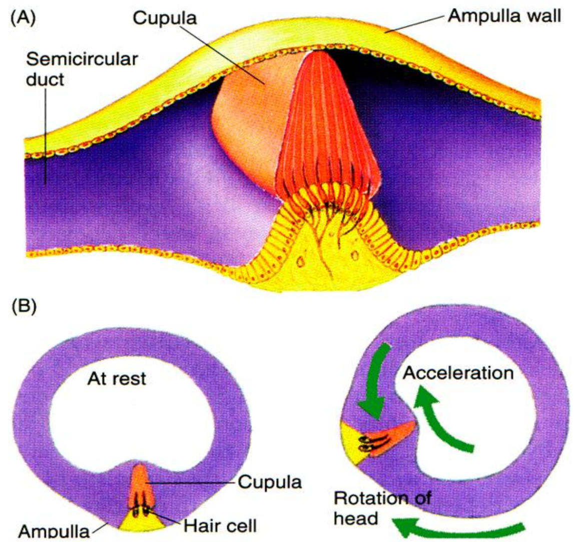
Lumbosacral organ (LSO=2)

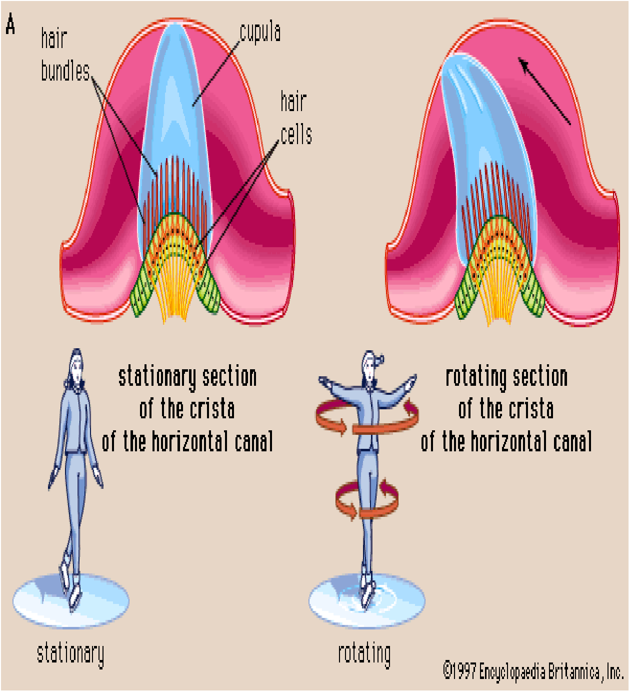
1-spinal cord 2- Lumbosacral organ (LSO)

3-vertebral canal 4-pneumatic bone

Lumbosacral organ (LSO)







Lumbosacral organ (LSO)is an **extralabyrinthine organ** of **equilibrium.**

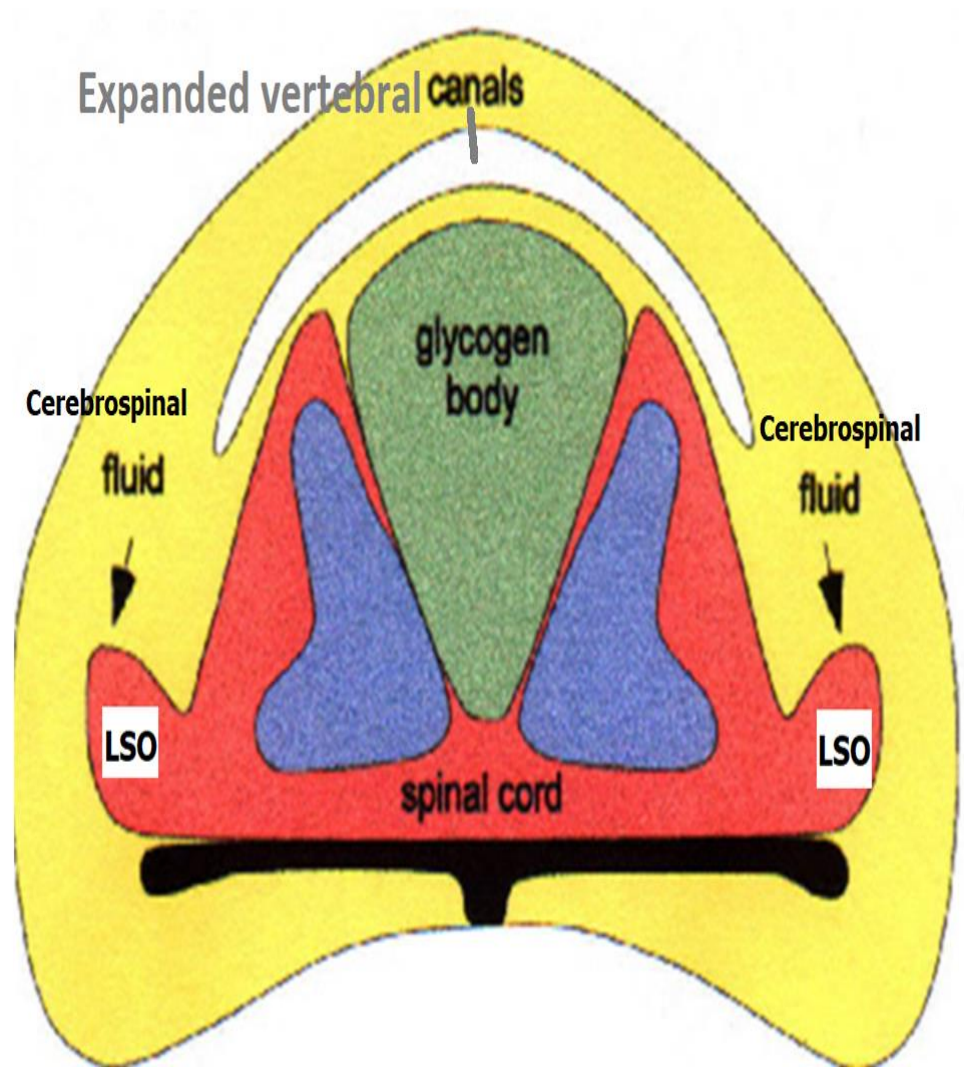
Bony labyrinth is **a series of bony cavities within petrous part of temporal bone**. It consists of 3 parts :

I-cochlea **II-vestibule** **III-3 semi-circular canals.**

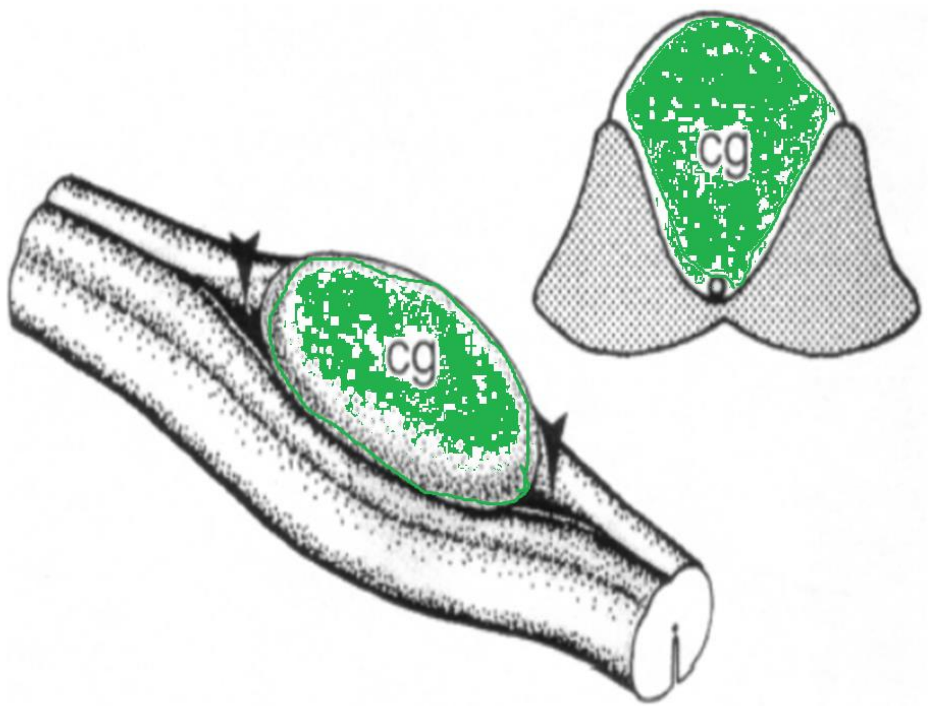
Lumbosracal organ= LSO(AccessoryLobes of Lachi=AL)

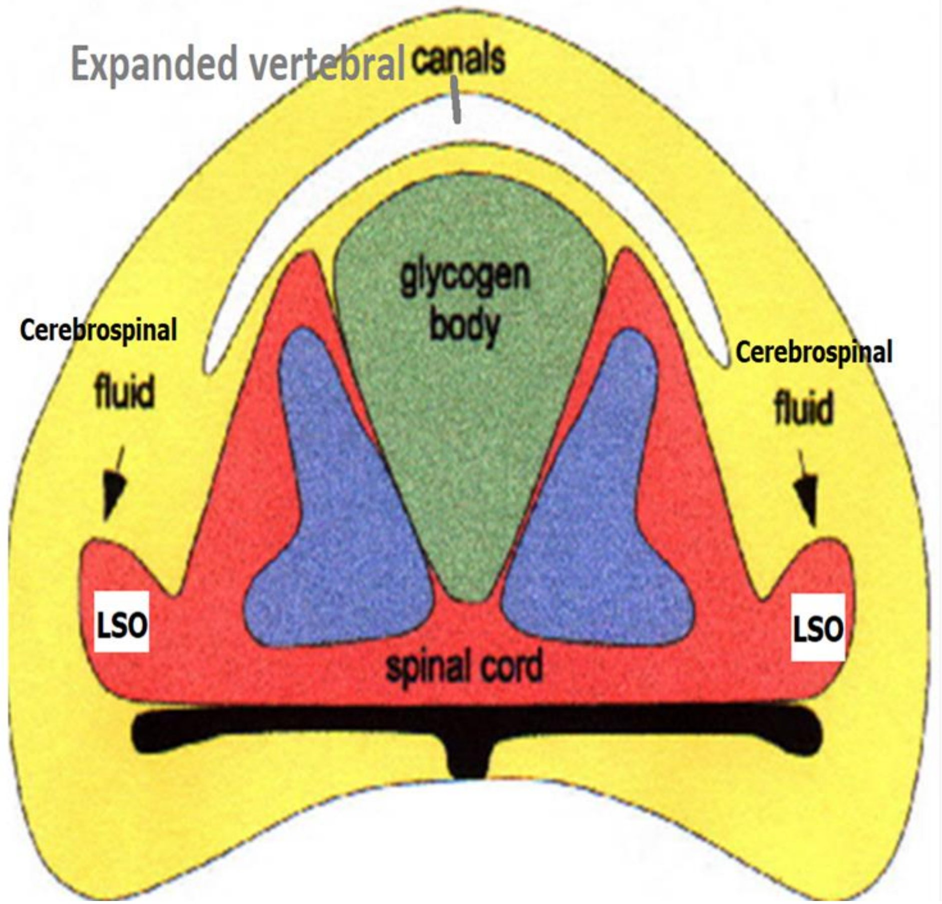
LSO is reminiscent (suggesting)**of a mass-spring accelerometer damped**(moist=slightly wet) **by spinal cord fluid**.

Previous work pointed out proximity of accessory lobes to neurocontrol units in spinal cord relevant for locomotion control



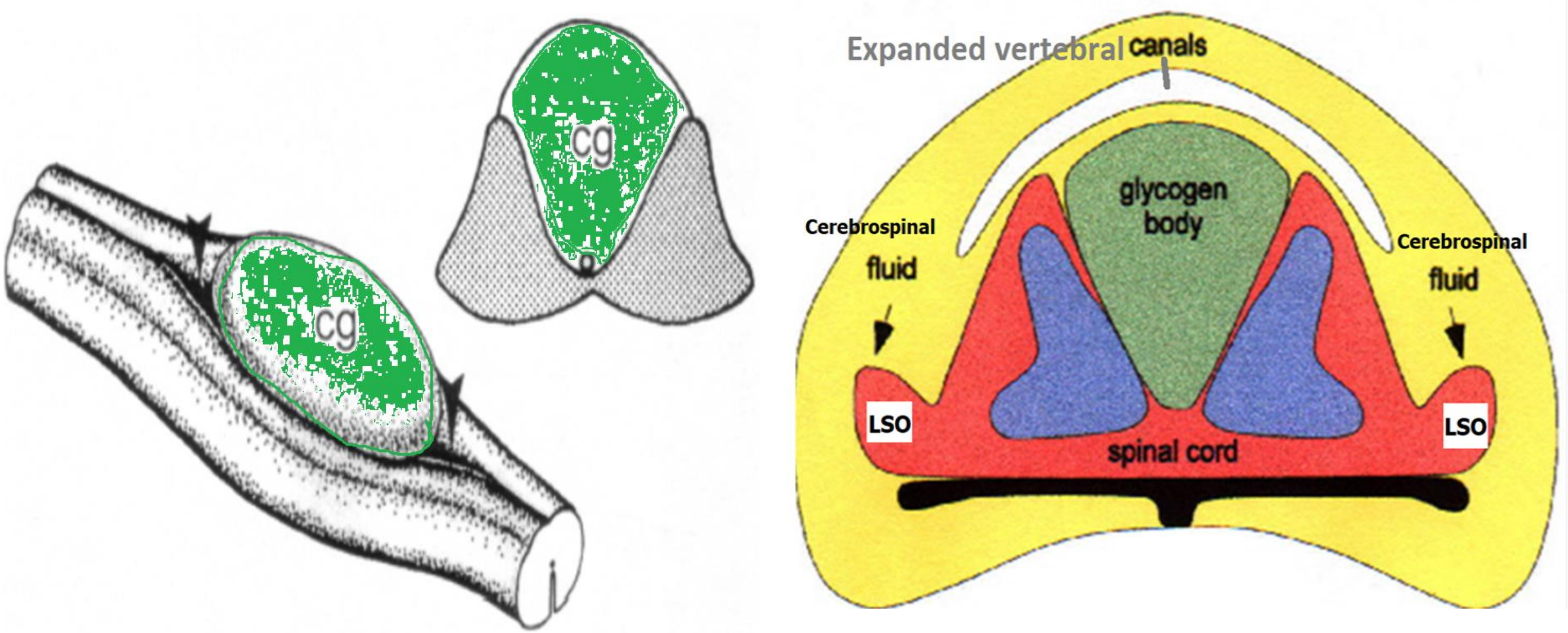
The glycogen body





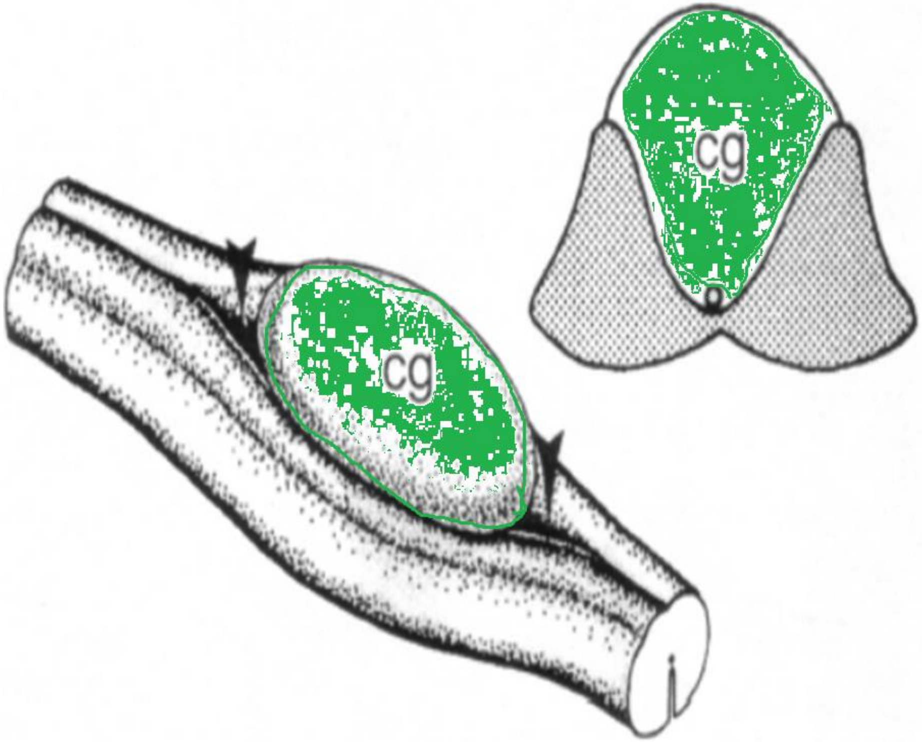
Egg-shaped, gelatinous mass of cells located at level of **lumbosacral enlargement**

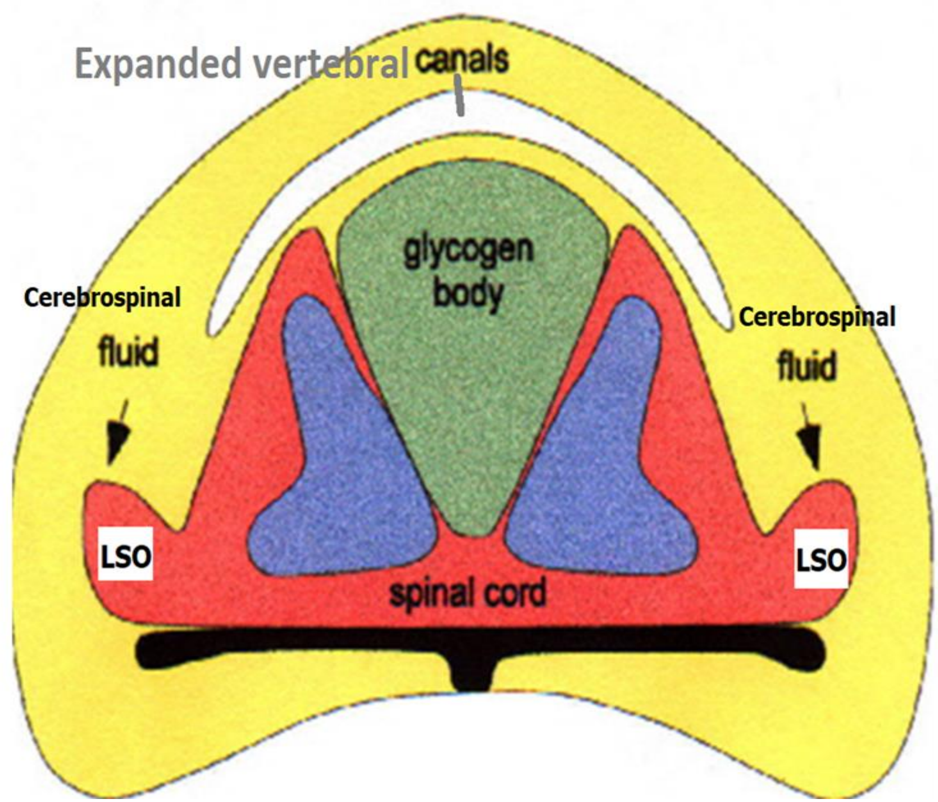
The glycogen body functions



Glycogen body is metabolically inert, may be to support the process of myelin formation in avian central nervous system (CNS).

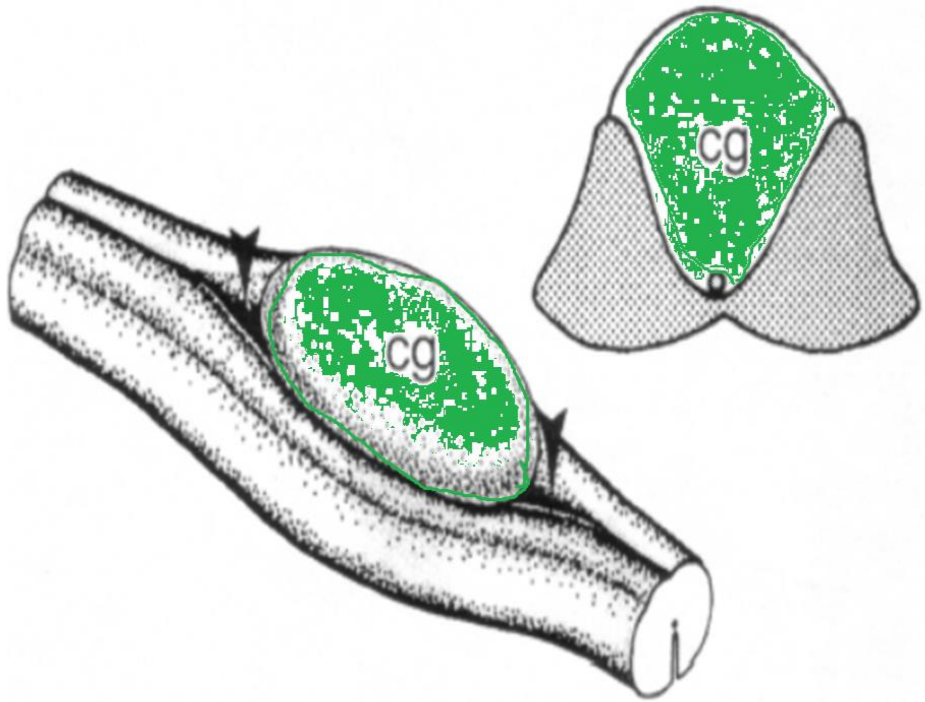
The glycogen body fuctions

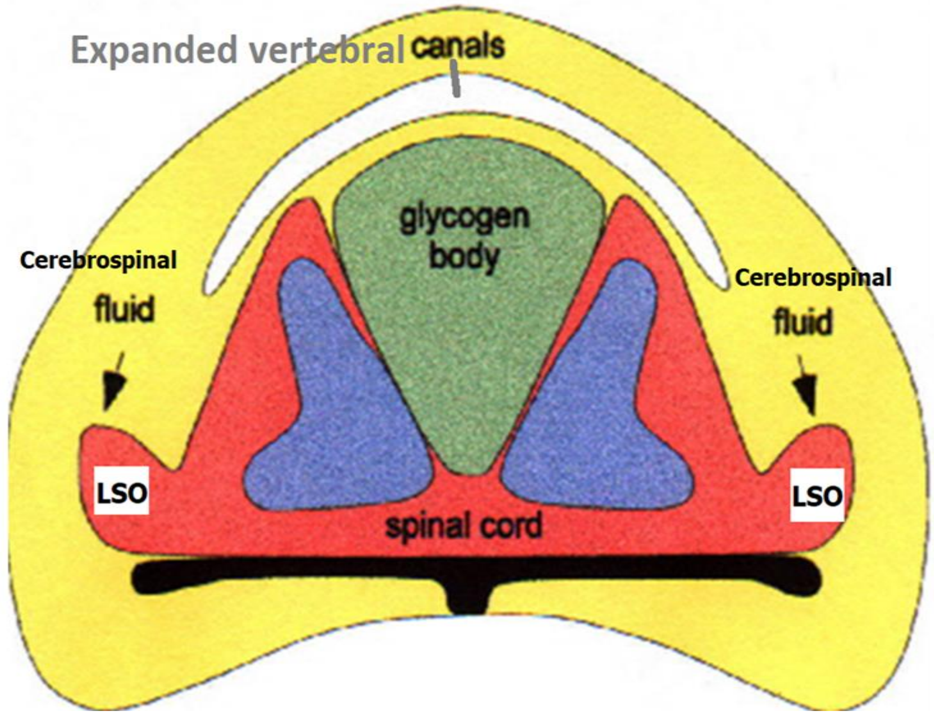




1-Serve as a **recyclable substrate** for production of reducing equivalents that would be available for synthesis of **myelin lipid cholesterol**.

The glycogen body functions





2-Serve as a **source of organic acids** which might provide **alternate substrates** to the CNS under conditions of **metabolic stress.**